



THE STATE OF INCLUSIVE INSTANT PAYMENT SYSTEMS IN AFRICA

SIIPS 2023

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About this report

The State of Inclusive Instant Payment Systems in Africa 2023 (SIIPS 2023) is in its second edition and will be repeated annually. The report aims to inform public-sector and private-sector players in Africa and beyond about the developments in the instant retail payment system (IPS) ecosystem in Africa, including an assessment of the inclusivity of such systems, both in functionality (accessible to all end-users) and governance (all licensed payment providers have fair

access and design input opportunities). For this report, only systems with live transactions and functionality as of June 2023 were included. The data in this report was gathered from publicly available resources and data shared by select central banks and switch operators in Africa from January to June 2023. It was supported by extensive stakeholder interviews during the same period. The consumer research was conducted between February and March 2023.

financially excluded. This points to the need to support the growth of DFS on the continent with investments in infrastructure, technology, technical skills, policy, and regulatory reforms, aimed at ensuring responsible access and usage of financial products and services.

AfricaNenda supports these needs through its mandate of driving financial inclusion in Africa through the deployment of inclusive instant payment systems (IIPS). AfricaNenda embraces a collaborative approach to working with other stakeholders within the payment ecosystem. Indeed, the words of the 'the Old Master' and philosopher Lao Tzu ring true, 'Do the difficult things while they are easy and do the great things while they are small. A journey of a thousand miles must begin with a single step'.

Payment Systems (SIIPS) in Africa 2023.. The SIIPS 2023 report aims to deepen insights and learnings around the development of inclusive instant digital retail payment systems in Africa. The first publication was well received by stakeholders in the payments industry and continues to be a critical resource for initiatives toward boosting payments infrastructure in the continent. It revealed priority areas in digital payments for Africa and promoted collaborative initiatives. The SIIPS 2022 map and data dashboard provided an easy way for IPS stakeholders to visualize instant payment system data from across the African continent, at regional and country levels. Insights from the SIIPS 2022 deliverables continue to inform decisions about enabling policies and regulations, and efficient IIPS implementation.

DO THE DIFFICULT THINGS
WHILE THEY ARE EASY AND DO THE
GREAT THINGS WHILE THEY ARE SMALL.
A JOURNEY OF A THOUSAND MILES
MUST BEGIN WITH A SINGLE STEP."

— Lao Tzu, philosopher

the state of the IP landscape. In this edition, we highlight changes in the availability of IPS at country and regional levels, the different types of IPS that exist, and how central banks, and the private sector each play key roles. The report emphasizes the inclusivity positioning of each operational IPS, which is critical if they are to provide the payment layer of digital public infrastructure in Africa, a necessary enabler of an inclusive digital economy (along with consent networks, digital ID, and trusted exchange). The report also includes findings from consumer and MSME end-user surveys on digital payments usage, which validate the importance of functionality and the extent to which IPS support use cases and trust as key IPS design considerations that drive inclusivity.

Lastly, the 2023 report showcases critical considerations to unlock cross-border payments through policy and regulatory harmonization. A harmonization framework for the continent could reduce the complexity and facilitate greater competition, which ultimately could lead to cheaper, faster, and more accessible cross-border digital payment options for customers. The resulting increase in scale and accessibility of cross-border payments would support crucial continental agendas, such as digital trade under the African Continental Free Trade Area (AfCFTA).

the central banks and IP operators of the Central African Economic and Monetary Community (CEMAC), Ghana, Madagascar, Malawi, Mozambique, Rwanda, South Africa, Zambia, and Zimbabwe. There remains a scarcity of available data and information, however. As a result, AfricaNenda had to rely on a mix of publicly available data and information instead of accessing these directly from institutions. We continue to advocate for the sharing of more data, more collaborations towards fulfilling Africa's digital public infrastructure needs, and enhanced cross-border transactions, for the benefit of all Africans.

AfricaNenda will continue to work with governments, central banks, Regional Economic Communities, development agencies, the private sector, and all DFS stakeholders to reduce the barriers to financial inclusion. We welcome collaborators to support the development of inclusive instant payment systems. We will continue our efforts to provide critical pre-project planning and program management support to expand IIPS projects, advocate for critical policy reforms, and enhance the capacity of African institutions and payments industry experts. We believe in our vision of universal financial inclusion in Africa by 2030, and we acknowledge the critical role that partnerships play in achieving it.

due to early barriers to adoption, these systems must be designed and deployed in a way that includes everyone, protects consumers, and bolsters competition and innovation. This report includes a framework to assess and guide the inclusivity of IPS, and highlights exemplar country and regional deployments that other countries can look to for inspiration and insight. It also highlights the critical opportunity for development and philanthropic organizations to support countries in these efforts through financial and technical assistance.

Much of the world is starting to focus on the incredible power of IPS as a foundational part of a digital public infrastructure (DPI). This is evidenced by the focus on DPI from the G20 this year, under India's leadership, as

so countries to design, build, and scale at least one DPI component in a safe and inclusive manner within five years. One of the most important and clearest ways to help achieve this goal is through IPS implementation in countries across Africa.

What would have seemed like only a dream a few years ago—interconnected inclusive instant payment systems across every country in Africa—is now within reach. The impacts of this development on financial inclusion, remittances, trade, and overall inclusive economic growth are starting to be realized, yet there is so much more to be imagined. We must all rise to the occasion to keep up this incredible momentum and the SIIPS 2023 report provides a clear roadmap of how we can make this happen.

CHAPTER 1 52

INTRODUCTION

CHAPTER 2 56

THE LANDSCAPE OF INSTANT PAYMENT SYSTEMS

2.1	The IPS landscape has seen modest change since 2022	57
2.2	IPS fall under one of four types.....	60
2.3	Geographic IPS coverage gaps persist in 2023	68
2.4	IPS operate across a range of transactions, channels, use cases, and participants	76
2.5	Standards and technical integration can facilitate trust	92
2.6	Most IPS offer only a basic level of inclusivity	94

usage, and habitual usage.....	112
3.3 Summary of consumer research findings	127
3.4 Consumer barriers can be partially considered in IPS design	130

CHAPTER 4 132

BARRIERS TO AND OPPORTUNITIES FOR INCLUSIVITY IN INSTANT PAYMENT SYSTEMS

4.1	The IPS business model requires scale to drive usage	134
4.2	IPS need a compelling value proposition to encourage PSP participation	141
4.3	IPS must increase digital financial inclusion for women	146
4.4	IPS must expand merchant and government payment use cases	151
4.5	Technology standards can facilitate IPS adoption and efficiency	156

5.1	Why is regulatory harmonization important?.....	162
5.2	Which regulation should be harmonized?.....	170
5.3	How to harmonize regulation?.....	176
5.4	In summary	184

CHAPTER 6 186

FUTURE PERSPECTIVES

6.1	Market trends	188
6.2	System trends	191
6.3	Consumer trends	194

CHAPTER 7 196

CONCLUSION

B.	Consulted stakeholders	199
C.	Landscaping data tables	220
D.	Dispute resolution in instant payments.....	225
E.	Instant payments fraud	228
F.	Customer research methodology.....	235
G.	Cross-border regulation tables	238
H.	Case studies.....	244
	Rwanda: eKASH	244
	Origin story	244
	Governance and operations	246
	Learnings related to inclusivity	251
	Zambia: National Financial Switch	252
	Origin story	252
	Governance and operations	254
	Inclusivity learnings	259
	Malawi: NATSWITCH	260
	Origin story	260
	Governance and operations	262
	Inclusivity learnings	267
	CEMAC: GIMACPAY	268
	Origin story	268
	Governance and operations	270
	Inclusivity learnings	274

BOX 2.11 Initiatives to grow the participant ecosystem.....	91
BOX 2.12 NFS in Zambia as of June 2023	96
BOX 3.1 Customer research methodology overview	102
BOX 3.2 User experience: The impact of limited consumer redress on trust in Morocco.....	117
BOX 3.3 User experience: Transaction journey experiences highlight usage drivers and barriers	120
BOX 3.4 User experience: Risks and challenges women face in using digital payments.....	121
BOX 3.5 User experience: Core digital payment drivers for merchants.....	122
BOX 3.6 User experience: Challenges faced by agents in Malawi.....	124
BOX 3.7 User experience: Discontinuing digital payment usage due to fraud experience	126
BOX 3.8 User experience: The benefits of accessible customer service and recourse mechanisms in Senegal.....	127
BOX 5.1 Operational barriers for cross-border payments include inconsistent messaging standards, stringent requirements set by correspondent banks, expensive liquidity management, and costly and opaque foreign exchange pricing.....	169
BOX 5.2 Definition of harmonization	170
BOX 5.3 Regional economic communities in Africa	175
BOX D.1 Examples of consumer protection guidelines for IPS participants.....	225
BOX D.2 Example of the Ombudsman for Banking Services in South Africa (OBSSA).....	227
BOX E.1 SIM card activation guideline examples	232
BOX E.2 Examples of ecosystem monitoring approaches adopted by IPS.....	233

FIGURE 2.6 Channels facilitated by IPS type, multiple mentions (n=30).....	81
FIGURE 2.7 IPS instruments supported, multiple mentions (n=31)	83
FIGURE 2.8 Enabled use cases by IPS type, multiple mentions (n=31)	84
FIGURE 2.9 IPS value chain	85
FIGURE 2.10 Median number of participants per IPS type (n=28)	88
FIGURE 2.11 Inclusivity spectrum	94
FIGURE 2.12 Mapping of IPS across the inclusivity spectrum	98
FIGURE 3.1 Cross-country analysis—frequency of digital payment usage	105
FIGURE 3.2 Average volume of weekly transactions	106
FIGURE 3.3 Frequency of digital payment usage per individual user groups.....	107
FIGURE 3.4 Frequency of digital payment usage per MSME user groups	107
FIGURE 3.5 Average number of transactions per week for the different user groups	108
FIGURE 3.6 Barriers and drivers based on the access, early usage, and habitual usage framework.....	114
FIGURE 3.7 Proportion of non-users of digital payments naming the respective barrier as their top reason	115
FIGURE 3.8 Proportion of respondents that indicated the respective early usage barrier as their top-most barrier	116
FIGURE 3.9 Proportion of respondents naming each as their top-most barrier to habitual usage digital payments	119
FIGURE 3.10 Transaction journey example for an app	120
FIGURE 3.11 Transaction journey example for USSD	120
FIGURE 3.12 Transaction costs as a barrier to users	125

FIGURE H.9 | Natswitch model overview262

FIGURE H.10 | Natswitch transaction flows263

FIGURE H.11 | Natswitch transaction volumes and values266

FIGURE H.12 | GIMACPAY timeline269

FIGURE H.13 | GIMACPAY model overview270

FIGURE H.14 | GIMACPAY transaction flows271

FIGURE H.15 | GIMACPAY transaction volumes and values273

LIST OF MAPS

MAP 0.1 | There are 32 active domestic and regional IPS in Africa as June 202336

MAP 0.2 | Domestic IPS in development (n=17)38

MAP 0.3 | Planned regional IPS (n=3)40

MAP 2.1 | Map of 29 active domestic IPS in Africa as of June 202368

MAP 2.2 | Map of countries without domestic IPS (size of adult population)69

MAP 2.3 | Map of active regional IPS in Africa as of June 202371

MAP 2.4 | Domestic IPS in development (n=17)73

MAP 2.5 | Regional IPS in development (n=3)74

MAP 2.6 | Cross-border IPS functionality summary75

and their level of digitalization113

TABLE 3.6 | The most common frequent payment use cases among MSME respondents and their level of digitalization114

TABLE 3.7 | Summary of customer research findings130

TABLE 5.1 | Barriers to cross-border retail payments for PSPs166

TABLE 6.1 | Market, system, and consumer trends189

TABLE C.1 | Data table222

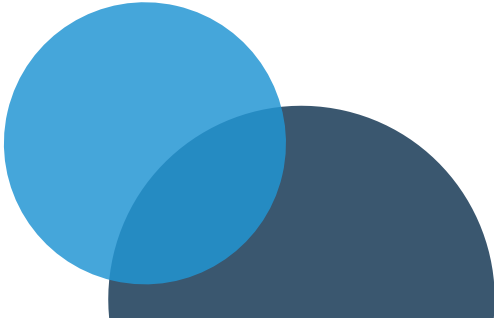
TABLE C.2 | IPS inclusivity level scoring226

TABLE F.1 | Detailed sampling breakdown239

TABLE G.1 | Regional bodies and respective focus areas241

TABLE G.2 | Regional divergence in key regulations242

TABLE G.3 | Existing regional harmonization initiatives underway244



ATM	Automated teller machine
AU	African Union
B2B	Business-to-business
BAM	Bankers Association of Malawi
BCEAO	Banque Centrale des États de l'Afrique de l'Ouest
BEAC	Banque des États de l'Afrique Centrale
BIS	Bank for International Settlements
BNR	National Bank of Rwanda
BOG	Bank of Ghana
BoP	Balance of payments
BOZ	Bank of Zambia
CBC	COMESA Business Council
CBDC	Central bank digital currency
CBN	Central Bank of Nigeria
CCAF	Cambridge Centre for Alternative Finance
CCBG	Committee of Central Bank Governors
CDD	Customer due diligence
CEMAC	Economic and Monetary Community of Central Africa
CTF	Combatting of financing of terrorism

DFI	Digital financial service
DIAL	Digital Impact Alliance
DNS	Deferred net settlement
DPI	Digital public infrastructure
DPO	Direct Pay Online
DRC	Democratic Republic of the Congo
EABC	East African Business Council
EAC	East African Community
ECCAS	Economic Community of Central African States
ECOWAS	Economic Community of West African States
EECA	Eastern Europe and Central Asia
EFT	Electronic funds transfer
e-ID	Electronic identity
eKYC	Electronic know your customer
EMV	Europay, MasterCard, and Visa
ERCA	ECOWAS Regional Competition Authority
ESAAMLG	Eastern and Southern Africa Against Anti-Money Laundering Group
EU	European Union

	Settlement System
GIABA	Inter-Governmental Action Group against Monetary Laundering in West Africa
GIMAC	Groupement Interbancaire Monétique l'Afrique Centrale
GIP	GhIPSS Instant Pay
GNI	Gross national income
HDCT	Human Development Cash Transfer
ICT	Information and communications technology
ID	Identity document
IDI	In-depth interview
IGAD	Intergovernmental Authority on Development
IIPS	Inclusive Instant Payment System
IMF	International Monetary Fund
IPS	Instant payment system
ISO	International Organization for Standardization
KYC	Know your customer
LCSF	Local Currency Settlement Framework
LEAP	Livelihood Empowerment Against Poverty

NFC	Near-field communication
NFS	National Financial Switch
NIBSS	National Inter-Bank Settlement System
NIP	NIBSS Instant Payment
NSSF	National Social Security Fund
OBSSA	Ombudsman for Banking Services in South Africa
OMAC	Office monétique de l'Afrique Centrale
P2B	Person-to-business
P2P	Person-to-person
PAN	Payments Association of Namibia
PAPSS	Pan-African Payment and Settlement System
PF	Proliferation financing
POS	Point-of-sale
PPP	Public-private partnership
PSD	Payment Service Directive
PSOC	Payment Service Oversight Committee
PSP	Payment service provider
QR	Quick response

	Community
SARB	South African Reserve Bank
SDG	Sustainable development goal
SEPA	Single Euro Payments Area
SIIPS	State of Inclusive Instant Payment Systems
SIM	Subscriber identity module
SIMO	Sociedade Interbancaria De Mocambique
SLA	Service-level agreement
SMAC	Société monétaire de l'Afrique Centrale
SS7	Social Signalling 7
SSA	Sub-Saharan Africa
SYGMA	Système de gros montants automatisé
SYRAD	Système de règlement automatisé de Djibouti
SYSTAC	Système de télécompensation en Afrique Centrale

VASP	Virtual asset service provider
WAEMU	West African Economic and Monetary Union
WAFEM	West African Institute for Financial and Economic Management
WAMA	West African Monetary Agency
WAMI	West African Monetary Institute
WAMZ	West African Monetary Zone
XAF	Central African Franc
XOF	West African Franc
ZAR	South African Rand
ZECHL	Zambia Electronic Clearing House Limited
ZIPIT	Zimswitch Instant Payment Interchange Technology



Computerized telecommunications devices that provide financial institution clients with access to financial transactions in a public place (World Bank 2020b).



A mobile app is a front-end, in-between service that authorizes and processes payments between a user's payment portal (mobile device) and a vendor's bank or financial intermediary, including non-banks. It performs the encryption of cardholder data, authorization of payment requests, confirmation of purchases, and so on (Slesar 2022).



A system is available for use 24 hours a day, 365 days of the year, excluding planned maintenance or system downtime.



The BoP is a statistical statement that summarizes transactions between residents and non-residents during a given period. The BoP statement provides a clear picture of the economic relations between different countries and consists of three main components: current accounts, capital accounts, and financial accounts (UNCDF 2022c).



Bill payments

A payment made by a person from their bank, mobile money accounts, or other financial stores of value, to a biller or billing organization via a digital payment platform in exchange for the services provided (GSMA 2021a).



Branch

A bank's storefront location with a bank teller that handles cash deposits, withdrawals, and payment for goods and services.



Browser

Access for a consumer to make a payment electronically via a web page, linking the payer to the account details of their bank or financial service provider.



Central bank IPS

Typology term for the purpose of this report. The IPS is governed by the central bank.



Cross border

The movement of funds between financial institutions within two distinct countries. The cross-border transaction, which can be a range of payment use cases, often requires intermediaries operating in multiple jurisdictions (BIS 2021).



Cross-domain IPS

Typology term for the purpose of this report. A system that provides for all-to-all interoperability where switching, clearing, and exchanging instruments is contained within one overarching system. Cross-domain systems provide access to banks and non-banks and support transactions from both bank accounts and mobile money accounts. All-to-all interoperability includes the ability for end-users to directly transact between wallet accounts at different mobile money operators (MMOs), between mobile money accounts and bank accounts, and across bank accounts. Within one system, there are different rules to accommodate various instruments. The single system provides the governance framework and coordinates the operational functions end-to-end for the various instruments (GSMA 2014).



Debit card

A payment instrument linked to a depository account, such as an on-demand deposit, savings, or transfer account. It can be used to make both debit and credit transactions between accounts, as well as between cards (PASA 2022b). Although technically a pull payment, the locus of control is often with the payer, meaning debit cards can essentially function as a push payment.

¹ Nostro accounts are accounts owned by one financial institution but housed within another, where the financial institution could be a bank, MMO, or other payment service provider with stored-value accounts.



Digital

IPS definition term for the purpose of this report. A system that is electronic with services that are accessible on digitally enabled devices.



Digital public good

Digital public goods are open-source software, open data, open AI models, open standards, and open content that adhere to privacy and other applicable laws and best practices, do no harm by design, and help attain the Sustainable Development Goals (SDGs; Digital Public Goods Alliance 2023).



Digital public infrastructure

Digital solutions and systems that facilitate essential society-wide functions and services such as:

- Identification: the ability for people and businesses to securely verify their identity, as well as complementary trust services such as electronic signatures and decentralized, verifiable credentials.
- Payments: transferring money between people, businesses, and governments.
- Data exchange: flow of data across government and the private sector, with safeguards for personal data protection, including consumer consent (World Bank 2023a).



Fintech company

Financial technology company aiming to replace or enhance financial services provided by existing financial institutions.



Harmonization

Regulatory bodies in two or more countries agree to a set of regulatory frameworks/standards and/or establish a similarity in processes/services.



Indirect system participant

Payment value chain participants who do not have a technical integration with the central switch or settlement services and participate in the system via an integrated PSP/direct system participant.



Inclusive instant payment system

Inclusive instant payment systems process retail transactions digitally in near real-time and are available for use 24 hours a day, 365 days a year, or as close to that as possible. They enable low-value, low-cost push transactions that are irrevocable and based on open-loop and multilateral interoperability arrangements. Licensed payment providers have fair access to the system, and participants have equal input opportunities into the system. The central bank has a role in system governance. End-users have access to a full range of use cases and channels, as well as transparent and fit-for-purpose recourse mechanisms.



International
Organization for
Standardization (ISO)
20022

Introduced in 2004, ISO 20022 has become the standard exchange for new instances of electronic messaging and is used by most financial service providers for payment as well as non-payment transactions (World Bank 2021h).



ISO 8583

The most common messaging standard for card payments, ISO 8583 was established by the ISO in 1987 (World Bank 2021h).



Jointly owned

Where the central bank and private participants own the infrastructure jointly (World Bank 2021f).



Low-value payments

IPS definition term for the purpose of this report. Transactions of less than \$5.



Mobile money
operator

A mobile network operator, or an entity that has partnered with a mobile network operator, that provides mobile money services, a pay-as-you-go digital medium of exchange and store of value that operates independently of a traditional banking network (IMF 2022b).



Multilateral
interoperability

The permission structure for payment instruments belonging to a given system to be used in platforms developed by other systems, including in different countries. Multilateral interoperability involves a situation in which payment instruments that belong to a given system may be used in platforms developed by other systems, including in different countries. Multilateral interoperability involves the coexistence of multiple attributes, which can be combined in various ways. These attributes fall into three broad dimensions: technical, semantic, and business interoperability (BIS 2021).² The nature of the business interoperability rules determines whether a payment system is multilateral, but does not dictate the number of providers, platforms, systems, or jurisdictions.



Near-field
communication

A standards-based, short-range (a few centimeters) wireless connectivity technology that enables simple and safe two-way interactions between electronic devices, allowing consumers to perform contactless transactions, to access digital content, and to connect electronic devices with a single touch (BIS 2020).

² Technical interoperability involves the technical connections and exchange of data, whereas semantic interoperability requires data to be interpreted and acted upon consistently (BIS 2021). Business interoperability involves commercial agreements that provide standing rules and assurances for the exchange of different commercial instruments and associated risks between different schemes, platforms, and participants, including in different jurisdictions (World Bank 2012).



Open loop

At least a multilateral or third-party transparent interoperability arrangement, excluding closed-loop, on-us systems.



Overseer

A person who continually monitors the system and assesses how safely and efficiently it is operating (BIS 2016). They are responsible for assessment and monitoring of the system and enforcement of laws and regulations to promote safe and efficient payments. The system overseer can enforce policy mandates and is the main arbitrator of fairness or application of the scheme rules (CGAP 2021).



Participant-owned

Where the system is owned privately by its participants (World Bank 2021h).



Partially owned

Where ownership of components of the system is split between the central bank and private participants (World Bank 2021f).



Point-of-sale device

A specialized device that is used to accept the payment (for example, a card reader) at a retail location where payments are made for goods or services (GSMA 2021a).



Private-association IPS

Typology term for the purpose of this report. The IPS is governed by an association made up entirely of private-sector participants.



Proxy ID

An identifier (for example, e-mail address, mobile phone number) that may be used in lieu of the payer's or payee's transaction account information. These allow the public and the business sector to transact in a seamless manner while initiating a payment (World Bank 2021e).



Public-Private Partnership IPS

Typology term for the purpose of this report. The IPS is governed by a partnership arrangement consisting of the central bank and a representative of private IPS participants.



Real time

The value transfer is assured to be instant (within seconds).



Real-time settlement

When transactions are settled continuously as they occur (World Bank 2021b).



Recourse mechanisms

The mechanisms in place for consumers using the IPS to raise grievances and have them heard and resolved or redressed (CGAP 2013).



Regulator-owned

Where the central bank determines the procedures, and it controls the associated technical infrastructure (World Bank 2021f).



Spoofing

Faking the sending address of a transmission to gain illegal entry into a secure system (NIST 2023).



Central bank e-money

A digital currency that is collateralized or guaranteed by the central bank. These classes of instruments can enable settlement processes and services across e-money and other PSP providers, including indirect access to central bank reserves. All other functions are the responsibility of private e-money providers underpinned by adequate regulation.



System governance body

Responsible for strategic direction, including any explicit inclusivity mandate (pro-poor governance), and accountability of IPS participants. Their function is related to control over scheme management (CGAP 2021).



System owner

Responsible for and entitled to receive all the benefits and risks associated with ownership of the system (BIS 2003).



Sovereign currency IPS

Typology term for the purpose of this report. CBDC IPS combines a sovereign currency instrument and value transfer system that can provide a unified digital value transfer mechanism between commercial instrument systems, institutional stakeholders, and individuals within an economy.



Social Signalling 7 (SS7)

A telecommunication signaling architecture traditionally used for the set up and tear down of telephone calls. It has a robust protocol stack that uses out-of-band signaling to communicate between elements of the public switched telephone network. In recent years it has been superseded by the Diameter signaling protocol (Techopedia 2017).



Taxes and fees

Obligations that individuals pay to central, regional, and local public administrations, such as tax payments or utility payments (World Bank 2021b).



Third-party interoperability

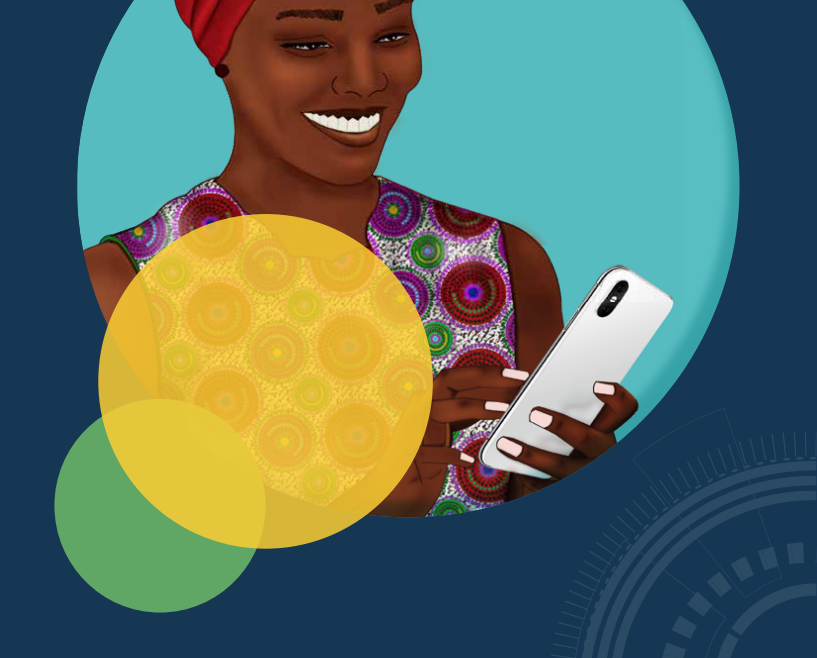
The foundation for the interoperability of IPS participants via a centralized switching or clearing layer, facilitated by a third party. In some, but not all, countries the third party is an aggregator (CGAP 2016). The third party can be a private entity or government owned. Interoperability is achieved when providers connect to the switch.



Transaction receipts

Notifications sent to consumers, via text, email, or other communication methods, that confirm the initiation or completion of a transaction. It should include information about the digital finance service provider, the location, the amount of the transaction, and identification detail, as well as details of the counterparty (World Bank 2021b).





SIIPS 2023 finds that the number of IPS systems and overall transaction volume has grown. There are now

across many adoption, risk, and usage; policy and regulatory harmonization can help to overcome some of them to enhance seamless cross-border payments.

What is an instant payment system and when does it become inclusive?⁴



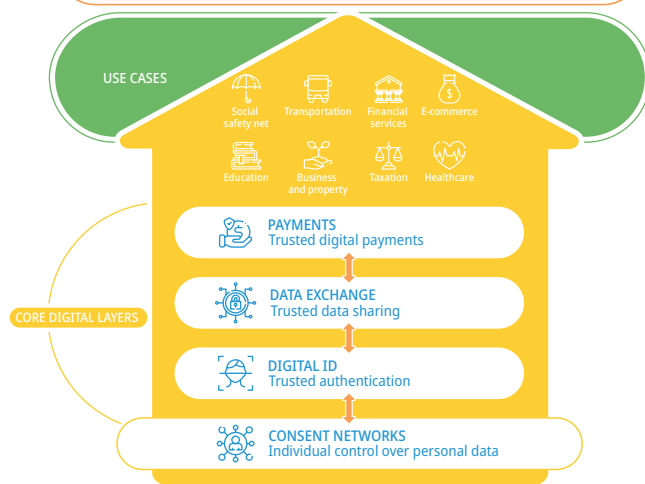
Instant payment systems (IPS) are retail payment systems that are multilateral—and open loop—and that enable digital push payments in near real time for use 24 hours a day, 365 days a year, or as close to that as possible.



Inclusive instant payment systems (IIPS) process retail transactions digitally in near real-time and are available for use 24 hours a day, 365 days a year, or as close to that as possible. They enable low-value, low-cost push transactions that are irrevocable and based on open-loop and multilateral interoperability arrangements. Licensed payment providers have fair access to the system, and participants have equal input opportunities into the system. The central bank has a role in system governance. End-users have access to a full range of use cases and channels, as well as transparent and fit-for-purpose recourse mechanisms.

³ CIMAOPAY in the Economic and Monetary Community of Central Africa (CEMAC) region, Netwitch in Malawi, etash in Rwanda, and National Financial Switch (NFS) in Zambia.

⁴ The definitions used in this report are, in principle, aligned with the definition of the Committee on Payments and Market Infrastructures (CPMI) but seek to emphasize a few specific aspects that are relevant from a financial inclusion context in several low-income countries—notably, mobile money accounts and push payments. Given this, even solutions that enable users of different mobile money providers to make and receive transfers in real time are considered under this definition, though the limitations of such arrangements are recognized in the different categorizations of IPS.



Source: Adapted from World Bank 2022g and Digital Impact Alliance (DIAL) 2023

LAUNCH YEAR		2022 TRANSACTION VALUES RELATIVE TO GNI
2006	RTC (South Africa)	19%
2011	ZIPIT (Zimbabwe)	3%
2011-2021	NIP (Nigeria), Nigeria mobile money, eNaira	186%
2012	SIMO (Mozambique)	23%
2015	NatSwitch (Malawi) (off-us) and NatSwitch (on-us)*	2.6%
2015-2016	GIP, Ghana MMI (off-us), Ghana MMI (on-us)*	129%
2016	Madagascar mobile money	72%
2016	Uganda mobile money	124%
2016	Taifa Moja (Tanzania)	89%
2017-2018	Kenya mobile money and PesaLink	93%
2018	NFS (Zambia)	10%
2019	MauCAS (Mauritius)	1%
2021	NamPay (Namibia)	0.2%
2021	Somalia Instant Payment System	<0.1%
2022	InstaPay (Egypt)	1%
2022	eKash (Rwanda)	0.1%
2022	EthSwitch (Ethiopia)	0.3%

● Cross-domain IPS
 ● Bank IPS
 ● Mobile money IPS
 ● Sovereign currency IPS
 ⊗ On-us transactions

*NatSwitch (Malawi) and Ghana MMI are the only IPS where information on on-us transaction data is available.

5 Note that measurement inconsistencies between countries affect comparability and aggregation accuracy of reported data. Some countries include on-us transactions (internal transactions between account holders of one provider), which increases values and volumes compared to other countries, and some only include mobile money cash-in and cash-out transactions, which understate values and volumes.



- Pricing models that can compete with cash and existing closed-loop solutions to incentivize uptake.
- Inclusive services for end-users, including effective agent channels and recourse mechanisms.



A compelling provider value proposition

- Continued roll-out of a portfolio of scale- and value-driving use cases to increase network touchpoints and keep digital value in circulation.
- An emphasis on value-added services including proxy IDs, centralized fraud and cybersecurity facilities, as well as centralized electronic know your customer (eKYC) and customer due diligence (CDD) facilities.
- Development of open APIs to promote open banking and foster a competitive landscape.



A conducive policy environment

- Continued improvement of the supporting ecosystem: risk-based and harmonized licensing of PSPs; network upgrades; sustained roll-out of agent networks; increased penetration of smartphones, broadened coverage areas for mobile data, and more affordable data access.
- Building out a principles-based regulatory framework for consumer protection and data privacy, and moving towards risk-based supervision.
- Emphasis on regional harmonization of policy and regulation for cross-border payments and transfers to enable IPS to catalyze digital trade and remittances.



Landscape snapshot

There are 29 live domestic systems across 21 countries and three live regional systems. This diverse landscape has relatively few new IPS, though that is to be expected since IPS take multiple years to implement and often require extensive industry and/or regulator consultation and technical expertise. Over the last year, the IPS landscape in Africa has evolved in the following ways (see Map 0.1):

- Three new systems, EthSwitch (Ethiopia), Virement Instantané (Morocco) and PayShap (South Africa) went live.

- Three systems, Meeza Digital (Egypt), MauCAS (Mauritius), and eKash (Rwanda), have been reclassified as cross-domain IPS. Meeza Digital and eKash had been classified previously as mobile money IPS and MauCAS was a bank IPS in SIIPS 2022.
- Seven countries have multiple IPS; only in Ghana are these systems interoperable with one another.⁶

⁶ Egypt, Ghana, Kenya, Morocco, Nigeria, South Africa, Tanzania.

- NIBSS Instant Payment (NIP)
- eKash
- Nigeria mobile money

GHANA *Interoperability*

- GhIPSS Instant Pay (GIP)
- Ghana mobile money interoperability (Ghana MMI)

ZAMBIA

- National Financial Switch (NFS)

NAMIBIA

- NamPay

SOUTH AFRICA

- Real-Time Clearing (RTC)
- PayShap

ZIMBABWE

- ZimSwitch Instant Payment Interchange Technology (ZIFT)

KENYA

- PesaLink
- Kenya mobile money

TANZANIA

- Taifa Moya
- Tanzania Instant Payment System (TIPS)

MALAWI

- NatSwitch

MAURITIUS

- Mauritius Central Automated Switch (MauCAS)

MADAGASCAR

- Madagascar mobile money

MOZAMBIQUE

- Sociedade Interbancária de Moçambique (SIMO)

IPS: Instant payment system

● **PAN-AFRICAN PAYMENT AND SETTLEMENT SYSTEM (PAPSS) PILOT**
IN WEST AFRICAN MONETARY ZONE (WAMZ)
The Gambia, Ghana, Guinea, Liberia, Nigeria, Sierra Leone

● **GIMACPAY IN CENTRAL AFRICA MONETARY AND ECONOMIC COMMUNITY (CEMAC)**
Cameroon, Central African Republic, Chad, Republic of Congo, Equatorial Guinea, Gabon

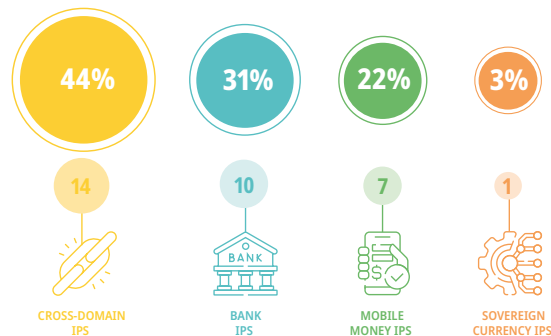
3 REGIONAL-LEVEL IPS

3 MORE UNDER DEVELOPMENT

● **TRANSACTIONS CLEARED ON AN IMMEDIATE BASIS (TCIB)**
IN SOUTHERN AFRICAN DEVELOPMENT COMMUNITY (SADC)

Angola, Botswana, Comoros, Democratic Republic of Congo (DRC), Eswatini, Lesotho, Madagascar, Malawi, Mauritius, Mozambique, Namibia, Seychelles, South Africa, Tanzania, Zambia, Zimbabwe

FIGURE 0.3 | Breakdown of IPS by type (n=32)



mostly focuses on providing domestic and regional IPS capabilities to close the gap for the CEMAC countries in Central Africa; the Pan-African Payment and Settlement System (PAPSS), which is in pilot in the West African

Several new domestic systems are on the horizon. Map 0.2 shows the 17 domestic systems under development across the continent:



MAP 0.3 | Planned regional IPS (n=3)

COMESA

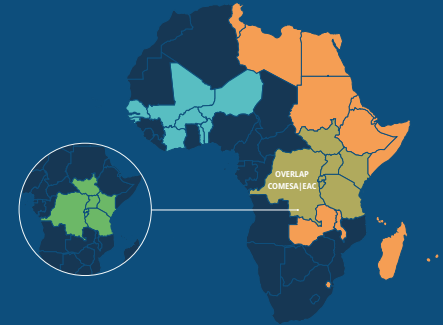
Burundi, Comoros, DRC, Djibouti, Egypt, Eswatini, Eritrea, Ethiopia, Kenya, Libya, Madagascar, Malawi, Mauritius, Rwanda, Seychelles, Somalia, Sudan, Tanzania*, Tunisia, Uganda, Zambia, Zimbabwe.

EAC

Burundi, DRC, Kenya, Rwanda, South Sudan, Tanzania, Uganda.

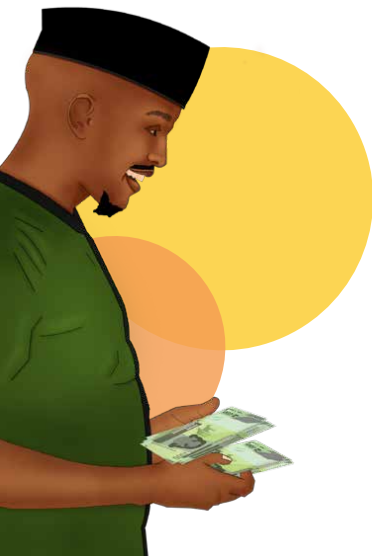
WAEMU

Benin, Burkina Faso, Côte d'Ivoire, Guinea-Bissau, Mali, Niger, Senegal, Togo.



*Tanzania is not a COMESA member state but will integrate with the COMESA regional IPS.

Although Nigeria, Madagascar and Morocco can access Buxa, a cross-border payment system supported by the Arab central banks, and spanning the Middle East, South Asia, and North Africa.



payments. P2B use cases are also on the rise, and three-quarters of domestic systems support both. In contrast, only 31% of IPS support business-to-business (B2B) payments. Critical use cases, such as government-to-person (G2P), person-to-government (P2B) or other government-linked services remain unavailable in most African countries. Only six IPS (GIP and MMI in Ghana, Madagascar mobile money, MarocPay in Morocco, NIP in Nigeria, and Uganda mobile money) support G2P payments, and NIP and the Ghanaian systems are the only IPS that enable the broadest range of use cases.



USSD remains the most prevalent channel, closely followed by apps. Seventy-five percent of IPS in Africa support USSD; most of these IPS are mobile money or cross-domain systems. App channels are the second most prevalent at 72%, but require smartphone functionality and internet connectivity, which continue to be a barrier for the majority of African citizens. Other channels, such as quick response (QR) codes, are on the rise.

The channels an IPS provides is often based on the type of participants in the system: bank IPS and cross-domain IPS support the largest variety of channels, while mobile money schemes are typically limited to mobile money operators (MMO) as participants. This explains the dominance of agent and USSD channels for these systems. Several systems have adopted a “channel agnostic” approach. This means that the system does not provide for a specific set of channels, but rather permits PSP participants to determine what channels they provide.



systems, notably by local participants (e.g. consumer MMOs dominate in mobile money IPS). Standard Bank Group and Ecobank participate in the most systems. The four MMOs with the largest IPS footprint are Airtel, MTN, Orange Mobile, and Vodacom (whose footprint includes Safaricom in Kenya). Fintechs play a vital role along the IPS value chain by providing innovative technologies to end-users, IPS operators, and payment system providers. They typically partner with direct participants to provide front- or back-end services, but as technical service providers they are facilitators rather than system actors.

Some progress toward inclusivity, but no systems yet qualify as “mature”

No IPS has reached the **mature** level of inclusivity yet as shown in the IPS Inclusivity Spectrum in Figure 0.4. Most systems either are unranked or meet only basic-level inclusivity criteria.

The five **progressed** IPS are all on their way to the **mature** level. These include four domestic systems covering three countries (Ghana, Malawi, and Zambia) and one regional system serving six (Cameroon, Central African Republic, Chad, Republic of Congo, Equatorial Guinea, Gabon). As a result, nine countries have access to progressed IPS functionality as of 2023. The two systems in Ghana, however, have plans to roll out an expanded set of use cases, which will bring them closer to maturity. Transparent consumer recourse mechanisms remain the most complex element to implement.

- Not ranked if the IPS does not fulfill basic inclusivity criteria where it does not enable P2B and P2P transactions or does not offer channels that are most used.
- If the IPS does not provide enough information/data on the public domain or via direct consultation that enables inclusivity assessment.

- **Minimum channel functionality:** most-used channel is supported
- **Minimum use case functionality:** P2P and P2B transactions are supported

- In addition to basic-level criteria:
- **Participation by all PSPs** cross domain IPS enabling all-to-interoperability
 - **Pro-poor governance** inputs possible by all PSPs or there is an explicit inclusivity mandate
 - **Central bank involvement** in governance

- In addition to basic- and progressed-level criteria:
- **Expanded use cases** supported
 - **Standards and monitoring** of provision of **consumer resource mechanisms**
 - **Low-cost for end-users** within a not-for-loss business model

- SYRAD (Djibouti)
- EthSwitch (Ethiopia)
- Kenya mobile money
- PesaLink (Kenya)
- Virement Instantané (Morocco)
- Nigeria mobile money
- eKash (Rwanda)
- Somalia National Payment System
- PayShap (South Africa)
- Tunisia mobile money
- PAPSS (Africa)
- TCIB (SADC)

- InstaPay (Egypt)
- Meeza Digital (Egypt)
- Gamswitch (Gambia)
- Madagascar mobile money
- MauCAS (Mauritius)
- MarocPay (Morocco)
- SIMO (Mozambique)
- NamPay (Namibia)
- eNaira (Nigeria)
- NIP (Nigeria)
- RTC (South Africa)
- Tafia Moja (Tanzania)
- TIPS (Tanzania)
- Uganda mobile money
- ZIPIT (Zimbabwe)

- GIP (Ghana)
- Ghana MMTI
- Natswitch (Malawi)
- NFS (Zambia)
- GIMACPAY (CEMAC)*

No IPS has reached the mature level although there are ongoing efforts to reach this state

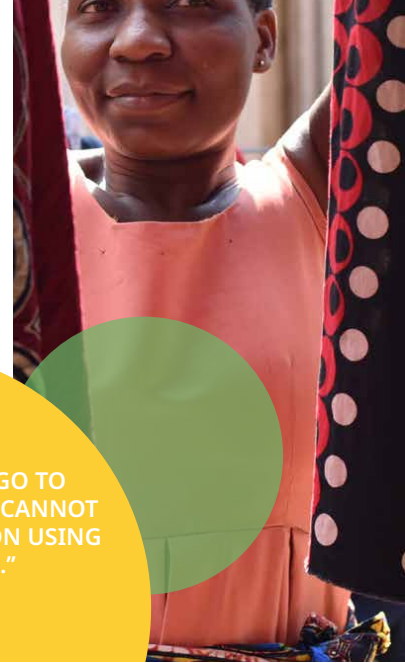
Assessment based on available data and information collected till June 2023. * GIMACPAY (CEMAC) enables domestic IPS functionality in six countries: Cameroon, Central African Republic, Chad, Republic of Congo, Equatorial Guinea, and Gabon.

● Cross-domain IPS ● Bank IPS ● Mobile money IPS ● Sovereign currency IPS □ Scheme interoperability between the two systems in Ghana



can drive digital adoption, including for women. Micro business owners and infrequent income earners were most likely to say digital payments hold limited value. Unreliable mobile networks, lack of widespread acceptance of digital payments by merchants, transaction costs, and complex user interfaces affect users' ongoing engagement with digital payments. Agents play a key role in educating and assisting users with specific transactions, managing issues that arise, and supporting consumer recourse.

Women continue to face more barriers than men, although the significance of the barriers depends on the context.



“MOST WOMEN DID NOT GO TO SCHOOL; THEREFORE, THEY CANNOT EASILY MAKE A TRANSACTION USING DIGITAL PLATFORMS.”

models	• Unilateral interoperability arrangements. • Sub-optimal interoperability arrangements.	• Share infrastructure for efficiency. • Disaggregate IPS on-us transaction data to bring more insights.
 IPS value proposition for PSPs	• Dominant PSPs and those that have invested heavily in their own infrastructure do not buy in to the value of interoperability with less-dominant PSPs. • Lack of transparency of scheme rules, performance data (volumes, and values). • Failure to address key regulatory barriers for PSPs, domestically and cross-border: compliance burden, foreign exchange limitations, operational challenges.	• Develop IPS designs through a participant-led consultative process. • Make scheme rules and data visible to improve PSP trust and buy-in, and to assist with compliance. • Endorse payment digitalization and implementation of regulatory change. • Allow new entrants to live-test products in a risk-controlled environment.
 Digital payment inclusion for women	• A persistent gender gap in digital payment usage, plus gaps in phone, internet, and legal identity documentation access. • Failure to tailor product design to women's needs.	• Establish effective recourse mechanisms to counter fraud and increase trust. Conduct financial education campaigns for women on their rights and responsibilities. • Analyze gender-disaggregated data to drive product and service design. • Leverage agent networks, including female agents, to onboard, build trust, and familiarize women with digital payments. • Integrate the G2P use cases. • Adopt electronic know your customer systems to lower access barriers.



	Barriers	Opportunities
 Merchant and government payment use cases	• Limited digital value circulation impacts IPS' ability to scale and reduce costs. • Lagging user adoption of merchant payments and persistent merchant informality. • G2P contracts selectively awarded or not digitalized/still reliant on cash distribution.	• Encourage cheaper and reliable P2B payments for merchants and end-users through scheme rules on pricing and transparency; incorporate value chain B2B payments with end-user incentives. • Digitalize G2P distribution and open distribution to various players.
 Technology standards	• International Organization for Standardization (ISO) 8385 is outdated but ISO 20022 remains expensive. • A lack of standardized QR codes. • Data sharing restrictions limit innovation.	• Adopt application programming interface (API) integration layers to enable integration with ISO 20022. • Adopt standardized QR codes to increase convenience. • Inform country strategies on Open Banking and Open Finance to propel technology standards forward.

cost, complex documentation, and reporting requirements drive end-users to use informal channels for both remittances and trade payments. Overall, cross-border retail payments on the continent are expensive and inaccessible, and hence still largely informal.

Why harmonize?

Providers cited the lack of risk-proportionate payments licensing framework, a high regulatory burden with significant penalties for non-compliance, inconsistencies between know your customer (KYC) and customer due diligence (CDD) requirements across countries, data localization requirements and complexities in tax and balance of payment reporting across jurisdictions, as the top regulatory barriers to cross-border digital payments implementation. These barriers increase risk and cost to providers, and challenge market entry by innovators who want to focus on inclusive cross-border payments. Harmonization of regulation across jurisdictions will reduce complexity and facilitate greater competition, which could lead to cheaper, faster, and more accessible cross-border payment options for end-users.



- Financial consumer protection provisions on complaint and dispute resolution processes, as well as disclosure and transparency.
- Foreign exchange access and reporting regimes.
- Data privacy, cross-border data sharing, and data protection principles, including compatibility of payment data standards and formats.

Who can harmonize?

Central banks are the key actors in driving harmonization. To enable them to do so, cooperation structures and agreements are needed, particularly related to financial integrity and data harmonization.

In addition to central banks, data protection authorities can play a vital role in facilitating cross-border trade by ensuring protection of data during international data transfers. Tax and monetary policy authorities play critical roles in addressing exchange control barriers, while regional economic communities (RECs), monetary unions,

considered, therefore are the building blocks for effective cross-border retail payment harmonization that can be developed concurrently and iteratively:

- 1 Policy formulation at the regional and domestic levels.** The first building block is for regional and domestic policymakers to establish a common goal—shared principles—and to equip regulators with mandates for cooperation.
- 2 Alignment of regulatory frameworks.** The second building block is domestic alignment of regulation, guidance, rules, practices and implementation according to regional principles, with approaches that are reasonably compatible within countries' respective technical and risk tolerances. Development organizations can contribute with technical support and capacity building.
- 3 Formal agreements.** The final building block is for payment service directives and trade agreements to be entered into as formal tools for the realization of longer-term harmonization outcomes.

Market trends

- Regulators are revising payments and e-money laws to foster innovation.
- Digital ID rollouts will increasingly allow for additional proxy ID options.
- Virtual assets for cross-border retail payments can divert scale from IPS.



System trends

- Banks remain crucial participants of IPS.
- Fintechs continue to provide front- and back-end services in partnership with established PSPs rather than becoming direct IPS participants.
- Open Finance is emerging.
- CBDCs are emerging as decentralized instant settlement and interoperability mechanisms but technical assistance support is oversubscribed.



End-user trends

- End-users are increasingly susceptible to and aware of fraud and cybercrime.
- End-users are persistently price sensitive.
- End-users are increasing their smartphone adoption, but growth in data access has been slow.





digital payments systems, ensuring the services are accessible on digitally enabled devices.

Available: The system is available for use 24 hours a day, 365 days a year, excluding planned maintenance or system downtime.

Open-loop: The system is multilateral and interoperable, and thus excludes closed-loop, on-us systems.

2000). Instant transactions generally cannot be reversed by the payer (with the exception of fraudulent or erroneous transactions).

Enables low-value payments: There is no minimum transaction amount.

IIPS process payments **digitally in near real time** and are available for use **24 hours a day, 365 days a year**, or as close to that as possible. They enable **low-value, low-cost, push** transactions that are **irrevocable** and based on **open-loop multilateral** interoperability arrangements. Licensed payment providers have **fair access** to the system, and system participants have **equal input** opportunities into the system. The **central bank** has a role in system governance. End-users have access to a **full range of use cases and channels**, as well as transparent and fit-for-purpose **recourse mechanisms**.

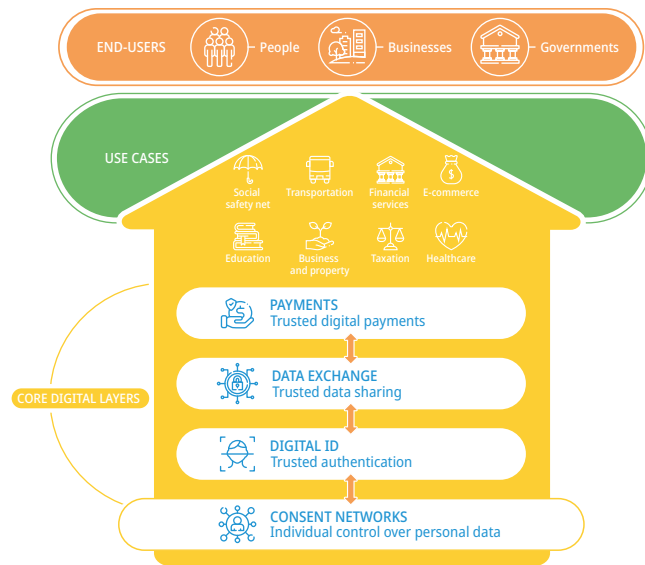
For an IPS to be an inclusive IPS (or IIPS) it must meet the following aspirational benchmark, which draws on the work of AfricaNenda (2021), CGAP (2021), the World Bank (2021b), the Bill & Melinda Gates Foundation (2019) and the Bank for International Settlements (2016).⁷

The development of IPS and IIPS in Africa is a critical enabler of financial inclusion, of digitalization, and of cross-border trade. Trends already point to rising digital payments adoption in Africa, fueling instant payment systems (IPS) usage.

⁷ Debit pull-only systems that do not support credit push transactions at a minimum are excluded. Instant debit pull transfers will likely play an important role in the future, especially for recurring person-to-business payments with trusted businesses and where convenience is at a premium, but they are currently not widely available.

⁸ See Chapter 2 for an overview of the definition terms.

FIGURE 1.1 | Digital public infrastructure layers



Source: Adapted from World Bank 2022g and Digital Impact Alliance (DIAL) 2023

This report offers insights into the progress that has been made on the availability of IIPS in Africa. It was developed using a mixed-method research approach combining government and private-sector sources and literature to create a comprehensive database of the IPS landscape in Africa; qualitative research with key stakeholders; quantitative and qualitative consumer and micro-, small-, and medium-sized enterprise (MSME) surveys in five countries; a deep dive into the issues related to cross-border regulatory harmonization using documented literature and reports and interviews with key stakeholders; and IPS case studies. For more details on the research methods, see [Annex A](#) for an overview, [Annex E](#) for the details on the consumer research, and [Annex H](#) for the documented case studies.

5

Chapter 5 provides an in-depth discussion on cross-border retail payments and their associated regulatory barriers. The chapter explains the importance of regulatory and policy harmonization on the continent, the progress of harmonization to date, and learnings from across the globe that policymakers can draw on.

6

Chapter 6 provides future perspectives from three angles: the market, the system, and the consumer.

7

Chapter 7 concludes the report and summarizes the recommendations for action.



The following sections offer detailed data on key elements of the IPS landscape.

2.1 | The IPS landscape has seen modest change since 2022

The IPS landscape has a few notable changes since 2022 (see Table 2.1).⁹ Three new systems—EthSwitch (Ethiopia), Virement Instantané (Morocco), and PayShap (South Africa)—went live since 2022; and three systems—Meeza Digital (Egypt), MauCAS (Mauritius), and eKash (Rwanda)—have since been reclassified from mobile money/bank IPS to cross-domain IPS. Overall, the number of countries

served by domestic IPS increased by one, as Ethiopia is now live. The slow rate of change since 2022 is in line with expectations since IPS take multiple years to implement and scale, often requiring extensive industry and/or regulator consultations and technical iterations. Table 2.2 shows changes to the IPS landscape from 2022-2023 (more detailed information available in [Annex B](#)).

⁹ While conducting the research for the SIIPS 2022, Africallenda built a comprehensive database of IPS, which was updated for this report (an interactive map is available on the Africallenda website). The database classifies and maps IPS based on their characteristics. Updates reflect new IPS that have launched, changes to reported data, data shared directly, new or revised information in the public domain, and foreign exchange adjustments.

IPS types

**Cross-domain IPS**

A system that provides for all-to-all interoperability where switching, clearing, and exchange of instruments is contained within one overarching system. Cross-domain systems provide access for banks and non-banks and support transactions from both bank accounts and mobile money accounts. All-to-all interoperability includes the ability for end-users to directly transact between wallet accounts at different mobile money operators (MMOs), between mobile money accounts and bank accounts, and across bank accounts. Within one system, there are different rules to accommodate various instruments. The single system provides the governance framework and coordinates the operational functions end-to-end for the various instruments (GSMA 2014).

**Bank IPS**

A system that provides access to banks only and that supports instruments associated with bank accounts.

**Mobile money IPS**

A system that provides access for mobile money providers only and that supports instruments associated with mobile money accounts.

**Sovereign currency IPS**

A central bank digital currency IPS. Such an IPS combines a sovereign currency instrument and a value transfer system that can provide a unified digital value transfer mechanism between commercial instrument systems, institutional stakeholders, and individuals within an economy.

BOX 2.1 | Three new systems launched since SIIPS 2022

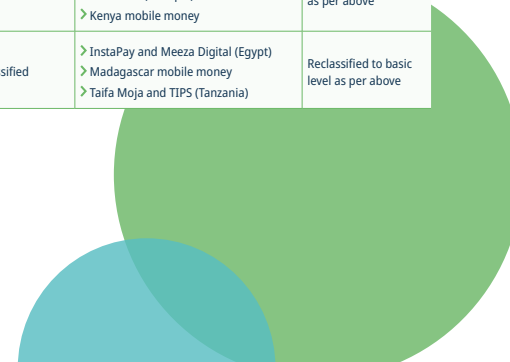
- **EthSwitch.** EthSwitch launched its Real-time Retail Payments Platform Project in Ethiopia in 2022 after receiving funding to modernize the country's retail payment system from the African Development Bank's Africa Digital Financial Inclusion Facility and the Bill & Melinda Gates Foundation. After a successful pilot between nine banks and one microfinance institution (MFI), the person-to-person (P2P) service went live in October 2022, enabling bank, MFI, and e-money issuers to instantly send funds between accounts and digital wallets via both USSD and apps (EthSwitch 2021). EthSwitch operates the system.
- **PayShap.** PayShap launched in March 2023 in South Africa as a bank IPS (BankservAfrica 2023). It is currently supported by the four largest banks: Absa, FNB, Nedbank, and Standard Bank. BankservAfrica, a central clearing house and the manager of the regional TCIB system, processes the payments in real time. The system was built for low-value payments with a transaction cap of \$167 (ZAR3,000) and a daily limit of \$278 (ZAR5,000). The initiative was developed as part of the South African Reserve Bank's Vision 2025, with BankservAfrica and the Payments Association of South Africa as the main implementors (Business Tech 2022). The launch includes only P2P payments, but P2B payments are also under development (Stakeholder interviews 2023; Jumar 2023). There are no public plans to date to add use cases beyond P2P and P2B or to extend the IPS to non-bank participants, such as mobile money operators.
- **Virement Instantané.** The Moroccan central bank, Bank Al-Maghrib, and the clearing house, Groupement pour un Système Interbancaire Marocain de Télécompensation (GSIMT), jointly launched the bank IPS, Virement Instantané, in June 2023. The implementors plan its eventual rollout to all banks in Morocco to increase transaction speed, increase availability, and to simplify payments. The service offers instant P2P payments. There is a no-cost mandate for end-users for the first three months, after which banks are free to set their own transaction fee. There is a daily cap of \$1,960 (MAD 20,000). Workshops with banks and proof of concept testing informed the system architecture, which runs on ISO 20022 (Bank Al-Maghrib 2023).

¹¹ IPS-specific websites were used as the next best source of scheme rules for gathering information. To aid the analysis, if the website specified mobile money operator (MMO) and bank participants while offering typical MMO and bank-based channels, the IPS is classified as cross-domain.

	10	9	1 reclassified to cross domain IPS	➤ MauCAS (Mauritius)	Now accessible by non-banks, moved to cross-domain
 Mobile money	7	9	2 reclassified to cross domain IPS	➤ Meeza Digital (Egypt) ➤ eKash (Rwanda)	Both now accessible by banks, moved to cross-domain
Transaction data					
 Values data collected	22 IPS	13 IPS	10 IPS value data points added	➤ eKash (Rwanda) ➤ NamPay (Namibia) ➤ InstaPay (Egypt) ➤ Madagascar mobile money ➤ SIMO (Mozambique) ➤ Taifa Moja (Tanzania) ➤ ZIPIT (Zimbabwe) ➤ GIMACPAY (CEMAC) ➤ Somalia National Payment System ➤ EthSwitch (Ethiopia)	Data publicly available or submitted by central banks/IPS operators
			1 IPS most recent value data not available in	➤ Gamswitch (The Gambia) No data from 10 IPS: S184D (Libya); MarocPay (Morocco); Gamswitch (The Gambia); TIPS (Tanzania); Meeza Digital (Egypt); Tunisia mobile money; PayShap (South Africa); Virement Instantané (Morocco); PAPPS; TCB)	Data for 2022 not available

10 Unlike with physical infrastructure, an IPS continues to develop after it has launched; the table shows changes of system names, participants, and functionalities that have taken place since the release of the 2022 report.

 Basic	15	11	2 reclassified	➤ Kenya mobile money ➤ eKash (Rwanda)	Kenya mobile money does not provide open-loop interoperability for person-to-person (P2P) payments yet, only for (P2B); eKash does not provide P2B payments yet
 Not ranked	12	13	4 added	➤ PayShap (South Africa) ➤ Virement Instantané (Morocco) ➤ EthSwitch (Ethiopia) ➤ Kenya mobile money	Three new systems; one IPS reclassified as per above
			5 reclassified	➤ InstaPay and Meeza Digital (Egypt) ➤ Madagascar mobile money ➤ Taifa Moja and TIPS (Tanzania)	Reclassified to basic level as per above



For an IPS to be a cross-domain system, it must have a switching capacity between commercial money instruments (such as debit electronic funds transfer (EFT), credit EFT, and domestic card instruments) and e-money instruments.¹² Operators use one of two approaches to achieve a cross-domain IPS.

The more common approach is to exchange e-money into commercial money instruments and then switch and clear them with other commercial money instruments on the same platform and between commercial money accounts. The value is then exchanged back into e-money and cleared to the recipient's account, such as a mobile money account. This is the case for NatSwitch in Malawi, for which the scheme operating guidelines set out the switching of e-money and commercial money instruments (NatSwitch Limited 2020; see Box 2.2).

Another approach uses a "quasi" cross-domain arrangement entailing the switching of e-money instruments and commercial instruments according to different scheme rules. In most cases, clearing happens in commercial money instruments, but the exchange of commercial money to e-money instruments on either

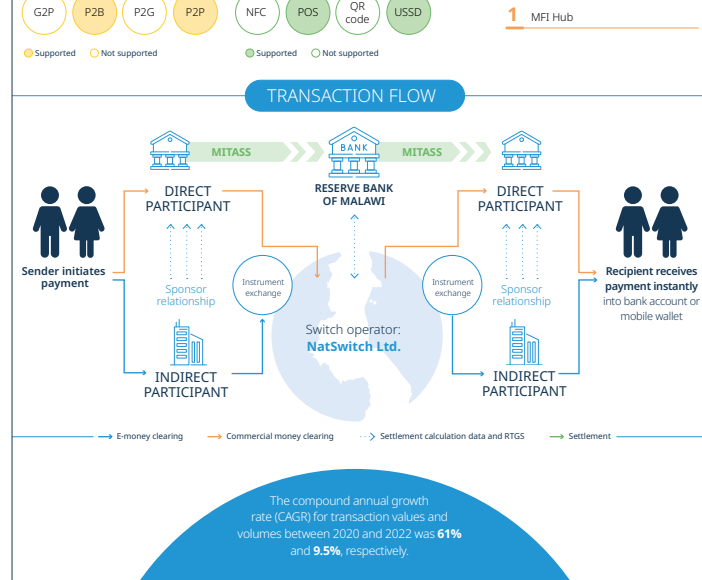
side of the transaction occurs outside of the system and mostly on proprietary platforms. MarocPay (Morocco) uses this type of arrangement, whereby it uses separate switches for e-money and commercial bank money (Bank Al-Maghrib 2020).

Not all cross-domain systems can be categorized based on the arrangement types, as scheme rules or comprehensive information is not available.

Box 2.2 is a snapshot of the use cases, functionalities, and transaction flow of the IPS in Malawi (NatSwitch). A detailed case study is available in [Annex H](#).

As noted previously, the type of interoperability arrangement is a key part of differentiating between IPS types. Third parties facilitate most interoperability arrangements (29 out of 32; see Figure 2.2). This third party is a central switch, such as RSwitch for eKash (Rwanda) and Zimswitch for ZIPIT in Zimbabwe. Three mobile money IPS interoperate through direct integration with one another (multilateral interoperability) instead of through a central switch. The three are Kenya mobile money, Madagascar mobile money, and Taifa Moja in Tanzania.

¹² Card instruments can be cross-border as well as issued by non-banks in some jurisdictions. In most, if not all, jurisdictions in Africa, card instruments are commercial money instruments within the local jurisdiction.

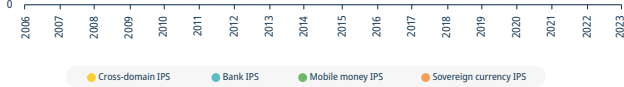


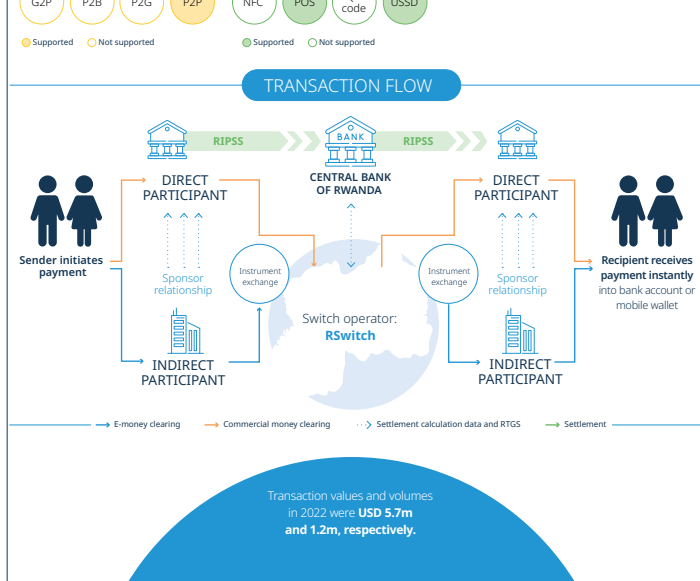
schemes; most of these mobile money IPS launched between 2013 and 2016. These systems typically came about in response to interoperability directives from the regulator, as in the case of Uganda, or to industry discussions, as in Tanzania (Figure 2.3; CGAP 2015; Onwuegbuchi 2017). No mobile money IPS have launched since 2018. The total number of mobile money IPS stands at seven systems.

Since 2018, governments in countries that lacked an IPS but had a dominant mobile money player have tended

to launch their own sovereign currency IPS. For example, eKash in Rwanda was established in 2022 as a cross-domain IPS, despite the near-monopolistic presence of MTN's MoMo (Box 2.3; Gilbert 2022). The number of bank IPS have been steadily increasing: six systems have launched since 2020.

Box 2.3 is a snapshot of the use cases, functionalities, and transaction flow for the IPS in Rwanda (eKash).





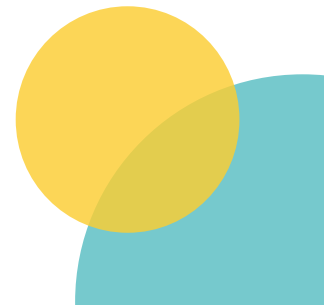
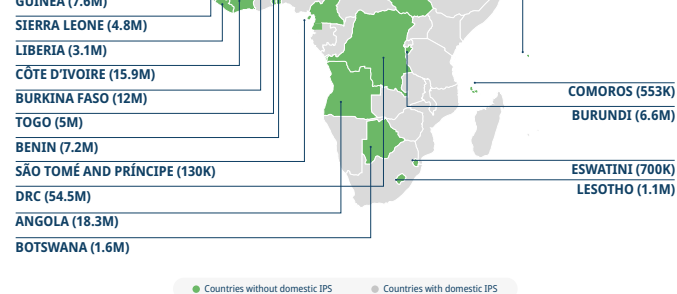
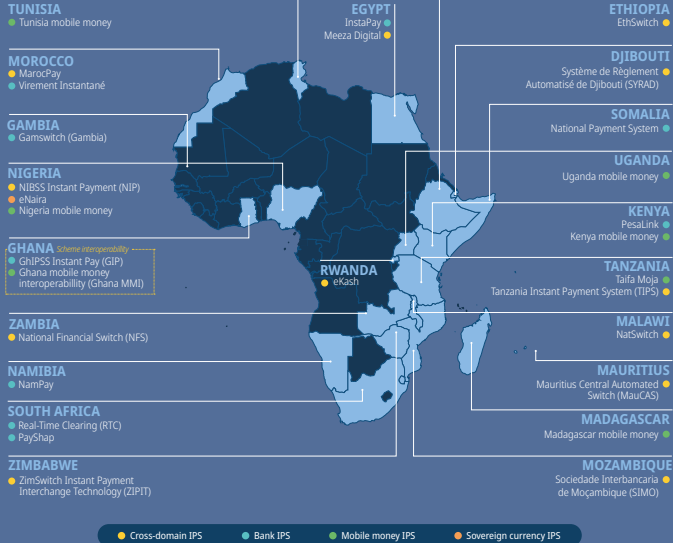
Rwanda (Stakeholder interview 2022). Examples include Ghana, which has finalized its eCedi testing. It includes an online/offline capability pilot in an isolated village

use cases, there is less emphasis on direct release cases such as consumer wallets and ecosystems (Stakeholder interviews 2023).

TABLE 2.3 | CBDC projects in Africa

Country	Status
Côte d'Ivoire	Research
Egypt	Research
Eswatini	Research
Ghana	Pilot
Kenya	Research
Madagascar	Research
Mauritius	Research
Morocco	Research
Namibia	Research
Nigeria	Launched
Rwanda	Research
South Africa	Research
Tanzania	Research
Tunisia	Research
Uganda	Research
Zambia	Research
Zimbabwe	Research

Source: CBDC Tracker 2023



where are also cross-border IP's alternatives:

- **MFS Africa**, a pan-African mobile payment gateway, connects over 400 million mobile money wallets and 200 million bank accounts across 35 countries (Nnamani 2023).
- **Buna in the United Arab Emirates** connects banks across the Middle East and North Africa region. Buna launched its instant cross-border payment functionality in March 2023. It plans to provide cross-border functionality capable of reaching 172 million African adults.



GIMACPAY

CEMAC COUNTRIES (34.4M ADULTS):

Cameroon, Central African Republic, Chad, Republic of Congo, Equatorial Guinea, Gabon.



PAN-AFRICAN PAYMENT AND SETTLEMENT SYSTEM (PAPSS)

WAMZ PILOT COUNTRIES (157.5M ADULTS):

The Gambia, Ghana, Nigeria, Sierra Leone, Guinea, Liberia.



TRANSACTIONS CLEARED ON AN INSTANT BASIS (TCIB)

SADC COUNTRIES (223M ADULTS):

Angola, Botswana, Comoros, DRC, Eswatini, Lesotho, Madagascar, Malawi, Mauritius, Mozambique, Namibia, Seychelles, South Africa, Tanzania, Zambia, Zimbabwe.

● Cross-domain IPS

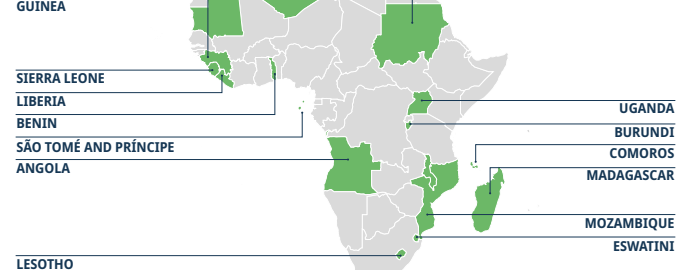
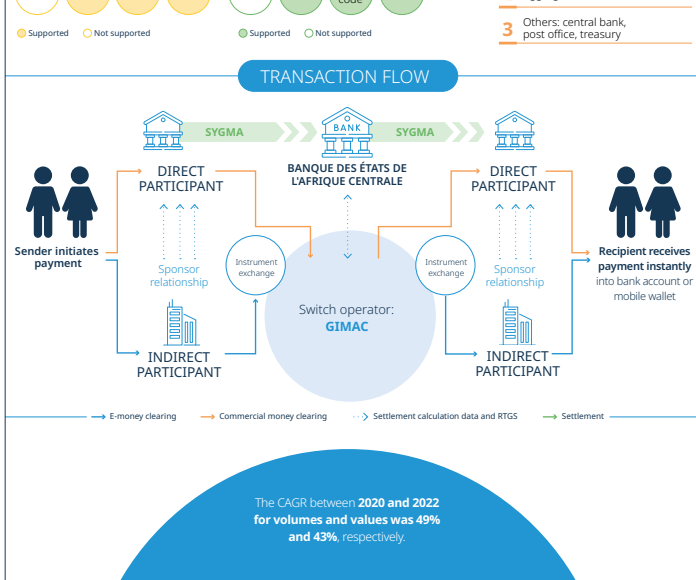
● Bank IPS

● Mobile money IPS

● Sovereign currency IPS

*Tanzania is not a COMESA member state but will integrate with the COMESA regional IPS.

The box on page 74 is a snapshot of the use cases, functionalities, and transaction flow for the IPS in the Economic and Monetary Community of Central Africa (GIMACPAY).



13 Countries include Burundi, Comoros, DRC, Djibouti, Egypt, Eswatini, Eritrea, Ethiopia, Kenya, Libya, Madagascar, Malawi, Mauritius, Rwanda, Seychelles, Somalia, Sudan, Tanzania, Tunisia, Uganda, Zambia, and Zimbabwe.

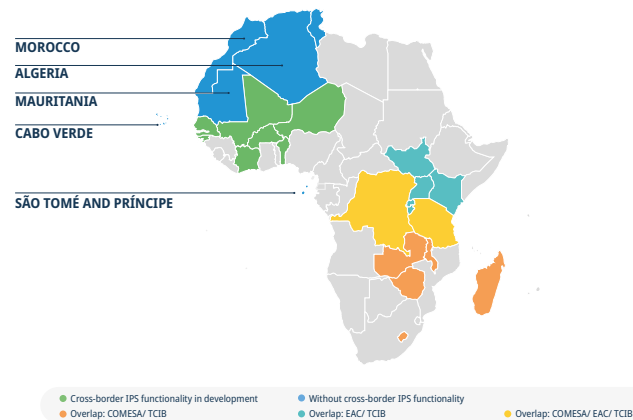
BOX 2.6 | The Common Market for Eastern and Southern Africa (COMESA) regional instant payment system



→ **The COMESA Business Council (CBC)** is developing an integrated, low-cost, and interoperable digital payment platform focused on low-value cross-border trade within the region. This regional IPS is motivated by the need to transform the cash-based MSME ecosystem into digital markets and increase the volumes of formal cross-border retail. The regional IPS will follow a cross-domain model, supporting both e-money and commercial bank money instruments, and allowing direct participation from domestic IPS in addition to mobile money operators and banks. It aims to support P2B and B2B use cases in addition to P2P (COMESA Business Council 2022).

→ **COMESA IPS** aims to follow a similar design approach to TCIB in terms of governance and business model. CBC is crafting the scheme rules to align with TCIB's as much as possible to assist the eventual integration of the two systems, according to stakeholder interviews conducted in 2023.

MAP 2.6 | Cross-border IPS functionality summary



14 Comoros, DRC, Eswatini, Madagascar, Malawi, Mauritius, Tanzania, Zambia, and Zimbabwe.

15 Burundi, DRC, Kenya, Rwanda, South Sudan, Uganda, and Tanzania.

2.4 IPS operate across a range of transactions, channels, use cases, and participants

The IPS already in the market, as described in this section, operate at various scale levels and support a range of channels, instruments, and use cases. These elements contribute to the inclusivity of the systems for end-users.

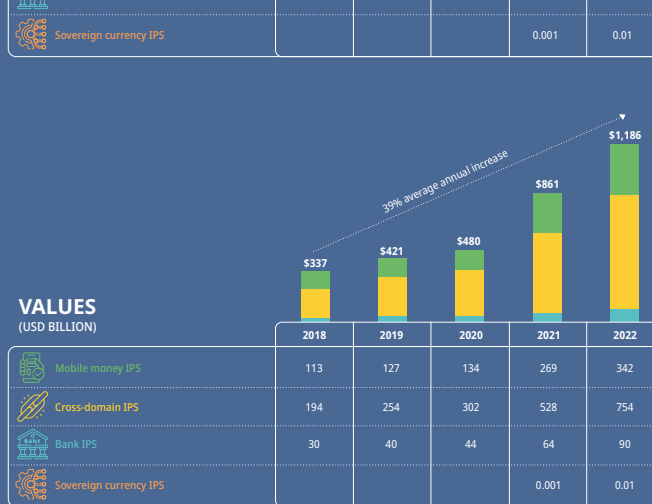
2.4.1 Transaction flows are increasing in volume and value

Over the past five years, the number of processed transactions has dramatically increased by an average annual rate of 47% (Figure 2.4).¹⁶ The rapid societal changes brought on by the COVID-19 pandemic drove acceleration in transaction growth. As existing systems have seen increased use and new systems have come online, the increase in transaction value has likewise grown by 252% between 2018 and 2022, reaching \$1.18 trillion in 2022.¹⁷

Mobile money IPS support the largest number of transactions while cross-domain IPS process the largest values. Bank IPS experienced the highest year-over-year growth in volumes since 2021, at 66%, compared to 48% for cross-domain IPS and 32% for mobile money IPS. The overall growth in volumes and values may be attributed to the steady increase in the number of cross-domain systems (see Box 2.8 for details on the transaction data).

¹⁶ One important aspect of updating the report from 2022 to 2023 involved ensuring that volumes and values are expressed in today's exchange rates. To achieve this, new central bank information was used where possible and adjusted accordingly. The respective exchange rates to the US\$ were applied on 31 March 2023 via oanda.com. Data from Gamswitch (The Gambia) was excluded in this graph due to 2022 figures not being available.

¹⁷ The following systems with information available have launched since 2020: InstaPay (Egypt); EthSwitch (Ethiopia); NamPay (Namibia); eNaira (Nigeria); eKash (Rwanda); Somalia National Payment System.



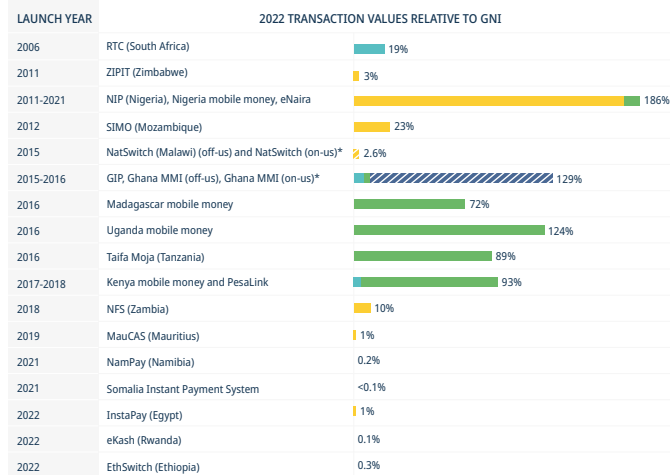
Note: The data in figure 2.4 above came from publicly available information on transaction flows or from the central bank or system operator in each respective country. This data is available for 22 IPS. The data for 10 IPS were unavailable. As a result, the actual transactions volumes and value may be underestimated. The data is missing from: SYRAD (Qibouti), Meeza Digital (Egypt), Gamswitch (The Gambia), MarocPay (Morocco), Virement Instantané (Morocco), PayShop (South Africa), TIPS (Tanzania), Tunisia mobile money, PAPSS and FCB.

➤ NamPay (Namibia)	Central bank source
➤ eNaira (Nigeria)	IMF
➤ NIP (Nigeria) and Nigeria mobile money	Central bank source
➤ eKash (Rwanda)	Central bank and operator submission
➤ Somalia National Payment System	Central bank source
➤ RTC (South Africa)	Operator submission
➤ Taifa Moja (Tanzania)	Central bank source
➤ Uganda mobile money	Central bank source
➤ NFS (Zambia)	Operator submission
➤ ZIPIT (Zimbabwe)	Operator submission
➤ GIMACPAY (CEMAC)	Operator submission

The different types of IPS support different purposes. For example, mobile money IPS support small value, frequent payments. For that reason, they account for 82% of IPS transaction volumes in Africa, despite representing only 29% of total IPS value. The average mobile money IPS transaction (\$17) is a far smaller value than the average bank (\$267) or cross-domain IPS transaction (\$142). Despite significant year-over-year

declines, Bank IPS continue to have the highest average transaction value (see Table 2.4). This suggests that customers are increasingly using bank accounts for lower-value transactions—though not as low as mobile money transactions.¹⁸ The average transaction size for cross-domain systems, in contrast, continues to fluctuate with no clear trend.¹⁹

FIGURE 2.5 | 2022 domestic IPS transaction values relative to GNI (n=21)



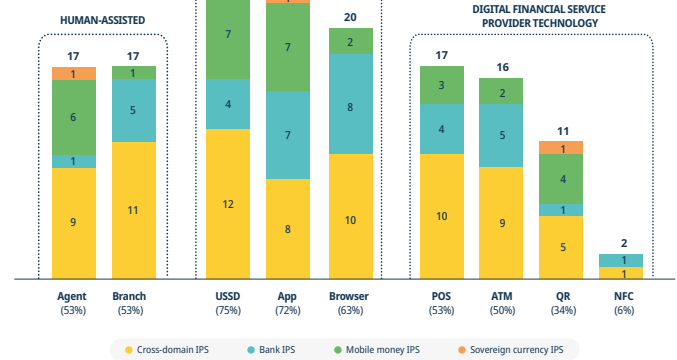
¹⁸ Adjustments due to increased data availability and exchange rate changes contributed to the change in average transaction values from SIIPS 2022.

¹⁹ Adjustments due to increased data availability and exchange rate changes contributed to the change in average transaction values in SIIPS 2022.

*NatSwitch (Malawi) and Ghana MMI are the only IPS where information on on-us transaction data is available.

dominance with on-us cross-switched transactions. In Ghana, mobile money on-us transactions have high transaction values relative to GNI (115%) in comparison to the off-us Ghana MMI (2.9%). Similarly, the value of off-us transactions processed by Natswitch in 2022 accounted for only 0.6% of the total transactions (Reserve Bank of Malawi 2022).

On-us cross-switched transactions. Consequently, the available data does not yet allow for a more granular assessment of the effect of the on-us/off-us transactions dynamic on IPS value proposition and scalability. More transparency around IPS transaction data availability and how it is calculated and aggregated is required, as is a greater understanding of end-user transaction drivers.



²⁰ Data is not available for SYRAD in Djibouti and PAPSS.

²¹ GIMACPAY (CEMAC), Ghana MMI and GIP, Kenya mobile money, MarocPay, MauCAS, eNaira (Nigeria), NIP (Nigeria), Taifa Moja and TIPS (Tanzania), and Uganda mobile money.



All mobile money and cross-domain instant payment systems support e-money instruments (Figure 2.7).²² Cross-domain IPS also support a range of commercial money instruments, such as credit and debit EFTs. As a result, the cross-domain IPS can exchange, switch, and clear e-money and commercial money instruments from payment initiation to termination within its own scheme rules and overarching governance. Mobile money IPS and bank IPS typically allow for different PSPs to interoperate only using the same instrument types. Mobile money IPS only process similar e-money instruments, and likewise, bank IPS exclusively process similar commercial money instruments, primarily credit EFT with a secondary focus on debit EFT. Sovereign currency IPS process currency via a CBDC instrument.

2.4.4 P2P use cases dominate, though P2B and B2B are growing

All IPS support the P2P use case; the P2B use case for merchant payments is also prevalent: 68% of IPS support both (Figure 2.8). Apart from one IPS (Kenya mobile money), all domestic IPS provide open loop, interoperable P2P payments as a core service.²⁴ Kenya introduced mobile money interoperability for P2B payments through multilateral integration, whereas the different MMOs negotiated the terms for interoperability for P2P use cases. The instantaneous aspect of P2B payments is important for merchants who expect electronic transactions to function the same way as cash (CGAP 2019a).

Few IPS enable use cases beyond P2P and P2B. Nine IPS explicitly support business-to-business (B2B) payments, including Meeza Digital (Egypt), Ghana MMI and GIP, Natswitch (Malawi), Virement Instantané (Morocco), NamPay (Namibia), NIP (Nigeria), RTC (South Africa), and NFS (Zambia). B2B is another key use case for aggregating scale and retaining digital currency in the economy. The three regional IPS support B2B cross-border transfers,

as does NIP in Nigeria (Stakeholder interview 2023). So far, only NIP in Nigeria and Ghana MMI and GIP support the broadest range of use cases.

There are critical gaps in the support of use cases beyond P2P and P2B, especially for government-to-person (G2P) payments (Box 2.10). Only six IPS support G2P use cases. Notably, none of them are bank IPS. Yet G2P payments are critical for recipients and for IPS to achieve scale, due to the sheer volume of adults who receive either social assistance payments, a pension transfer, or a government wage. Specifically, in 2021 more than 10% of Africa's adult population received a government transfer (World Bank 2021c). During the COVID-19 pandemic in particular, at least 117 million adults across 33 countries received social transfers (Kazeem 2020). Economically vulnerable households rely on government payments as a major source of weekly or monthly income (World Bank 2022b). Further research is needed on the G2P-IPS integration nexus in Africa.

²² Mobile money-based channels and instruments were only developed in the past twenty years when telecommunication companies realized that end-users were transferring airtime to one another (Piper 2020). The relatively new introduction of e-money instruments, as well as the difficulty innovating around e-money channels, explains why mobile money IPS support relatively fewer channels than other schemes.

²³ Data is not available for SYRAD in Djibouti.

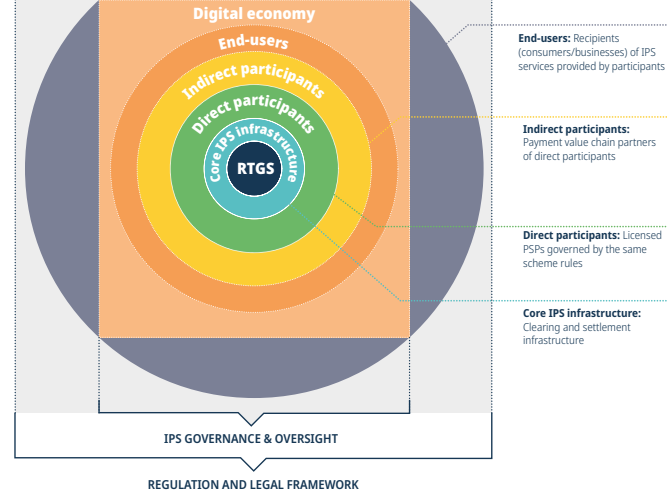
²⁴ Ibid.

BOX 2.10 | G2P value chains in instant payment systems

The following examples highlight ways that IPS can facilitate G2P payments:

- The Government of Morocco uses **MarocPay (Morocco)** to distribute education subsidies through the Tayssir program (Hadri 2022).²⁵
- **Leveraging Uganda's mobile money IPS, Airtel and MTN** (Ministry of Gender Labour and Social Development 2022) disbursed COVID-19 relief funds to teachers. The National Social Security Fund (NSSF) of Uganda also transmits payments to the PSP of the beneficiary's choice—the payment process required beneficiaries to submit their preferred provider among the MMOs operating in the country (NSSF 2017; Delliah 2022).
- **Madagascar mobile money** is being utilized for education grant transfers, the Human Development Cash Transfer (HDCT), and the Flavota Cash Programme for drought-affected households. About 60% of HDCT beneficiaries receive payment through mobile money accounts (Women's World Banking 2016).
- **NIBSS in Nigeria** hosts and validates payments for all of the government's social intervention programs in the country. The Central Bank of Nigeria (CBN) is planning on dispensing welfare payments in eNaira under the National Social Safety Net Program-Scale Up (Tunji 2023). As of March 2023, four million eNaira wallets were created for the purpose of social disbursements (Idowu 2023).
- **GIP and Ghana MMI** facilitate the transfer of G2P payments into an account associated with the biometric e-zwich card system. Recipients of social assistance payments can move the money into mobile wallets or bank accounts at ATMs through the two Ghanaian systems (Stakeholder interview 2023).

²⁵ Note that this data point is surmised from secondary research, as no data was available on MarocPay. As such, it is subject to confirmation.



Egypt's e-Naira and Nigeria's Kowal, each of which have two settlement cycles per day; Rwanda's eKash can settle funds multiple times within a day.

The central banks serve as the settlement agent since they hold the accounts. Third-party switches consolidate the settlement instructions for participants. Afreximbank is the settlement agent for PAPSS, while TCIB settles through the South African Reserve Bank or via correspondent banks. In the case of e-Naira in Nigeria, settlement is instant—that is one of the main value propositions of CBDC systems. However, recourse and risk mitigation are more complex with instant settlement. In four systems, namely Kenya mobile money, Madagascar mobile money, Uganda mobile money, and Tanzania mobile money (Taifa Moja), participants hold pre-funded accounts with one another to allow for settlement, which can increase trust and speed among participants. However, this approach also ties up large sums of capital in these accounts and can become complex to manage as the PSP pool grows.



26 Information on the settlement modality is not available for Meeza Digital (Egypt), MaroPay (Morocco), ETHSwitch (Ethiopia), and SIMO (Mozambique).

Bank of Somalia (National Payment System), and Bank of Tanzania (TIPS).

In the case of regional IPS, clearing either occurs through a centralized payment hub to which participants are directly integrated (hub arrangement) or through a domestic financial switch that is linked to a central hub (hub-switch arrangement). Two of the three regional IPS—PAPSS and GIMACPAY—use hub arrangements and TCIB follows a hub-switch arrangement.

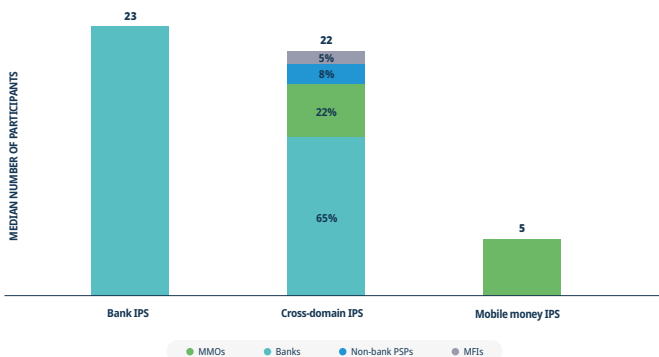


Direct participants

Direct participants are PSPs that sign participation agreements with the IPS system and fulfill criteria laid out in the scheme rules. Direct participants include commercial banks, MMOs, MFIs, and other non-bank PSPs that link to and utilize the IPS' core clearing infrastructure (CGAP 2021). Direct participants can also include those PSPs that have settlement accounts at the central bank, but there is no uniform definition across the IPS.

27 The switching and routing function includes routing tables, accounts, and the proxy management and updates of transactions. The switch operators' functions also pertain to the handling of anomalies, return transactions, and tracing and clearing of disputes.

FIGURE 2.10 | Median number of participants per IPS type (n=28)



28 Information is not available for SYRAD (Djibouti) and EthSwitch (Ethiopia). NIP and e-Naira (Nigeria) were excluded for this calculation due to being outliers. The e-Naira system in Nigeria has 33 bank participants so far while NIP in Nigeria has 450 participants, including 225 commercial banks, 200 microfinance banks, 7 MMOs, and 18 non-bank PSPs.

29 This information is available for 25 schemes.



Indirect participants

Indirect participants can fulfill two possible roles. First, they can be PSPs that access the IPS indirectly to provide payment services to end-users, these may include smaller PSPs, MMOs, fintechs, and MFIs in systems that only allow banks to be direct participants. Indirect participants typically require a sponsor PSP that is directly integrated, especially for settlement but also for clearing. For example, PesaLink in Kenya and GIP in Ghana allow PSPs, MMOs, fintechs, and savings and loans to indirectly access their respective bank IPS. In Nigeria, microfinance banks, fintechs, and super agents can indirectly access NIP.

The second role for indirect participants is that of a technical service provider offering key IPS services to direct participants. Technical service providers partner with direct participants to provide front- or back-end services. Fintechs, for example, provide direct participants with an opportunity to collaborate with providers of emerging technologies.

Current fintech examples in the IPS ecosystem include Famoco, which enables the Bank of Ghana to distribute salaries and pensions through their hybrid POS that includes biometric authentication of citizens. Moreover, the regional IPS in CEMAC, GIMACPAY, has recently partnered with Maviance, a digital financial service provider, to offer online e-commerce payments and agent banking services, among others (GIMAC-AFR n.d.)—these services have not yet been rolled out. In Rwanda, eKash supports the integration of third-party overlays service providers, such as Pivot Access and mVend (Bank of Rwanda 2021).³⁰ Other providers such as internet technology companies and marketing/advertising firms provide specialized services—for example, Malawi's NatSwitch has involved Mitra Systems for computer hardware support, Globe Internet Ltd. to manage networking and telecommunications, and BPC Banking Technologies Group for technical support on the switching infrastructure, according to stakeholder interviews conducted in 2023.

30 mVend is a technology solutions provider of software and applications; Pivot Access builds and manages software platforms.



It is important to call out the role of ownership, governance, oversight, and intermediation in an IPS ecosystem, even though these do not fit neatly within the participant value chain.

31	Regulator-owned: SYRAD (Djibouti), GPT and Ghana MNC, MarCoPay (Morocco), MAUCS (Mauritius), eNaira (Nigeria), Somalia National Payment System, TSPS (Tanzania), TCIB, Tunisia mobile money. Publicly owned: ESWITCH (Egypt), Kenya mobile money, PesaLink (Kenya), Madagascar mobile money, NamPay (Namibia), Natswch (Malawi), eKash (Rwanda), PayShap and RTC (South Africa), Tafa Moja (Tanzania), Uganda mobile money, and ZPITI (Zimbabwe). Ownership information was not available for ESWITCH (Ethiopia) or Virement (Antarctica/Morocco).
32	No information was available for EHSWICH (Ethiopia). Private associations: Kenya mobile money, PesaLink (Kenya), Madagascar mobile money, MarCoPay (Morocco), NamPay (Namibia), Natswch (Malawi), eKash (Rwanda), PayShap and RTC (South Africa), Tafa Moja (Tanzania), Uganda mobile money, and ZPITI (Zimbabwe). PPP model: GIMACAPD (CENAC) and PAPPS, Interswitch (Egypt), Gamswatch (The Gambia), Nip and Nigeria mobile money, SIMO (Mozambique), Meza Zingdi (Egypt), TCIB, and Tunisia mobile money. Central-bank model: SYRAD (Djibouti), GPT and Ghana MNC, MAUCS (Mauritius), eNaira (Nigeria), Somalia National Payment System, TSPS (Tanzania), and Tunisia mobile money.

Separate from governance is IPS regulatory oversight. All domestic IPS and GIMACPAY in CEMAC are regulated and supervised by their respective central banks. The arrangement is slightly different for the other two regional systems: the SADC Payment System Oversight Committee oversees TCIB and the PAPSS Governing Council oversees PAPSS.

(GIMAC-AFR n.d.; Mwareya & Simango 2022; Association of African Central Banks 2021).

Payment service providers and financial institutions may resist joining regional IPS without an incentive to integrate with the system—this hurdle is elaborated on further in Chapter 4. Several initiatives, highlighted in Box 2.11, have been put in place to attract more participants and enhance the value proposition of the systems.

BOX 2.11 | Initiatives to grow the participant ecosystem

be accurately and efficiently understood and processed (World Bank 2021d). Of the 21 systems about which AfricaNenda could access transactional data, 10 domestic IPS as well as PAPSS and TCIB use of the ISO 20022 messaging standard.³⁴ Six IPS use the ISO 8583 standard.³⁵ Nigeria's NIP and e-Naira, and Tanzania's TIPS developed proprietary messaging standards. NIP and NFS Zambia (currently using ISO 8583) plan to migrate to ISO 20022 in the future (Stakeholder interviews 2023).

As it relates to data security and end-user identification, IPS may use proxy IDs to identify end users. Proxy IDs can be more secure than requiring an end-user to input their account number, because fraudsters could use financial account numbers to perform unauthorized withdrawals. Proxy IDs are also more convenient because they are typically simple to remember and use (World Bank 2021e). For this latter reason, proxy IDs drive inclusion. Of the 21 systems for which information was available, 16 offered proxy IDs.³⁶ Thirteen of these offer mobile phone numbers as proxy IDs.³⁷ QR codes are particularly

Apart from mobile phone numbers and QR codes, there are three IPS that have developed IPS-specific aliases:

→ **InstaPay (Egypt)** uses an instant payment address, which is a simplified address of the end-user's account number (name@InstaPay, InstaPay n.d.).

→ **MauCAS (Mauritius)** allows for payment using a pseudonym rather than through an account number to facilitate ease of transfer. These proprietary IDs are centralized at the IPS level.

→ **PayShap in South Africa** assigns individual end-users a ShapID based on a bank-registered mobile phone number; a ShapID for a business is based on its bank account details, e.g. [mobile number]@[bank name] (PayShap 2023).

2.5.2 About half of IPSs rely on APIs for technical integration and access

Open APIs allow providers to access a system or to provide additional services to IPS operators. APIs are available for 15 IPS to date.³⁸ Participants can access Nigeria's NIP only through its open API structure, which enables participant integration, processing, and monitoring, as well as overlay services such as payee confirmation. For financial institutions that have not yet upgraded to ISO 20022, an API layer can provide integration with the system—this is the case with eKash (Rwanda). PesaLink (Kenya) is developing their switch to allow banks to integrate via APIs. The development of APIs for overlay services outside of Nigeria depends to some extent on the regulatory position with respect to open banking and open finance (discussed in Chapter 5).



34 Information was not available for SYRAD (Djibouti), Meeza Digital (Egypt), EthSwitch (Ethiopia), Kenya mobile money, Madagascar mobile money, MarocPay (Morocco), SMO (Mozambique), Taifa Moja (Tanzania), Tunisia mobile money, and Uganda mobile money.

35 ISO 20022: InstaPay (Egypt), PesaLink (Kenya), Gamswitch (The Gambia), GP and Ghana MML, MauCAS (Mauritius), Virement Instantané (Morocco), NamPay (Namibia), eKash (Rwanda), PayShap (South Africa), ISO 8583: GIMACPAY (CEMAC), Natswitch (Malawi), Nigeria mobile money, RTC South Africa, NFS Zambia, and ZIPTT (Zimbabwe).

36 Information was not available for: SYRAD (Djibouti), Meeza Digital (Egypt), EthSwitch (Ethiopia), Gamswitch (Gambia), SMO (Mozambique), NamPay (Namibia), Tunisia mobile money, GP and Ghana MML, Kenya mobile money, PesaLink (Kenya), Madagascar mobile money, MarocPay (Morocco), Nigeria mobile money, PAPSS, Taifa Moja and TIPS (Tanzania), TCIB, Uganda mobile money, NFS Zambia.

38 If transaction consent was obtained fraudulently or under duress, the transaction can be withdrawn, provided it has not yet been processed or settled. From the end-user perspective, there may also be issues related to transaction errors, should it happen that consumers send money to the wrong number or mobile money merchant code in error and then find that the transaction cannot be simply reversed without following due process and/or obtaining the consent of the receiving party. Erroneous transactions that cannot easily be recalled negatively affect end-user trust in digital payments. However, if transactions are too easily recalled, merchants may not use the IPS of fear of fraudulent chargebacks. These occur when transactions are recalled by an end-user after leaving a place of business even though they did get the goods, or a service was performed.

39 GIMACPAY and PAPSS, as well as InstaPay (Egypt), Gamswitch (The Gambia), GP and Ghana MML, PesaLink (Kenya), MauCAS (Mauritius), NIP and Nigeria mobile money, eKash (Rwanda), RTC (South Africa), TIPS (Tanzania), ZIPTT (Zimbabwe), and Virement Instantané (Morocco).



IPS ranked according to the **basic level** of inclusivity include two key criteria regarding basic functionality. These criteria are essential for the inclusion of all end-users in Africa. IPS are not ranked if they fail to meet the basic level of inclusivity, namely that they:



Enable the primary local channel: The IPS enables the channel or channels that are most used by the population within its geography. For example, it supports mobile money transactions via the IPS in markets where e-money is the preferred channel currently.



Enable P2P and P2B use cases at a minimum: The former provides speed and safety in addition to scale for personal P2P transactions. The latter provides scale for an efficient business model and to rival cash in merchant payments.

IPS that achieve a **mature level** of inclusivity have fulfilled the basic and progressed level criteria, as well as three additional functionality and governance conditions:



Enable all use cases: End-users can access a full range of use cases, including P2P, P2B, G2P, P2G, B2B, B2P, B2G, and G2B, for a holistic digital payment ecosystem that enables the full circulation of liquidity, as occurs in the analog economy. Interlinking use cases enhance digital utility for end-users and prevent capital from stagnating. Instead, capital can more easily and efficiently flow between actors in the economy.



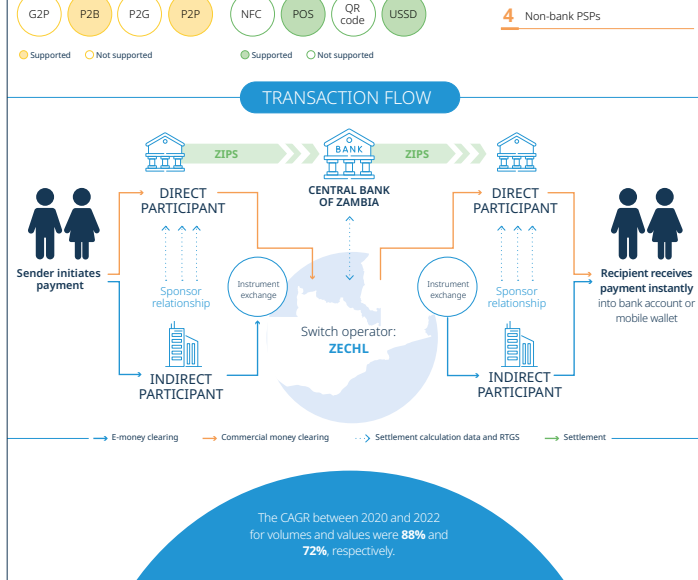
Provide recourse: The IPS sets standards for participants to ensure end-user recourse is in place, consistent with consumer protection, data privacy, and cybersecurity laws. The IPS effectively monitors the provision (by participants) of end-user recourse mechanisms, mitigating risks such as fraud. Ideally, the system also provides an additional layer of recourse for end-users should provider channels prove insufficient.



Serve end users at low cost: The cost for the end user of a digital payment transaction is as low as feasibly possible and delivered within a not-for-loss business model. Participants continuously monitor their pricing for non-compliance with any system-wide pricing conditions, such as caps or zero-fee requirements.

Box 2.12 on page 98 is a snapshot of the use cases, functionalities, and transaction flow for the IIPS in Zambia (NFS), one of the progressed IPS.



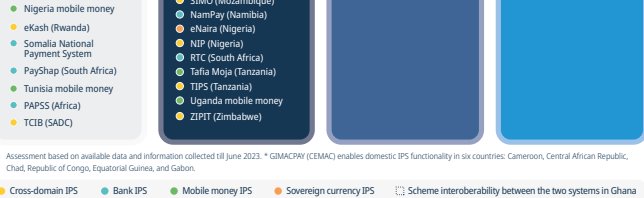


Based on the definitions of inclusivity within the AfricaNenda Inclusivity Spectrum, most IPS in Africa deliver only a basic level of inclusivity (Figure 2.12).⁴⁰

Twelve IPS do not fulfill the basic criteria of inclusivity. Of these, seven systems do not support the minimum use cases (P2P and P2B); another does not offer the preferred digital channel. The remaining four enable P2B but do not offer the preferred digital channel. For example, though SADC's TCIB and eKash are not currently ranked, they both plan to introduce P2B payments. When these plans are realized, they will improve their inclusivity.

Fifteen systems meet the basic level criteria, while five IPS fall within the progressed inclusivity category. These IPS systems cover nine countries through three domestic IPS in Ghana, Malawi, Zambia and one regional IPS GIMACPAY enabling domestic functionalities in 6 countries: Cameroon, Central African Republic, Chad, Republic of Congo, Equatorial Guinea, Gabon. Africa has no mature IPS to date. The distribution between the different inclusivity levels will likely change in the coming years given the ongoing developments and integrations within the individual IPS.

⁴⁰ For more information on how the IPS scored against the different inclusivity criteria according to each inclusivity level, refer to [Annex C](#). The fulfillment of ranking criteria is based on information available via online sources and stakeholder interviews. Access to more information may allow IPS inclusivity to be re-categorized.



At the basic level, NIP in Nigeria is the only system that offers a broad range of use cases. Yet it does not allow equal input by all participants into the scheme. Should that change, the IPS would achieve a higher inclusivity ranking.

The five IPS classified as progressed are on their way to achieving a mature state of inclusivity. They include GIMACPAY (CEMAC), and GIP and Ghana MMI, Natswitch (Malawi), and NFS (Zambia). In the case of Ghana MMI and Ghana's GIP, though they are two separate systems, they are integrated, and so jointly function as a cross-domain system operating at the progressed level.

The IPS ecosystem in Ghana supports a broad range of use cases. However, it does not enable effective oversight of recourse mechanisms provided by participants. Effective consumer recourse mechanisms are the most complex element to implement—and thus progress to a mature ranking. Only ZIPIT in Zimbabwe offers a

separate end-user recourse channel. Systems can go further by including shared recourse standards in the scheme rules and monitor participant PSPs to gauge the extent to which they are complying with consumer protection and/or data privacy regulations. In addition, they can follow Zimbabwe's lead to include an additional recourse channel for end-users who have exhausted recourse with their respective financial institution.

Having laid out a detailed picture of the IPS landscape in Africa, in Chapter 3, we will turn to the issue of digital payments usage. Our consumer research for the 2023 SIIPS widens the aperture on IPS usage on the continent with a deep review of a new set of countries: Cameroon, Malawi, Morocco, Rwanda, and Senegal. Beginning with patterns of digital payments use for both individuals and micro, small, and medium enterprises (MSMEs), we go on to explore barriers to digital payment adoption and conclude with implications of the consumer research for IPS design.





to understand payments usage and the drivers and barriers that individuals and micro, small, and medium enterprises (MSMEs) face. These perspectives can provide insights for IPS design.

AfricaNenda conducted the customer research in five countries: Cameroon, Malawi, Morocco, Rwanda, and Senegal.⁴¹ Researchers interviewed a sample of over a hundred individuals and MSMEs in each country to understand their experiences (see Box 3.1 and [Annex A](#) and [Annex E](#) for more details on methodology). This year's research complements that done in seven countries in the 2022 SIIPS report using a similar methodology.⁴²

While the sample is not nationally representative, it nonetheless helps to identify patterns, constraints, and drivers of use to unlock the value of digital payments. Our research focuses particularly on the “emerging market segment,” lower-income people and MSMEs based in urban and peri-urban areas, who are core to informing inclusive IPS design. The sampling strategy was set up to ensure that approximately 75% of participants are current or former digital payment users, to provide sufficient insights into the early usage and habitual usage constraints that exist beyond access.⁴³

The chapter is structured as follows:

- 1** The first section explores the most frequent payment needs and opportunities for digitalization by examining digital payments usage per country and per user group.
- 2** The second section assesses the drivers and barriers to digital payment access, early usage and habitual usage.
- 3** The third section summarizes the core customer research findings.
- 4** The fourth section concludes with implications of the customer research insights for IPS design in Africa.

⁴¹ AfricaNenda works at a pan-Africa level. The countries each year have been chosen based on existing relationships with AfricaNenda and to reflect the different context of diverse economies and regions in Africa: East, West, Central, North and Southern Africa.

⁴² The 2022 countries surveyed were the Democratic Republic of Congo, Egypt, Ghana, Kenya, Nigeria, Tanzania, and Zambia.

⁴³ The actual proportion of digital payment users per sample country can be found in [Annex D](#). The average percentage of people that have made or received a digital payment across the five consumer research countries based on the World Bank Findex data is 42% (World Bank 2021c).

systems, the study sample focuses on the emerging market segment, the low-income and peri-urban payments. Thus, researchers sampled only low-income earners and MSMEs in urban and peri-urban areas. Among sampled MSMEs, the study includes those that also conduct DFS agent activities. The sample focused on digital payment users: 79% of quantitative respondents and 90% of qualitative respondents made or received at least one digital payment within the prior month. The sampling strategy was set up to ensure that approximately 75%⁴⁴ of participants are current or former digital payment users, to provide sufficient insights into the early and habitual usage constraints that exist beyond access.

The customer research sample is not nationally representative, and any inferences made on a country-by-country basis are with respect to the sampled respondents, to help identify patterns, constraints and drivers in the usage of digital payments. The sample includes a range of respondents of different genders, ages, and income regularity.

See [Annexes A and E](#) for more details on methodology.



⁴⁴ The actual proportion of digital payment users per sample country can be found in [Annex D](#). The average percentage of people that have made or received a digital payment across the five consumer research countries based on the World Bank Findex data is 42% (World Bank 2021c).

Emerging Cluster					Nascent Cluster	
Cameroon	Malawi	Rwanda	Senegal	Morocco		
Share of adults using digital payments						
Proportion of population using digital payments over the past year [Global Findex 2021]	50%	40%	39% ^(2017 data) ⁴⁶	53%	30%	
Super users						
Proportion of the SIIPS sample using digital payments weekly [including agent-assisted payments]	Individuals	83%	57%	90%	94%	17%
	MSMEs	76%	77%	98%	94%	8%

Sources: World Bank 2021c; Author calculations

⁴⁵ Assisted transactions are defined as transactions that are cash along the first-mile—meaning, the customer brings cash to an agent or bank branch staff member and initiates the payment—and digital along the middle-mile from the sending institution to the receiving institution.

⁴⁶ Up-to-date information is missing for Rwanda as they did not participate in the Findex 2021 study.

category for emerging category super users. The emerging category super users are more prevalent than in 2022, as most 2023 countries fell into the emerging cluster (AfricaNenda, 2022a):



Leading category: None in 2023 (Kenya and Ghana from SIIPS 2022). A wide range of use cases has been digitalized and are frequently used by a growing portion of the population. A small sub-set of users in the countries sampled in 2023 are already using a range of use cases on a frequent basis (for example, Cameroon, Rwanda, and Senegal).



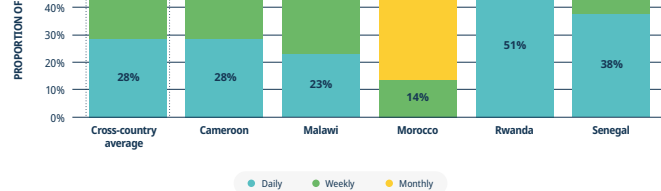
Emerging category: Cameroon, Malawi, Rwanda, Senegal (Nigeria, Tanzania, and Zambia from SIIPS 2022). Certain users make or receive digital payments for some use cases. Cameroon, Rwanda, and Senegal have a small group of users who perform a variety of digital payments on a frequent basis. However, large portions of consumers have still not adopted digital payments. Respondents from Rwanda faced access barriers (lack of phone and internet), but limited early usage barriers. Cameroon and Malawi faced both access and early usage barriers (lack of internet, phone, and trust). The main barriers in Senegal were trust (early usage barrier) and the lack of a reliable network (habitual usage barrier).

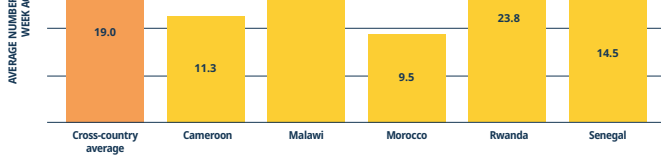


Nascent category: Morocco (the Democratic Republic of Congo and Egypt and from SIIPS 2022). The ecosystem of digital payment is immature, with low levels of adoption and infrequent use. Access and early usage barriers are typically high—either due to lack of physical access (i.e., agent points) or availability of digital payment instruments. In Morocco, the main constraint was limited network effects, followed by lack of trust and understanding of how to use the instruments.

Countries that fall into the emerging category based on digital payment adoption, both in the 2022 and the 2023 sample groups, have wide variations in their share of super users. In Tanzania, for example, half of the sample use digital payments, but only 21% of them are super users—meaning, those who use digital payments do so infrequently, perhaps by sending a remittance or

receiving funds on an irregular basis. In Cameroon and Senegal, in contrast, 94% and 83% of digital payment adopters are super users, respectively. Countries with a high share of super users in a place with relatively low overall digital payment adoption, such as Rwanda and Senegal, highlight opportunities to extend digital payment access to additional market segments.





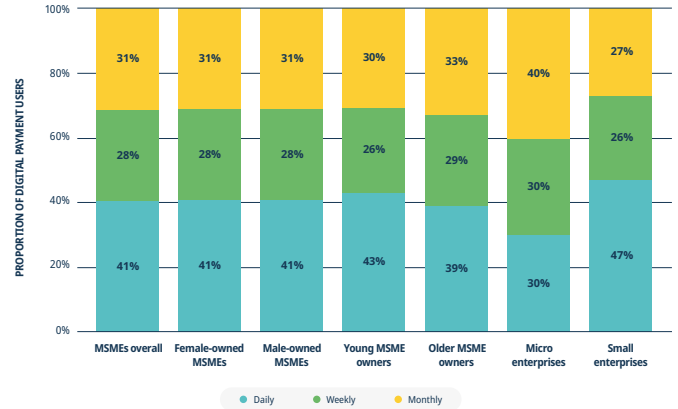
Frequency and volume of use: USER GROUPS

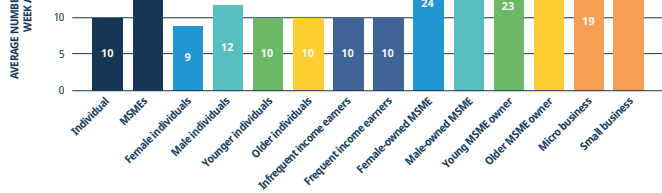
The data shows similar levels of digital payment usage between younger and older adults, and between adults with more or less frequent income streams. Larger differences exist between women and men users of digital payments (Figure 3.3). Surveyed men were 33% more likely than women to leverage

digital payments daily, for example. Women instead are more likely to utilize digital payments on a weekly basis.

Between individuals and MSMEs there are bigger differences, with individuals showing less digital adoption than MSMEs, especially small merchants: 17% of individual respondents, on average, use digital payments daily versus 41% of surveyed MSMEs (Figure 3.4). This is consistent with the findings from SIIPS 2022.

FIGURE 3.4 | Frequency of digital payment usage per MSME user groups





The averages displayed in Figure 3.5 obscure underlying differences between user groups within countries. Table 3.3 provides a more nuanced view of within-country differences by user group. Cameroon and Morocco have the most pronounced differences, particularly for gender and age, whereas Rwanda and Senegal have the least differences. Gender differences among MSMEs are most pronounced in Morocco, with an 18-percentage point gap between women-owned and men-owned MSMEs. The opposite was true in Cameroon and Malawi

where women-owned MSMEs outpace men-owned counterparts in digital payment use. Younger individual and MSME respondents are more likely to use digital payments in Cameroon and Morocco, whereas in Malawi and Rwanda, the opposite is true. Qualitative interviews reveal that the younger respondents who are active users of digital payments do so because they feel comfortable with technology. Older respondents adopting digital payments, in contrast, have more transactions overall and thus more opportunity to pay digitally.

Note: The findings regarding age have been consolidated for MSME and individual respondents.

3.1.2 Apps, USSD and mobile money agents are the most-used payment channels

The researchers analyzed the use of digital channels versus cash and found that respondents used nine digital channels across the five sample countries: app, ATM, bank agent, mobile money agent, near field communication (NFC), QR codes, SIM toolkit, unstructured supplementary service data (USSD), web browser.

Apps are most prevalent among digital channels in Morocco and Senegal, reflecting comparatively high smartphone and internet penetration in these countries (GSMA 2022; ANRT 2017; Pew Research Center 2018; see Table 3.4). USSD dominates in Malawi and Rwanda (used by 35% of respondents in Malawi and 50% of

respondents in Rwanda). In Cameroon, mobile money agents are the primary channel for 19% of respondents, closely followed by USSD at 18%. Though other channels are more prevalent than USSD in three of the five sampled countries, USSD is the only channel that has reached 30% as a share of digital transactions in any country. This is a reflection of the share of people in these countries who own a mobile phone with basic functionality, as compared with smartphone adoption, which drives app usage. Morocco is particularly fragmented in its channel use, with the top three channels only accounting for a combined 24% of respondents' primary usage, compared to a range of 35%-56% for the top three in the other countries.

Despite frequent and growing digital payment use, agents remain important due to the continued prevalence of cash. This is particularly true in Senegal, where 74% of respondents use both self-service digital channels and agents at least once a week. In Malawi, respondents use

agents for utility payments, and in Morocco, to receive pension payments. In Cameroon, agents play a critical role in facilitating digital transactions for women and the elderly, while in Rwanda, agents help those who lack access to digital devices.

3.1.3 Use cases

The research explored use cases for individuals and for MSMEs. For individuals, the most common digital payment use cases were receiving money, saving money, and purchasing mobile minutes (airtime). Yet the findings suggest that opportunities exist to increase the use of digital payments for other use cases, such as purchasing household goods and for salary payments. For MSMEs, the use of digital payments for airtime and transport for staff are common, while digital payments are less common for paying suppliers—an open opportunity.



How individuals use digital payments

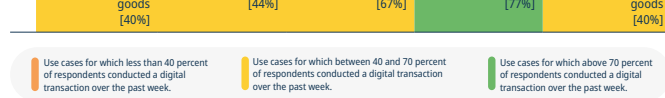
Individual respondents report receiving and sending money, saving money, and

purchasing airtime using digital payments; these use cases have been more widely embraced than others (Table 3.5).⁴⁷

In contrast, most person-to-business (P2B) payments—for purchases such as household goods, services, and transport payments—and business-to-person transactions—such as payroll—continue to be made in cash. In Senegal, for example, 75% of respondents conducted at least one digital transaction for airtime over the past week but only 40% of respondents did the same for household goods.

Lower relative use of digital payments for these transactions is partly a function of the fact—highlighted in Chapter 2—that most IPS launch with P2P use cases and then roll out other use cases, such as P2B, B2P, etc., later. There are opportunities to further digitalize these payments to benefit both merchants and end users.

⁴⁷ The research looked at the following use cases: save money, send money to family and friends, receive money from family and friends, receive salary, pay for government services, receive money from the government, utility payments, charity payments, pay for household goods, pay for a service, pay for a large household item, settle recurrent payments, loan repayments, airtime, pay for insurance premiums, and pay for medical services.



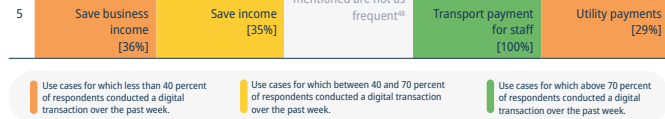
*Sample size <5 respondents

Other discrepancies shown in Table 3.5 suggest additional areas of opportunity for expanding digital payments. For instance, in Morocco, 53% of respondents conducted a digital transaction to buy household goods, but only 11% for payments for services, suggesting merchant payments are ripe for expansion compared to bill payments. Similarly, salary payments are well digitalized only in Rwanda, where 70% of respondents receive their salary payments digitally; salary payments are thus another area of opportunity for expanding payments digitization in many countries.



MSMEs

Table 3.6 shows the clear potential to further digitalize business-to-business (B2B) use cases for MSMEs. Payments to staff for airtime and transport are well digitalized, as are customer payments. In contrast, B2B payments to suppliers and utility payments to private sector companies remain largely cash based. Transactions related to loan or savings products are also still commonly conducted in cash by MSMEs across all markets in the sample.



Note: The researchers assessed the following use cases: save business income, staff salaries, pay for government services, receive money from the government, utility payments, send staff money for transport, send staff money for airtime, supplier payments, receive customer payments, settle recurrent payments, and loan repayments.

* Sample size < 5 respondents

MSMEs in Rwanda and Malawi show the deepest early use of the primary digital payment uses cases for MSMEs, with more than 80% of MSMEs in these countries reporting digital transactions for three out

of the five payment types. In contrast, less than 50% of respondents conducted digital transactions for several common payment use cases in Cameroon, Morocco, and Senegal.

48 In Morocco, MSME respondents indicated that they only conduct transactions on a weekly basis for three use cases, consistent with the low volume of transactions for Morocco. The qualitative research suggests that this is because the surveyed MSMEs in Morocco generally do not conduct as many transactions as in other countries, including transactions with suppliers conducted on a less frequent basis. This is likely the case because Moroccan MSMEs have a higher monthly revenue than the MSMEs in the other four analysed countries which implies that they can afford to conduct less frequent, higher-value transactions.



Access: Before consumers can use a digital payment product, they must have a financial account. This requires physical access to agent and bank locations (for both mobile money and financial institution account opening) and any necessary account-related documentation. Language barriers can prevent access to the institutions, tools, and information needed to open an account.



Early usage: Once registered, the consumer must have a compelling reason to use a new digital payment method instead of cash. The decision to use a digital payment instrument depends on the perceived balance between the costs and benefits of use, which reflects behavioral biases, comfort, and preferences. Awareness, user capability, and trust are critical factors to drive once access constraints are removed.



Habitual usage: Over time and through habituation, digital payments become embedded into daily life, as consumers move from ad hoc transactions to consistent and frequent use of digital payments for a variety of use cases. Among the range of factors that impact whether digital payments are habitually used, five stand out as most significant: ease of use, network effects, reliability, recourse, and speed.⁴⁹

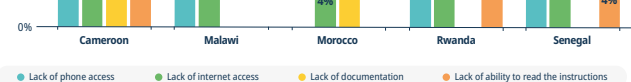
49 Network effect is defined as a phenomenon by which the utility of digital payment products and services for a user depends on the number of users using it: the more users using a product, the more value each user will get (Guliani 2022).



Access to smartphones and to the internet are a major digital payment constraint

Consumers face several constraints that limit their ability to access digital payment products. The foundational features of a digital financial service (DFS) ecosystem include network connectivity, ownership of mobile devices, possession of the necessary documentation, and access to agent or branch networks. As described below, AfricaNenda finds that access to phones and the internet are the primary barriers in the countries we surveyed; our research also draws attention to the critical role played by agents in promoting digital payments usage (Figure 3.7).

Lack of phone and internet access were cited as the primary access barriers in most of the countries surveyed. This finding underscores prior research indicating that moderate smartphone adoption rates (49%) and data/internet coverage (60%) among adults in Africa may limit end-users from using services that require a smartphone (GSMA 2022). Access to phones was most pronounced as a barrier for respondents in Rwanda and Senegal. While Senegal has relatively high phone penetration, a lack of smartphones limits individuals' ability to use apps and QR codes, both common in the Senegalese ecosystem.⁵⁰ Internet penetration is a leading barrier in Rwanda and Malawi, as well as in Cameroon. A lack of access to documentation was reported as a barrier only in Cameroon and Morocco. Language- or literacy-related barriers affected a small proportion of respondents in Cameroon, Rwanda, and Senegal.



For both individuals and MSMEs, agents represent an important means to access digital payments. This is especially true in Cameroon, where agents were effective in raising awareness of digital payments, assisting with onboarding, enabling transactions, and resolving queries (see adoption and usage factors below). Consumers perceive agents as their link to service providers and as a key contact point when they are unsure how to conduct a digital transaction or resolve a transaction issue. This is notable, as some respondents expressed concerns about the accessibility of agents. In Malawi, for example, 39% of respondents reported access to agents as a challenge. Indeed, research by the International Monetary Fund found that there were 553 active mobile money agents available per 100,000 adults in Malawi in 2020, far fewer than in Senegal, which had 2,282 active mobile money agents per 100,000 adults as of 2021 (International Monetary Fund, 2022).

use drivers identified in our qualitative research included: meeting a specific payment need that is more suited to digital payment rather than cash, safety, and security.

Several other factors influence usage as well, including possessing the skills and capability to use digital payment instruments, privacy concerns, and trust in the digital transaction process (Figure 3.8).



I PAY USING CASH BECAUSE
I DON'T HAVE MEANS YET TO
BUY TELEPHONE."

— Male, 18-29, Non-user,
Micro enterprise owner, Rwanda

⁵⁰ Smartphone adoption in percentage of total connections was reported to be 48% in 2018 (GSMA 2019).



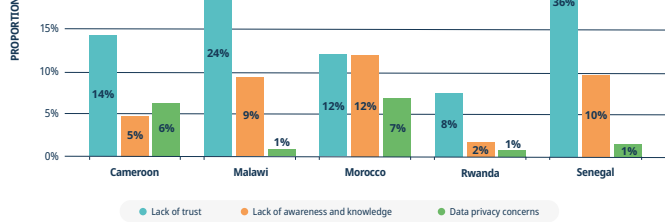
Willingness to use digital payments requires trust

Willingness to use digital payments for the first time is influenced by a range of factors. Primary



PEOPLE SHOULD NOT BE
WALKING LONG DISTANCES
TO FIND AGENTS."

— Male, 18-29, Non-user, Malawi



The lack of trust stems from both prior negative experiences and respondents' lack of confidence in their own ability to manage digital payments. In Cameroon, widespread fraud has resulted in distrust among both non-users and users of digital payments. Low literacy levels affect exposure to digital payments, a foundational input to acceptance, leading non-users to distrust and avoid digital payments. Female and older respondents, in particular, fear that their poor understanding of digital payments will result in error or theft.

Other drivers of mistrust are more country specific. Where social networks play a strong cultural role, such as in Morocco, respondents trust a face-to-face interaction more than remote digital payments. Fraud experiences, especially in an environment where trust is paramount, diminishes digital payment usage (see Box 3.2).



proposition that is superior in terms of convenience, payments are more time and cost efficient and therefore more convenient than cash. For example, individual respondents acknowledged the benefit of digital long-distance P2P payments and recurring payments. MSMEs see the clear value of digital customer payments. In contrast, value-added services such as credit or finance planning tools were not viewed as compelling because they were not very developed or were otherwise nascent in their market. The exception was business respondents in Rwanda, who pointed to the benefit of partitioning savings and accessing loans as key early and habitual usage drivers.

The value proposition of digital payments is linked to the level of ecosystem digitalization, including the number of digital-ready endpoints. Consumers who have started using digital payments may find it challenging to maintain cashless habits if their social network and local businesses predominantly operate in cash.

Marketing activities by providers, such as marketing campaigns and rewards and incentive schemes, can help increase digital payments usage within an ecosystem and thereby unlock network effects. Examples in the field include point-based reward systems, where consumers receive fee reductions and/or gifts once they earn sufficient points, earned by making digital transactions. Digital salary payments can also drive early usage by reinforcing a digital ecosystem, supporting network effects. Government intervention can spur early usage as well. For instance, during the COVID-19 pandemic, the Rwandan government introduced fee waivers for digital

WHEN I PAY FOR THINGS LIKE BANANAS OR FRUITS, I PAY IN CASH BECAUSE THE VENDORS DO NOT HAVE PHONES."

— Male, 30–44, User, Rwanda

I STARTED USING IT FREQUENTLY DURING THE COVID[-19] PANDEMIC. BEFORE THAT I USED TO USE CASH."

— Male, 30–44 years, Rwanda

payments to incentivize the use of digital channels and contactless payments (Rwagombwa 2020). The fee waiver covered charges for push and pull services, money transfers, and merchant point-of-sale transactions. This resulted in a 4.5X increase in digital payment usage in Rwanda over a four-month period from January to April 2020.

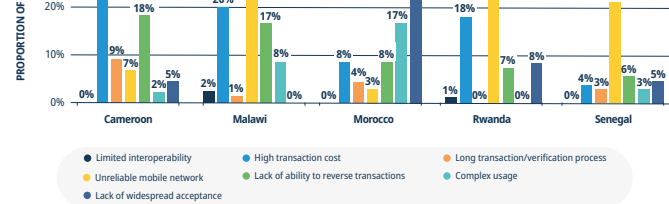
their usage becomes more sophisticated, transitioning from occasional transactions such as receiving salaries or sending remittances, to more frequent transactions such as household purchases. Eventually, digital payments become habitual and an integral part of daily life.

Barriers to habitual usage vary considerably between the different countries (Figure 3.9). In Morocco, where cash remains the most prevalent mode of payment, a lack of widespread acceptance of digital payments is a significant barrier. Infrastructure barriers can also create a barrier. For example, telecommunications infrastructure is fundamental for the processing of transactions, and unreliable mobile networks adversely affect the user experience in a variety of ways. Surveyed digital payment users in Malawi, Rwanda, and Senegal noted unreliable mobile networks as a challenge. In Malawi, the lack of reliable network coverage causes delays in initiating and processing transactions. In Cameroon and Senegal, network disruptions delay the receipt of confirmations of completed transactions.



I DISLIKE HOW SLOW THE NETWORK IS... A TRANSACTION CAN TAKE TOO LONG."

— Male, 18–29, User, Rwanda



As with initial usage of digital payments, convenience and time savings are the most compelling drivers of habitual use of digital payments. Respondents noted both the benefits from cost savings on transportation for long-distance transfers and from the convenience of conducting transactions remotely. Respondents also use digital payments to minimize cash-related risks, such as incorrect change, fake currencies, or theft, and value the traceability and visibility that comes with using digital payments (see Box 3.3 for a representation of the customer payment journey).

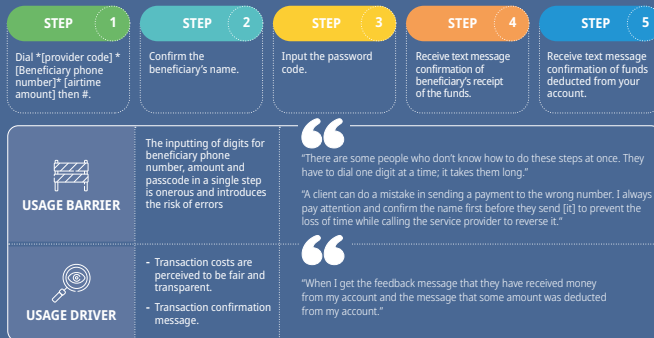


I USE [THIS APP] BECAUSE I DON'T HAVE TO WORRY ANYMORE ABOUT RECEIVING FAKE CASH [BILLS]."

— Male, 30–44, User, Senegal

 USAGE DRIVER	Users are able to have visibility into their transactions.	“The advantage of the application is that even after making the transaction, I continue to have visibility on the elements of the transaction. That is, the reference, phone number, amount, and so on.”
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FIGURE 3.11 Transaction journey example for USSD



In Morocco, for example, a lack of independent income sources limited women's digital payment usage, as men are more likely to be responsible for household expenses. In Malawi and Rwanda, illiteracy was cited as a significant challenge for women's access to and usage of digital payment services. Illiteracy can compound challenges related to low self-confidence and limited exposure to digital payments, as noted by respondents in Cameroon, Malawi, and Senegal. As a result, women respondents in these countries are more likely than men to rely on agents to transact on their behalf. Women respondents from Cameroon and Senegal reported that women risk exposure to fraud and threats to data privacy, curbing their willingness to engage in digital payment transactions (Box 3.4).

Surveyed micro businesses and individuals with irregular income do not see the benefits of digital payments when compared to larger businesses and those with regular incomes. In the markets surveyed, respondents with unpredictable income often do not consider themselves potential digital payment users, perceiving that digital payments are only for salaried employees. Similarly, owners of micro enterprises surveyed may believe their business is too small to realize advantages from digital payments.

go to school; therefore, they cannot easily make a transaction using digital platforms.”





Merchants are embracing digital payments to increase convenience and reduce exposure to payment risks

Customer demand, agent outreach, and fraud prevention are among the top catalysts for businesses to adopt digital payments. MSMEs may initially explore digital payments at the prompting of customers,

payment providers, or mobile money customers, ranging from network effects, insufficient availability of agents, fraud, and unreliable mobile networks.

MSMEs can earn additional revenue by becoming agents. Surveyed MSMEs who are also agents reported that they were recruited by mobile network operators (MNOs) or introduced to the agent business by their social circle. The main attraction for business owners to become agents is the additional source of revenue, which can diversify their businesses and enable them to increase their total income. Moreover, businesses experience liquidity benefits from having both cash and digital-based inflows and outflows.

Handling more cash increases the risk of carrying fake currency, theft, and robbery. Additionally, customers often expect agents to resolve transaction issues. To mitigate these risks, agents record transactions accurately for verification. A few ways they do this is by maintaining booklets of handwritten records of transactions or by taking screenshots of transactions. In Senegal, agents close their businesses early to minimize crime. Thus, while incorporating agent services within a business offers rewards, owner-agents must manage and mitigate risks effectively as further explored in Box 3.6.



THE 10% DISCOUNT ON THE PROVIDER'S BILL ENCOURAGES ME TO USE [DIGITAL PAYMENTS]."

— Male, 30–44,
Micro enterprise, Morocco

I WAS INITIATED TO DIGITAL PAYMENTS BY MY CLIENTS. THEY GAVE ME THE DESIRE TO GET INVOLVED IN IT."

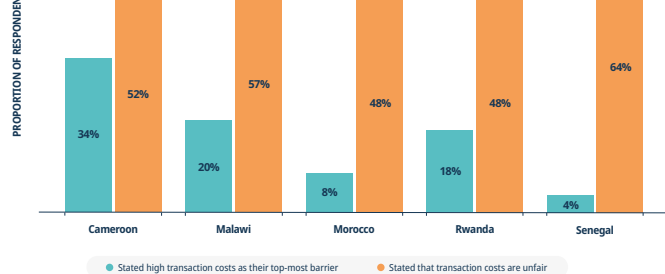
— Male, 30–44, User,
Micro enterprise, Cameroon



IT'S A BUSINESS THAT HELPS ME MAKE SOME MONEY IN ADDITION TO MY SMALL BUSINESS IN COSMETICS AND PHONE ACCESSORIES."

— Male, 30–44, Agent and micro enterprise,
Senegal





3.2.4 Fraud and the lack of recourse mechanisms is a significant barrier

Incidences of fraud primarily involve suspicious messages and calls attempting to obtain an individual's personal details to access their accounts. Among surveyed digital payment users, fraud appears to be most prevalent in Cameroon and Rwanda and the least prevalent in Morocco (Figure 3.13). For users in Cameroon and Rwanda, the most frequent fraud cases involve receiving calls attempting to access an individual's PIN or inform

consumers that they mistakenly sent money to their wallets. After those calls, consumers reported that their mobile money accounts were debited. In Morocco, respondents reported physical thefts at automated teller machines and the physical theft of wallets, which affect card users. Respondents in Morocco and Senegal, the two countries with the highest adoption levels of payment apps, reported the lowest level of fraud.

The availability of recourse mechanisms influences the impact of fraud on payment use. Although fraud incidences are higher in some countries, the impact of fraud experiences is more severe in countries with weak recourse mechanisms. For example, 47% and 27% of respondents in Cameroon and Senegal, respectively, cited a lack of resolution for their transaction issues. As a result of this high level of fraud, digital payment users in Cameroon reported abandoning digital payments entirely or immediately cashing out funds following digital transactions (Box 3.7). In contrast, in Rwanda, less than 20% of respondents reported being unable to resolve transaction issues, and as a result, few mentioned any impact of fraud on their usage behavior.



BOX 3.7 | User experience:
Discontinuing digital payment usage
due to fraud experience

Charlotte, 30-44 years old, is a women MSME owner in Cameroon who discontinued her digital payment usage due to past incidences of fraud: *"I used to have mobile money however, I stopped using it. When I had my money in my mobile money wallet, scammers would call and ask to confirm your code. When you confirm [the] code, they withdraw all your money so that's why I like when my money [is] in cash; because nobody can steal it from me."* One of her experiences with fraud involved multi-channel contact attempts: *'Another instance was when they first sent me a message informing me that I have money inside my phone, then someone called me saying they made a mistake by sending me money. On checking, he instead dragged [withdrew] my money, I didn't have money in my account.'*

customer service appears to be the most accessible, with respondents noting that agents are either unable or unwilling to assist. Users must instead visit the MNO office in person to resolve their issues.

3.3 | Summary of consumer research findings

Despite differences across the countries the sample, there were some common conclusions from the digital payment user data. **They are:**



Countries face common usage drivers and pain points. While variances exist across countries, an overarching view shows that lack of trust is the most significant barrier to digital payment, followed by unreliable mobile networks, and high perceived transaction costs (Table 3.7).



Women face additional barriers compared with men although the severity is country specific. Limited financial independence, inadequate financial literacy, lack of confidence in their ability to use digital payments, and concerns about potential fraud and data privacy are the main barriers that fortify the gender gap in digital payment use. In most countries, women transacted less frequently than men, though the impact on adoption was limited. Respondents highlighted the important role of agents as core facilitators to women's use of digital payments.

Usage behavior	users that use digital payments at least once a week [including assisted transactions]					
	Main payment providers used	MTN & Orange Money	Airtel Money, TNM, NBS Bank	MTN, Equity Bank Rwanda, Airtel Money, Bank of Kigali	Wave, Orange Money, Free Money	CIH Bank, Wafacash, Cash Plus
	Main payment channel used	Mobile money agents	USSD	USSD	App	App
	Largest user group differences based on at least weekly usage [size of gap in percentage points]	Older > younger MSME owner [17 pp]	Small enterprise > micro enterprise [32 pp]	Small > micro enterprise [9 pp]	Small > micro enterprise [9 pp]	Male-owned > female-owned MSME [18 pp]
	Most digitalized use case for individuals overall	Send or receive money	Pay for government services	Airtime	Save money	Receive salary
	Most digitalized use case for MSMEs overall	Recurrent payments	Transport money for staff/receive customer payments	Transport money for staff	Loan repayments	Receive customer payments



Drive	outreach	transactions	outreach	
	• Network effects • Marketing promotions • Free transactions or lower transaction fees	• Employer influence • Marketing promotions • Minimize risks related to cash use	• User rewards and incentives	• Minimize risks related to cash use • Employer influence • Helps with personal financial planning



inadequate service delivery.



Preconditions for success. Phone and internet access remain significant impediments to digital payment use, outside the control of the IPS and direct participants. National digital strategies are key to improving these areas.

IPS design implications:



Trust in provider and importance of recourse. Trust is critical for early and habitual use of digital payments. Scheme rules and operations have to build and retain consumer trust by enabling effective recourse mechanisms and consumer protection. These should scale with the dynamics of the market and can range from simplified rules to complex, centralized real-time analytics.



Consumer capability and awareness. Distribution, marketing, and targeted education initiatives are needed to onboard and empower consumers, especially women and the elderly, and to demystify risks of digitalization. Industry-wide initiatives could be adopted toward this goal in collaboration between IPS and participants.



Reliability and ease of use are key drivers or barriers to use. Complex USSD menus, failed transactions, and apps that malfunction are particularly detrimental to use.



Prices are perceived as high across all countries. IPS and participating PSP that generate revenue through scale and volume rather than transaction costs are more likely to drive usage, given the untapped market of the financially excluded population.

- These suggested design principles were consistent with last year's consumer insights. Additional design elements to consider in countries with more significant degrees of usage include the need for value-added services to further deepen digital payments and limit cash-out transactions where usage becomes more mature.

With the combined insights from Chapter 2 on the landscape of IPS in Africa and the Chapter 3 overview of end-user adoption, Chapter 4 will examine the barriers and opportunities to scale, sustainability, and inclusivity for IPS in Africa.





must cross to be successful and achieve inclusivity. They are:

- 1 Achieving scale to drive usage
- 2 Developing a compelling value proposition to encourage PSP participation
- 3 Increasing digital financial inclusion for women
- 4 Expanding merchant and government payment use cases
- 5 Employing technology standards to facilitate IPS adoption and efficiency

This chapter provides an overview of these hurdles and the associated opportunities that IPS and their stakeholders could pursue to drive instant payment inclusivity.



- 
- Costly interoperability arrangements that are not fit-for-purpose.
- For end-users:**
- High interoperability fees.
- For private-sector players:**
- Disaggregate data from on-us transactions to better understand implications on IPS business model.
 - Embed enhanced value propositions and additional use cases into system design.
 - Develop value-added services to improve the business model.
- For public-sector players:**
- Mandate interoperability in the absence of sufficient private-sector advancement.
 - Evaluate the potential to consolidate infrastructure with regional partners.
 - Integrate government disbursements (G2P) and collections (P2G) in the IPS.
- For development partners:**
- Undertake more research to inform optimal IPS design and business models.
 - Provide early-stage funding throughout the IPS value chain investment needs, including for Proof of Concept, development, implementation, and scaling.

model, most often operated by a central bank, has to cover operational expenses. That means that it needs to achieve scale. Yet there are a number of barriers IPS face to reaching scale. They include high prices relative to cash; and competition, including by private payment solutions and regional IPS; and a lack of payment data transparency in markets with dominant PSPs running on-us transactions with significant consolidation. Below, we examine each of these issues.



Perception of digital payment transactions as expensive

As seen in Chapter 2, instant digital payment volumes and values have risen in many countries in Africa. This suggests that IPS have the potential to scale sufficiently to drive costs down and encourage widespread and inclusive use. Some countries show faster progress toward that end than others. Ghana, Kenya, Madagascar, Nigeria, and Uganda process a significant number and value of transactions relative to their gross



In regions where proprietary systems are already established, IPS need a clear value proposition to achieve scale

In certain countries in Africa, proprietary systems preceded the entrance of IPS to serve the demand for digital payments. These proprietary systems—such as Mpesa in Kenya—have played a vital role in advancing digital payments. However, these proprietary system providers are servicing the same payment corridors that the three regional IPS in this study (GIMACPAY, PAPSS, and TCIB) are counting on for significant volume to subsidize lesser-used corridors (Stakeholder interview, 2023). The regional IPS will struggle to reach scale if they try to enter these markets as direct competitors, instead of as complementary providers with a unique and compelling offer.

⁵¹ The solid foundational work is often overlooked and accounts for many misaligned systems. It is a costly exercise but ensures a more fit-for-purpose outcome. The foundational work includes research diagnostics, understanding the market and development objectives, setting the strategic objectives, building the governance and system framework, and, ultimately, system specifications, roles, and procurement. The governance framework needs to be designed upfront and not while the system is being built as it should be the driving force behind the IPS.

⁵² Users of digital payments are overrepresented in our sample of survey respondents. Therefore, while respondents indicated the expense of payments is not a barrier to use, current non-users or infrequent users of digital payments may be more inclined to choose cash over digital payments if fees are perceived as being too expensive.

Examples of proprietary solutions include:

- **MFS Africa**—This proprietary system is expanding its network of providers by offering an application programming interface (API). It self-reports connections to over 400 million mobile money wallets and 200 million bank accounts across 35 African countries (MFS Africa 2023). It currently partners with most major mobile money operators (MMOs) in Africa, and has integrated MoneyGram, PayPal, Western Union, WorldRemit, and Xoom for many corridors, including those that the three regional IPS are targeting.
- **TerraPay, Thunes, and Zeepay**—These growing payment network providers link PSPs with one another across borders. They developed in the absence of regional IPS, giving each a first-mover advantage in their respective geographies. They already have integrations with PSPs and have established themselves as trustworthy service providers. For IPS to grow, it will thus be important to identify a distinctive value proposition in markets served by TerraPay, Thunes, and Zeepay, rather than duplicating the service already available in the market.

Similarly, many private providers today are well-established in cross-border, person-to-business (P2B) merchant payment transactions. They enable these use cases through mobile money, and through local and international cards. These fintech providers offer critical services for the payments ecosystem. As newer entrants to the market, IPS operators will need to consider how their products and services interact with those already in wide usage in the market.

Examples of existing providers and services include:

- **Direct Pay Online (DPO)**—With operations in more than 20 African countries and connections to over 100,000 merchants, DPO Pay offers a payment gateway for businesses to accept credit and debit cards, mobile money transfers, and cross-border currencies in real-time, especially targeting e-commerce transactions (DPO 2023).
- **Cellulant**—With a focus on e-commerce merchant payments in 35 African countries, Cellulant collates payments through its Tingga platform, which allows end-users to pay through mobile money, local and international cards, or direct bank transfers.

confiscating, how and appropriate ownership, and lack of

Competition between regional systems presents another barrier to scale for individual IPS—although a less acute one than overlap with the private sector. There is an overlap in functionality between the planned COMESA system and TCIB. PAPSS also has continent-wide ambitions for some of the same use cases. PSPs will be faced with the choice of which systems to join and may select them on a per-transaction basis. This can translate into significant scale fragmentation. Depending on the interoperability fees charged, however, the competition could also lead to lower costs for end-users.



Payments processed “on-us” or within closed system involving multiple PSPs limit the market value of an IPS

Across Africa, there are also a significant number of transactions processed as “on-us” within the same provider system, which obscures the full picture of the market for digital payments. Instant payment transactions in Ghana, Kenya, Malawi, and Uganda are done mostly via mobile money through one or

processed by a single provider limit the scale potential for an IPS, which can lead to sustained high costs in markets where on-us transactions dominate. IPS owners and operators must therefore be diligent in examining the business case for entering different countries. If the IPS determines that there is an open opportunity it can fill, it can explore ways to incentivize the dominant players within a country to join the IPS.

On-us transactions are not the only challenge to IPS scale. Some PSPs also integrate directly with one another following a standard set of rules. This is particularly the case in Kenya (for P2B transactions), Tanzania, and Uganda. In these multi-lateral arrangements, no third-party switch is required to facilitate interoperability. Only mobile-money IPS use this model, where it is in place, it is more difficult to transition to a cross-domain model. Nonetheless, these IPS achieve lower operational costs compared to a third-party interoperability model.

This approach gives a competitive advantage to entrenched providers, which limits opportunities for innovation and enhanced customer products, especially by neo-banks, MFIs, and fintechs. A multi-lateral IPS also typically requires pre-funding from the different

⁵³ The dominant market position by one or two providers was ultimately created through telecommunications business models and policy imperatives with unintended consequences in financial services. This has stifled competition and innovation in areas required to establish a digital ecosystem. Competition regulators, together with financial regulators and supervisors, should monitor these institutions, particularly where innovation is being limited by a regulatorily supported dominant market position.

including by providing the supporting tool for all PSPs, for customers. This can increase payment service quality, end-user interfaces, and recourse.

IPS can pursue scale in several ways:



Encourage PSP adoption

IPS operators can motivate PSPs to join their system by establishing open, adequate, transparent scheme rules, continuous stakeholder engagement, and central bank support. In markets that lack interoperability, the IPS value proposition to providers is grounded in the quality and pricing of the services provided by incumbents. In markets where the providers already established interoperability—for example through bilateral arrangements—the IPS business model depends on whether the provider decides to leverage it over established channels. Incumbents want to protect their market share and would rather route their transactions through bilateral mechanisms than on the switch.

payment needs, corridors, and include banks or non-banks to be part of the IPS, creating greater choice for end-users. While these efforts require time and investment, they can create a fit-for-purpose IPS ecosystem that is available to and used by all PSPs in a region. A comprehensive stakeholder consultation process also allows IPS operators to adjust their value proposition to compete with profit-maximizing systems.



Support use cases beyond P2P

P2P transactions alone cannot sustain an IPS, as P2P involves frequent cash in and cash out transactions that remove liquidity from the system. Furthermore, even if most P2P payments were digitalized, it would still represent a limited share of the total payments value chain. Adding government-to-person (G2P), person-to-business (P2B), and business-to-business (B2B) payments is key to achieving low-cost transactions and inclusivity; as these payments are recurring, they are integral to driving lasting behavior change. (The imperatives for successful merchant and G2P payment integration are further



Seek subsidies, philanthropic funding and patient capital to support IPS during the start-up and scale phases

IPS design factors such as the type of system (bank, mobile money, cross-domain, or sovereign) and interoperability arrangements should be aligned with the size of the digitally active population, the number of PSPs in the market, and the underlying regulatory environment. A system should be self-sustainable and financed by its participants to ensure longevity.



Share infrastructure

The current capacity of many IPS exceeds the volumes of domestic transactions. Rather than building new independent systems in every instance, an alternative approach may be to leverage excess capacity in the existing switching infrastructure in regional or underutilized domestic systems. This approach, which helps to keep costs low, has been adopted by GIMACPAY and is envisioned by the WAEMU IPS—both regions are also monetary unions, which makes this cooperation easier.

54 These nostro-vostro accounts are prefunded and it is essentially the MNO's own funds sitting in another MNO's e-money account. There would need to be a special dispensation, security, or liquid guarantee to cover the funds or overall exposure of the scheme participants to any one of the other participants. Operationally, cash management becomes intensive with each additional participant. It can quickly reach a point requiring automation, depositing increasing amounts on competitor platforms, or require the implementation of a dynamic collateral management platform across the scheme.



→ High compliance burden, foreign exchange limitations, and operational challenges.

For end-users:

→ Lack of all-to-all interoperability.

- Improve trust and buy in, and to assist with compliance.
- Endorse payment digitalization agenda and implement regulatory updates.
- Allow new entrants to live-test products in a risk-controlled environment.

For private-sector players:

- Encourage participation in the consultation processes to articulate operational challenges.

For public-sector players:

- Convene all relevant players and ensure equal access for all PSPs.
- Embrace reforms to mitigate policy and regulatory barriers (especially around CDD and licensing) and foreign exchange challenges.
- Mandate greater transparency in scheme rules and detailed, disaggregate reporting of data related to volumes and values.
- Enable concept testing through regulation around innovation frameworks.

For development partners:

- Evaluate IPS design effectiveness and value proposition in the market context.

IPS may continue to exist in isolation through existing bilateral arrangements rather than through a domestic or regional switch (Stakeholder interviews 2023).

A lack of transparency surrounding scheme rules and IPS volumes and values exacerbates this prevailing sense of competition between IPS and the PSPs. Only one IPS (Rwanda's eKash) has made their scheme rules openly accessible (National Bank of Rwanda 2020). The lack of access to scheme rules deters new and established PSPs and limits their buy-in. Confusion about rules, risk mitigation measures, fee structures, and governance modalities can fuel perceptions of unfairness and a lack of balance in power between different PSPs (Stakeholder interviews 2023). Some aspects of the scheme should remain proprietary to mitigate risks—for example, the technical specifications and fraud/risk rules. Where possible, however, IPS should pursue transparency to drive higher engagement by PSPs. This includes transparency about transaction volumes and average values. Few central banks and IPS operators publish and publicize this data, which can limit participation by new entrants to a country or to a scheme. PSPs must prioritize integrations and are likely to place those with more transparency higher on their list.

In addition, PSPs face many regulatory requirements from which IPS participation does not shield them (regulatory barriers are further discussed in Chapter 5). While IPS can help address certain regulatory requirements, such as interoperability or the existence of inclusive digital rails, many countries and regions still have constrained digital payment markets.

2 Risk proportional customer due diligence regulation and guidance. Again, when non-bank PSPs are subject to bank regulations, the know-your-customer (KYC) requirements are unnecessarily burdensome given the PSPs' lower fiduciary responsibility relative to banks. Tiered KYC rules facilitate wider participation in digital payment markets.

3 Modern and adaptable cybersecurity and data protection principles and guidance.

4 Competition rules/regulations around channel access, such as the quasi monopoly by MMOs of USSD, digital connectivity access (telecom market liberalization), and the provision of DFS, among others.

5 Policy and regulatory harmonization and common payment system directives.

These challenges are major cost drivers for PSPs. IPS that are not actively working with regulators to address these issues may not offer a sufficient value proposition.



Recent work has been conducted on transaction performance (e.g., service levels). Nigeria (for NIP) and South Africa (for RTC) already have dashboards in the public domain with IPS performance data. Making this data available raises awareness of the IPS and creates visibility of the system for stakeholders, development partners, and investors. Africa needs more countries to take this step using common measurement standards. This will lead to better insights, and richer data analytics, which will, in turn, improve the common knowledge of IPS. With more, and more accurate, information, ecosystem stakeholders will be better equipped to argue for inclusive access to financial services for the poor.



Increase PSP input

IPS should provide an avenue for PSPs to provide feedback on design principles, rules, and risk mitigation measures. This feedback could inform IPS design in a way that encourages PSP participation. It can also strengthen communication between central banks, operators, and industry stakeholders.

The experience of setting up the IPS in Malawi, Rwanda, and Zambia demonstrates the value of engaging in intensive and ongoing stakeholder consultations and participant-led design principles to meet PSP needs. With stakeholder input, the IPS volume and value projections can be based on solid foundational research, including demand- and supply-side

by the industry, as it guides all payments stakeholders toward a common and mandatory goal. Endurance and consistency in policy guidance, including a strong focus on public-private partnerships, can be core to the future success of an IPS. As a case in point, Rwanda's National Payment System Strategy 2018-2024 strongly endorsed a transition to a cashless society and laid out a gradual approach, including an implementation plan and performance indicators to track progress (National Bank of Rwanda 2018).



Seek regulator endorsement

The central objective of many of the IPS highlighted in this report—such as the regional GIMACPAY system as well as Egypt's InstaPay, and PayShap in South Africa—is to increase the inclusivity of digital payments. Where central banks

have been successful in setting up innovative hubs, which are often central contact points to provide support, and guidance to either regulated or unregulated FSPs to help them navigate the legal and regulatory environment, without involving the testing of products or services. There are regulatory sandboxes, which allow financial providers to innovate and test new products in the market prior to achieving full regulatory approval or licensing but can be expensive and logistically complex to manage. Regulators can use letters of no objection, limited licenses, and provisional licenses. IPS implementors can establish approaches that allow new entrants to test products in a live environment without significant risks to the market. This could be enabled through risk-proportional licenses. The data insights that IPS transactions produce can be used to create robust identity proxies or aliases, customize consumer and business products, and foster open banking and open finance.

of IPS participants. The following details more detailed incorporation into scheme rules:

- **Centralized insights into consumer protection and cybersecurity threats**—Using transaction pattern data, IPS can supplement PSPs' existing transaction monitoring systems. Having access to this kind of data would be especially relevant for mobile money providers and MFIs, who might lack the capital and capacity to implement advanced fraud monitoring tools. Delivering these insights requires

creating a robust technical assistance center/**central banks, financial intelligence centers/units, and PSPs**—IPS can serve as a central resource across its market for a range of technical capabilities, for example, by strengthening PSP risk assessment models and implementing concepts such as simplified due diligence. This is feasible only if all licensed PSPs are IPS participants. Having a central source of methodology and knowledge at the IPS can help PSPs conduct risk-based assessments of their product, channels, and clients (Stakeholder interviews 2023).



⁵⁶ In South Africa, there was a growing digital divide between those that adopt digital services and those that withdrew all their money at an automated teller machine. RTC was geared to large transactions and wealth management clients due to the pricing and as a way of not cannibalizing existing product and revenue lines. Predictably, commercial strategies rendered RTC largely underutilized. The national Treasury, which is the financial inclusion policymaker, had initiated discussions across the industry years ago and formulated the concept of IPS for the mass market using proxy identifiers. The concept languished for years until it was revisited, given persuasion from the central bank and other regulators and supervisors due the numerous private attempts to establish a viable e-money ecosystem that had ended and given a change in strategic thinking on retail revenue lines.

→ Gender-specific designs are not yet being embraced as a standard in IPS development.

For end-users:

→ Women's digital payments needs are not adequately met.

- Analyze gender-disaggregated data to identify product and service design opportunities for women.
- Establish effective recourse mechanisms to counter fraud and increase trust.
- Integrate G2P use cases.

For private-sector players:

- Leverage agent networks, including female agents, to onboard, build trust, and familiarize women with digital payments.
- Adopt eKYC processes for improved access.
- Improve go-to-market activities and features for women, taking into account lower smartphone ownership and appropriate proxy IDs.
- Improve recourse mechanisms for female clients.
- Collect and analyze more gender-disaggregated transaction data.
- Leverage technology to improve security of personal information and physical safety for women.
- Develop gender-inclusive strategies to reach more women.

For public-sector players:

- Remove regulatory access barriers by reforming documentation requirements for KYC and explicitly allowing eKYC.

- Develop public- and private-sector women's financial assistance around gender intentional and transformative strategies.
- Support public- and private-sector strategies around increased smartphone roll-out and adoption.

4.3.1 Can IPS address barriers to women's financial inclusion?

The gender gap in digital payments remains significant. In some countries in Africa, it has widened in recent years. Global Findex data highlights that women in Sub-Saharan Africa (SSA) are still less likely than men to own and use either a financial institution account or a mobile money account. Specifically, the gender gap in bank accounts nearly doubled from 7% in 2011 to 12% in 2021 (49% of women have a financial institution account versus 61% of men). The gap is smaller for mobile money accounts: 30% of women own one compared with 36% of men. Individual countries see different dynamics as it relates to the widening or the narrowing of their account ownership gender gap. For example, in Côte d'Ivoire and Mozambique, significantly fewer women than men opened new accounts since 2017, increasing the gender gap to 27% and 22%—from 11% and 18%, respectively

(World Bank 2021c). While account ownership in Ghana and Senegal is relatively high, women are 10% and 15% less likely than men to have an account, respectively (GSMA 2023a). Across the sampled countries in this report, digital payments usage is also subject to a gender gap. Indeed, women conduct fewer digital transactions than men (15 transactions for women per week compared with 22 transactions for men).⁵⁷

Several challenges will need to be overcome to narrow gender gaps in account access and digital payments, including the lack of financial literacy and independence for women, constraining social norms, and systemic lack of access to social infrastructure such as telecommunications networks, phones, government-issued IDs, and the like.



57 This finding is corroborated by GSMA in Ghana, Kenya, and Senegal (GSMA 2023a).

and Ledger-wood 2023). IPSs that want to address these systemic issues and develop women-centric outreach or product offerings may also be stymied in that effort by a pervasive information gap—gender-disaggregated data remains largely unavailable at the system level.

4.3.2 Opportunities for IPS to promote gender-inclusive digital payments

While many of the key barriers to women's use of instant digital payments lie outside the purview of an IPS, there are actions IPS and other market stakeholders can take to mitigate some of them. They include the following:



Establish effective recourse mechanisms to counter disruptions and fraud

IPS should be pro-active in monitoring and enforcing recourse mechanisms across digital channels and providers. Though PSPs usually function as the first-line contact for end-users seeking recourse, IPS can provide an additional recourse channel between the PSP and the central banks or consumer protection agencies. More information on recourse can be found in [Annex D](#) and more information on fraud can be found in [Annex E](#).

a mobile money account (GSMA 2023a). Finally, social norms play a role: in six countries in Africa, women are legally prohibited from opening a bank account without their husbands' consent (World Bank 2023a).⁵⁸

In the event of service disruptions that create the need for recourse, IPS can set rigorous standards for cyber resilience and business continuity to ensure rapid recovery of services after disruptive events. Operational risk management policies established by the IPS should also include transparency clauses mandating that participants share data related to operational incidents with the participant group. Within the system design, participant PSPs can be required to perform security screenings to ensure dispute resolution timelines and comparable standards for all payments in an IPS. Centralized transaction limits and other rules-based flags are potential measures to cap the exposure of end-users to fraud (BIS 2016). IPS may also require upgrades to the infrastructure for USSD to allow end-users, and particularly women who frequently use USSD, to transact safely in high velocity settings (e.g., a market).

Multi-lateral organizations can also influence policies that exist across multiple countries. In 2020, the Group of Seven (G7) launched its Partnership for Women's Digital Financial Inclusion in Africa, bringing together five institutions to focus on building gender-inclusive, sustainable, and responsible digital financial systems (G7 2022).⁵⁹ The mainstreaming of gender across all of the Alliance for Financial Inclusion (AFI) working groups has the potential to embed gender awareness into policy (AFI 2020). To date, AFI has recognized 32 member institutions for their progress in promoting gender inclusive finance, including through policy changes. Of these, 17 are African central banks.⁶⁰ IPS designers, supervisors, and regulators can apply the learnings from these institutions in their respective jurisdictions.

governance by a central bank, which collects data from financial institutions using the national ID (World Bank 2023c). Insights from this data are fed back to the market.

IPS, especially those that run on ISO 20022 or proprietary messaging systems with enhanced data fields, can guide participants to capture sex-disaggregated data in real-time while applying data privacy and data protection principles. Alternatively, an IPS in collaboration with a central bank could mandate in its scheme rules that participants report gender-disaggregated data on a monthly or quarterly basis. This latter approach may be more feasible in places or systems that face technical barriers to data collection and reporting.

58 Cameroon, Chad, Equatorial Guinea, Eswatini, Guinea-Bissau and Niger.

59 "The African Development Bank's Africa Digital Financial Inclusion Facility is working to catalyze gender-intentional development of infrastructure, policies, regulations, and product innovation. The World Bank's Identification for Development Initiative is working to help countries build inclusive and trusted ID and civil registration ecosystems that increase access to—and the quality of—services and economic opportunities, promote the realization of rights, and empower people with more control over their personal data. The United Nations Capital Development Funds Africa Policy Accelerator is accelerating inclusive policies for digital financial services. The Abdul Latif Jameel Poverty Action Lab is funding and generating policy-relevant evidence on the effects of digital identification and e-payment systems. The Oxford Blavatnik School of Government's Digital Pathways Initiative assessed countries' ecosystems for digital development." (G7 2022)

60 Central banks of Angola, Burundi, Egypt, Eswatini, Ghana, Lesotho, Liberia, Malawi, Morocco, Mozambique, Nigeria, Rwanda, São Tomé e Príncipe, Tanzania, Uganda, Zambia, Zimbabwe.



- End-users have been awarded selectively or not digitalized.

For end-users:

- The argument for convenience is less persuasive due to lack of full use-case integration.

For private-sector players:

- Facilitate the distribution of G2P payments, especially in rural areas.
- Cater to various P2B acceptance technologies supported by both smartphones and feature phones.
- Develop and adopt adequate pricing strategies for merchants and end-users to drive adoption of P2B payments.
- Adjust P2B strategies in line with longer-term vision of full interoperability and consumer choice.
- Run targeted awareness campaigns around G2P and P2B payments to drive adoption.
- Develop B2B and other use cases to drive digitalization of the full value chain.

For public-sector players:

- Pursue G2P integration with IPS rather than with individual providers.
- Digitalize payments to and from businesses and within the public sector for full value chain digitalization.
- Run awareness campaigns around G2P and merchant payments digitalization with the private sector.
- Develop pricing strategies with the private sector and IPS to drive G2P and P2B digital payment uptake.

other types of payments including P2P, P2B, and G2P. Yet, as Chapter 2 pointed out, most IPS prioritize the P2P use case as a starting point. P2P payments alone are not enough to sustainably scale an IPS, however. Furthermore, focusing on P2P payments without adding other payment types in a timely way creates ecosystem blockages, because end users must cash out their money if they want to spend it with merchants or use it to pay bills. This happens even in countries with expanding access to financial services. For example, in Malawi, which is successfully expanding penetration of mobile money, a large share of transactions in 2022 were cash in/cash out, airtime top-ups, and P2P transfers (Reserve Bank of Malawi 2022). This limits an IPS' ability to scale and thereby bring costs down. It also prevents the development of network effects and the incorporation of non-payment providers into the IPS ecosystem (e.g., governments). In sum, scale and convenience dictate that IPS offer a wide range of use cases, particularly merchant and government payments. Yet they face challenges achieving that end.

on the rise. According to research conducted for this report, 20 IPS already support P2B merchant payments. The uptake and use of P2B payments lags far behind P2P, however, even though merchant payments represent a significantly larger addressable market. As shown by the consumer research from Rwanda (the most digitalized of the five focus countries), 56% of transactions for airtime are digital, compared to only 34% of transactions for household goods and 33% of transport payments.⁶¹

A lack of trust is one reason P2B adoption lags P2P. End-user individuals and merchants are concerned about payment transactions failing because of network quality issues. These can come about from electricity outages or poor network connections. Digital platforms also make it easier for fraudsters to operate. Each of these issues introduces trust concerns that may drive end-users and merchants to opt to use cash.

IPS must offer a compelling value proposition that these other market players do not have in order to complement their offerings. For example, IPS may promise interoperability and spare merchants the cost and effort of joining multiple systems and investing in duplicate hardware or complicated operational workarounds. Failure to land on a compelling IPS value proposition may result in continued market fragmentation and, as a result, non-inclusivity in digital P2B payments.

Some additional issues that prevent the further uptake of merchant payments include:

- Transaction fees for P2B payments can range from 0.5% to up to 10% per transaction. Even in geographies that have a transaction maximum to prevent excessive charges, the expense can decimate the margin for an MSME. In Nigeria, for example, the cap is \$2.15 (NGN1,000). Compared to the perceived low cost/no cost of cash, these charges are substantial.

G2P payments, which are often recurring, and which reach marginalized population segments, can be an important driver of scale for IPS. For some beneficiaries, they can be the only digital source of funds. According to the Global Findex 2021, 37% of banked adults in developing economies opened their first account to receive a government or private sector wage payment (World Bank 2021c). This finding highlights the significance of digital payments in expanding financial inclusion.

Only six IPS currently integrate G2P payments, however (Ghana GIP and MMI, Madagascar mobile money, MarocPay, NIP in Nigeria, and Uganda mobile money). Though the G2Px Initiative at the World Bank is supporting the modernization of G2P programs in Africa, most are not yet digitalized and instead rely on cash-based distribution. Where they are digitalized, they are often disbursed exclusively through one or two providers, limiting the distribution network, and requiring beneficiaries to cash out. Digitalization of government payments should be a priority of IPS as part of offering a holistic set of payment use cases to support the broader economy.

61 Despite the drive for digital-only transport payments in Rwanda, most people still pay moto and bus drivers in cash (Stakeholder interview 2023).

62 TNM Mpamba, a Malawian mobile money service provider, launched a digital Mastercard debit card called Khadi Mbambande in 2021 to increase financial inclusion among the underserved population in the country. The solution links with a customer's TNM Mpamba mobile money wallet and can be used by customers to make in-store digital payments, withdraw money, and make online purchases without needing a bank account (Chingape 2021; Mastercard 2021b). Early evidence shows significant take-up of Khadi Mbambande, which is now expected to bring more than 9 million new people into the payment system, approximately half of the Malawian population (Chingape 2021).

Merchant Network and more providers in place, data from Kenya and Rwanda shows increased merchant adoption of digital payments during the COVID-19 pandemic, when temporary fee reductions were in place.

Global Gateway in Africa have into operating, moving to digital P2B and B2B payments lessens the risks and operational costs of cashing out digital funds for both agents and merchants.



Encourage G2P distribution

IPS can encourage choice in the distribution of G2P payments. Including a larger provider base can lead to higher rates of digitalization and broader use cases.

Other governments could integrate digital or indirectly. NatSwitch in Malawi is planning such an integration layer for the coming year to facilitate digital G2P payments and drive scale through an e-payment gateway (Stakeholder interview 2023).

63 Additional research is needed on the G2P-IPS integration nexus in Africa, its challenges, and best practices, to better inform inclusivity opportunities.

64 TymeBank did not use an IPS to facilitate the payment processing.



→ Data sharing restrictions limit innovation.

For end-users:

→ Disjointed payments experience is inconvenient.

→ Informal currency strategies of financialization and open banking.

For private-sector players:

- Proactively develop APIs to enhance collaboration with value chain actors.
- Provide input into domestic or regional strategies on open banking and open finance.
- Develop and implement forward-looking strategies around data sharing between institutions.

For public-sector players:

- Develop an adequate domestic or regional approach to open banking and open finance.
- Develop a domestic or regional mandatory QR code standard.
- Revise regulation and guidance around data sharing where needed to drive innovation.
- Safely share data with the private sector and across public sector departments.

For development partners:

- Support private sector transition to ISO 20022 where required.
- Provide technical assistance to adopt adequate QR code standards.
- Elevate global examples of data sharing to advance development outcomes.



ISO 8583 adoption increases complexity and expense

Consider the ongoing fragmentation in the payments market between ISO 8385 and ISO 20022, the most common standards among IPS. ISO 20022 is an especially powerful standard as it provides a common language and syntax for financial transactions and for international payment systems. Large banks have already migrated to ISO 20022 to align with international counterparts and international systems, such as SWIFT. ISO 20022 is the most supported standard among IPS for which AfricaNenda has scheme information. Non-bank participant PSPs have not adopted ISO 20022 as widely, however, possibly due to costs, especially for smaller PSPs (World Bank 2021h).

ISO 8583 was designed, in contrast, for a pre-internet era and does not easily support the use of modern communication technologies. Many banks have nonetheless implemented the standard to accommodate card processing. ISO 8583 is also relatively inflexible: it has a fixed message format and limited support for



The largest PSP controls a concentration of data

Data bestows market power, especially for customer segmentation and targeting. Key market players, such as large banks and dominant mobile money providers, can wield their extensive client data for competitive advantage. This increases the entry-costs for start-up innovators and fintechs limiting competition and consumer options.

technology the cost burdens certain standards impose on participants.



Standardized QR codes

Standardization of QR codes, a low-cost and simple technology, is widely accepted as a promising means to increase convenience and unlock scale in P2B payments. Early implementations are largely using static QR code protocols.⁶⁵ Most use a shared QR code model implemented with EMVCo standards.⁶⁶ Several countries have released domestic QR code standards. The Ghana Interbank Payment and Settlement System, for example, was the first to launch and implement a domestic QR code standard, the Universal QR Code, in partnership with Hightech Payment Systems. Merchants can receive payments via a static or dynamic QR code. The Central Bank of Nigeria has also developed a domestic QR code standard, the Nigeria Quick Response Code. In 2022, the Bank of

from the smallest to the most complex micro-merchants, them an attractive option for expanding the digital payments ecosystem.

Governments and regulators also have a role to play in promoting standards. Country strategies on Open Banking and Open Finance (broadening responsible access to financial data to promote innovation and competition), for example, can propel technology standards forward. Open finance has been introduced in the UK, the European Union (EU), and Australia, among other countries. Nigeria and South Africa have taken steps toward open banking through regulatory standardization of API access. Rwanda is also considering open banking based on the European Union's Payment Service Directive 2, which requires informed customer consent for payment initiation and any use of personal data. The Rwanda Payment Systems Strategy 2018–2024 introduces APIs in the financial sector and also supports the implementation of technical standards by 2024.

⁶⁵ Static QR codes contain a fixed or static set of data that, once generated, cannot be overwritten. QR code payment systems usually start out with static QR codes, especially in emerging markets and for small merchants, as these are cheaper. Merchants can print the QR code sticker and end-users with feature phones can manually key in the number printed next to the merchant's QR code sticker. Dynamic QR codes, in contrast, can be customized, and their data changed for each scan (e.g., the transaction amount and transaction cryptogram). Such QR codes can be scanned at a point-of-sale terminal or smartphone to initiate push payments. Dynamic QR is more secure, and cryptographic techniques and time stamps can be applied for verification (World Bank 2021g).

⁶⁶ EMVCo is a global standard for payment card transactions that was jointly developed by Europay, Mastercard, and Visa. The EMV standard was created in the 1990s to provide a more secure method for processing credit and debit card payments, and it has since become the dominant standard for card-based payments worldwide. EMVCo has also developed a specification for QR code payments for mobile.





5

Chapter 4 discussed four main barriers to sustainability and inclusivity that IPS face in the current market. In this chapter, AfricaNenda aims to shine a spotlight on a concrete issue in which many of these barriers come into play. Namely, cross-border payments and the regulatory context in which they operate.

Payment regulations and policies are meant to safeguard financial system stability and integrity and provide clarity to the market. As noted throughout this report so far, regulation and regulators play a key role in IPS formation, operation, and governance, forming the foundation on which payment systems can thrive. Regulations can also inadvertently introduce friction and misalignment, however if the countries into or out of which cross-border payments pass operate under different regulations. Should this friction cause end-users to avoid leveraging digital channels for cross-border payments, it can have a negative impact on trade and remittances, and by extension on the scale potential of regional IPS.

This chapter delves into actions that may break down cross-border regulatory barriers and enable remittances (P2P transfers), MSME trade payments (B2B), and cross-border merchant payments (P2B). These are the core of cross-border retail payments. The chapter outlines the current state of cross-border payments, the challenges that exist with them, and the opportunities and approaches to regulatory harmonization that could increase digital retail payments. Exemplars from African and international regions and their respective regulatory bodies are also included.



expensive, inaccessible to many users, and are largely informal. We explore each of those issues in greater detail below.



Formal remittances corridors remain expensive

SSA is the most expensive region in the world for sending remittances, which are personal transfers by migrants living abroad; they have an average cost of 7.8% (World Bank 2022a). Remittances are a direct lifeline for many households on the continent. In addition to supporting families and communities, these flows are a way for many people in the diaspora to maintain community connections while contributing to development back home.

For some countries, formal remittances account for a large proportion of GDP—for example, in 2021, remittances to The Gambia and South Sudan were equivalent to 28% and 25% of GDP, respectively (World Bank 2021f). These P2P transfers support day-to-day household expenses, education, health care, investment, real estate, insurance, and life events in households and communities across the continent (Gupta and Pallito 2009; Hassan and others 2017). Expensive intra-Africa remittances are particularly damaging to household finances, as more migrants stay



Retail B2B and P2B payment options across borders remain inaccessible even though regional trade is on the rise

Few cross-border payment systems enable B2B and P2B payment options. Cash is still the preferred channel. For example, 80% and 75% of cross-border traders sampled in the COMESA region use cash when purchasing and selling goods, respectively (AfricaNenda 2022a). Roughly 80% of retail cross-border traders operating between Eswatini, Mozambique, South Africa, and Zimbabwe pay their suppliers in cash (FinMark Trust 2021c).

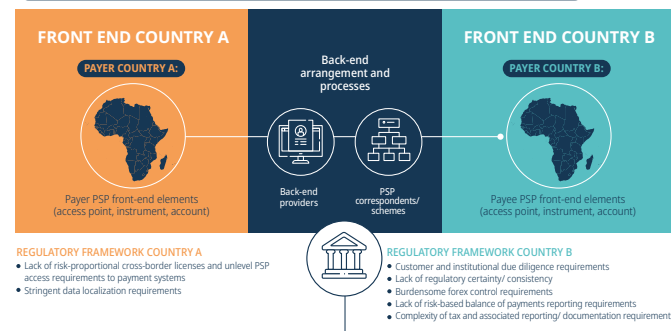
Regional trade links nonetheless are steadily gaining strength. In 2016, intraregional trade in SSA represented 20% of total exports versus 4% in the 1990s (IMF 2018b). Intra-regional trade is expected to continue to increase, aided by the Africa Continental Free Trade Area (AfCTA) agreement. Digital trade is poised to play a pivotal role in achieving the AfCTA's goals, as it is one of the catalysts to increase intra-African trade from its present level of 18% to an estimated 50% by 2030 (United Nations 2020). IIPS will play a key role in providing the necessary digital rails for intra-Africa digital trade enablement.

intelligence, and foreign exchange restrictions.

The complexity of cross-border payments is not limited to the jurisdictions involved in sending and receiving

products supply or take precedence, especially when laws between jurisdictions contradict one another. Regulatory barriers occur throughout the cross-border retail payments value chain (Figure 5.1).

FIGURE 5.1 Regulatory barriers along the cross-border retail payments value chain



Sources: World Bank 2022e; Cnfrfi 2018a

67 Hawala is an informal way of transferring money, without physical movement, through a network of dealers known as hawaladars. Hawala's distinctiveness lies in the trust-based relationship between hawaladars, often based on family village, or ethnic connections (Corporate Finance Institute 2022).

prohibited)				
e. Lack of simplified and proportionate balance of payments reporting requirements				
f. Differing tax regimes and associated reporting/documentation requirements				
g. Lack of regulatory certainty/consistency				

Sources: Cenfri 2018; Stakeholder interviews 2023



The regulatory barriers summarized in Table 5.1 create the following challenges:



Lack of risk-proportional cross-border licenses and level playing field in payment system access

The licenses given to PSPs, regardless of whether they are deposit-taking banks, or non-deposit-taking fintechs, are often not tiered according to the risk the institution poses to the financial system. A lack of proportional licensing results in onerous requirements for institutions involved in cross-border payments. While stringent prudential requirements are appropriate for large banks and deposit mobilization institutions that have multiple revenue streams and high complexity, PSPs with simpler

products and no involvement in customer deposits do not pose the same risk to the financial system and thus should not be subject to the same requirements.

A blanket approach to licensing can cement the dominance of banking institutions in the cross-border payments space and limit competition because non-bank PSPs cannot afford to bear the prohibitive costs of compliance. Supervisors often cite a lack of resources and skills on their part as the reason for blanket requirements, but using a restrictive licensing approach discounts the importance of competition and risk-based supervision.⁶⁸ Domestic regulators in different countries also have varying approaches and standards around requirements for PSP access to payment systems (World Bank 2021a). This results in an uneven playing field for non-bank and bank PSPs.

of address, source of income, import and trade licenses for B2B transactions, and business licenses for MSMEs, among others, are exclusionary requirements for many prospective customers.

Not only do cumbersome CDD requirements lead to higher operational costs and limit access to end-users, but in the absence of a risk-based approach, PSPs are effectively focusing on compliance rather than on monitoring money-laundering or terrorist-financing risk. Whereas compliance risk refers to risk of failing to comply with regulations and legislation (with non-compliance leading to fines or other disciplinary measures), money laundering, terrorist financing, and proliferation financing risk relates to the extent to which a product, client, institution, or sector can be exploited for illicit activities.⁷⁰

Many countries are on the FATF grey list because they do not effectively measure and understand AML/CFT/CPF risks at a country level. Grey listing

for non-deposit-taking institutions, means cross-border payments need prior review by regulators or government authorization before execution, depending on factors such as the destination country or the send amount. For instance, money in WAEMU cannot be sent outside the region without going through a local bank and providing supporting documentation (Cenfri 2018b). Many transactions must be conducted through bank branches and/or authorized foreign exchange dealers. Some jurisdictions have adopted tight foreign exchange rate management regimes, which can have expensive consequences for formal cross-border PSPs. For example, Nigeria's tight exchange rate controls, sparked by a plunge in oil prices and foreign exchange revenue, led to large parallel foreign exchange markets and significant impacts on the formal cross-border payment industry (World Bank 2017a).

⁶⁸ In the Intergovernmental Authority on Development region, the regulatory framework for non-bank PSPs imposes similar obligations as they do for bank institutions—this is the case for Djibouti and Uganda (UNCDF 2022a). The example highlights the case where domestic regulatory regimes do not have a distinguished licensing system for non-bank and bank PSPs respectively.

⁶⁹ In the SADC region, the requirements around timing and verification of a customer's identity differ across countries—e.g., in Angola, AML laws permit PSPs to complete the verification of a customer's identity as soon as reasonable after the establishment of the relationship, whereas Botswana, the Democratic Republic of Congo, and Lesotho do not allow for this approach (FIMark Trust, 2014).

⁷⁰ While non-compliance can lead to fines or other disciplinary measures, money laundering or terrorist financing risk can lead to serious threats to the financial system and end-users. These two risks are different and should be understood separately. An institution can have low compliance risk because it follows all the regulatory requirements (onboarded users have proof of address documents and identity documents) yet their AML/CFT/CPF risk can be high if rules-based risk management approach makes it easy for criminals to circumvent the system.



Stringent data localization requirements and prohibited cloud data storage

Stringent data localization requirements affect PSPs' ability to operate in different jurisdictions. In addition to higher costs due to duplicative data servers, a poorly installed local data server can introduce data security risks for a jurisdiction compared to cloud computing or shared data centers, which often have more rigorous risk containment capabilities (Yayboke and others 2021; Kugler 2021). Cloud storage instead of physical data storage in servers within a country's borders can still ensure data privacy and sovereignty for customers and nations, respectively. For example, South African PSPs must comply with a stringent set of requirements to use offshoring and cloud computing services (South African Reserve Bank 2020).



Lack of simplified and proportionate balance of payments reporting requirements

The balance of payments (BoP) is a vital source of information for a country—it specifies information on

important economic indicators, including remittances, to allow for comparison across countries (UNCDF 2022c). The process of submitting BoP reports to central banks can be burdensome for payment providers, however, due to the lack of standardization of remittance codes associated with transactions. Therefore, manual consolidation of different remittance codes is needed for the different supervisors, including reason for transfer.

An overly detailed BoP list can lead to data inaccuracies, as the reasons for sending a transfer can be overwhelming and not mutually exclusive. As a result, some PSPs choose default/blanket codes, such as “family support,” that are imprecise or incorrect and thus distort BoP data accuracy (IMF 2022a). In addition, the reporting requirements vary across jurisdictions, which makes process automation challenging. Some partner institutions require detailed BoP declarations, and some receiving remittance cash-out payments will not approve a transaction until the BoP is reported and formal trade documents (e.g., bills of lading, invoices) are provided. This disproportionately affects smaller PSPs who do not have the capacity or funds to streamline the process through automation. Requirements, such as for invoices that are tied to BoP declarations, can discourage end-users from conducting formal transactions.



Differing tax regimes and associated reporting/documentation requirements

Cross-border PSPs must abide by the varied tax systems of multiple jurisdictions, including submission of numerous documents, all of which raises the cost of providing cross-border payments services.⁷¹ As an additional barrier, tax authorities in different jurisdictions are increasingly seeking to pre-emptively tax trans-national digital transactions and proceeds, including remittance receipts for family support, which are considered taxable income for recipients. In some countries, cross-border transactions to a mobile money wallet result in additional tax costs to the recipient. These taxes reinforce the appeal of cash, as they do not apply to over-the-counter cross-border transactions. For example, in Uganda, a 1% levy imposed on all mobile money withdrawal transactions, including remittances, in 2018 was quickly cut to 0.5% following public pushback and a 24% drop in transaction values (UNCDF 2021a).

The 1.75% e-levy imposed on all electronic transactions, including remittances, in Ghana was reduced to 1.5% and then 1% in January 2023 (Ghana Revenue Authority 2023). Cameroon introduced a 0.2% mobile money tax in 2022 while Zimbabweans have paid since 2018 the highest money transfer tax (2%) in Africa.⁷²

The US Foreign Account Tax Compliance Act requires foreign financial institutions to report all accounts owned by US citizens and other covered individuals (i.e., green card holders, permanent residents) directly to the United States Internal Revenue Service (IRS). The PSPs must register with the IRS, irrespective of whether they receive payments directly from US sources. Subject to certain limits and circumstances, PSPs may be required to report KYC/CDD, private customer data, detailed balances, and transaction information (SARS 2023). These taxes and their associated compliance requirements can severely undermine PSPs' ability to reach scale, as remittance senders and receivers alike are driven to the informal market to avoid transaction-based taxes. (World Bank 2017b).

⁷¹ The East African Business Council highlighted how the harmonization of EAC domestic taxes are key issues for PSPs (East African Business Council 2021).

⁷² The Intermediate Money Transfer Tax is a direct tax chargeable on whenever a financial institution mediates the transfer of money except through check. It includes US Dollar nostro accounts as well as the transfer of money from mobile money transfer agents to recipients and thereby will be incurred by all cross-border IPS that terminate in a bank account or mobile money wallet (RPMG 2022).

highlighted in Box 5.1, place further constraints on the potential of cross-border payments to scale.



banking. Reasons include changes in the regulatory and enforcement landscape, economic and trade sanctions, AML/CFT/CPF concerns, and increasing compliance costs (IMF 2017). The scarcity of correspondent bank relationships has led the banks to leverage their market power by unilaterally dictating the terms of relationships with PSPs and enforcing stringent requirements in terms of reporting, capital, and processes. For instance, a correspondent bank may require that transactions be settled in the US dollar; this adds a layer of extra-territorial US regulatory requirements, such as monitoring of trades and accounts for compliance with US Foreign Corrupt Practices Act and Office of Foreign Assets Controls, regardless of the country of origin and receipt. This increases a PSP's operational costs amid a reduction in the availability of foreign exchange arrangements.



Expensive liquidity management—Liquidity management imposes disproportionate costs on PSPs due to partnership costs and challenges with cash flow guarantees. Furthermore, there is competition for liquidity between domestic payment systems and cross-border systems. The liquidity costs arise because of the number of correspondent accounts requiring pre-funding to facilitate instant payments. This results in unproductive liquidity, which is often the largest asset on the balance sheet of a new PSP (Stakeholder interviews 2023). This unproductive use of capital makes PSPs engaged in cross border retail payments less competitive compared to other PSPs. In addition, the preference for cash by many end-users makes managing liquidity costly for PSPs. An effective cash float system that allows agents and other access points to manage liquidity effectively relies on partnerships with cash-heavy businesses or a concentration of access points closer to bank ATMs or branches.



Costly and opaque foreign exchange pricing—Currency exchange rates are a significant contributor to high remittance prices. Cross-border retail payments, especially remittances, are a major source of hard currencies for African economies. There is limited demand, and thus trade, between illiquid African currencies, partly due to the fixed exchange rates and foreign exchange controls that negatively impact the demand (BIS 2019). The actual foreign exchange spread is often much higher than the wholesale rate (official rate) given the lack of a foreign exchange market where currencies can be traded at competitive rates. The result is a dual exchange of currencies that are converted into and then out of hard currencies such as the US dollar. Partner institutions, who take on-the-spot currency risk, inflate the spread to improve average profitability, especially where foreign exchange is at the core of their business model. Foreign exchange pricing practices also lack transparency, which prevents the partner as well as the end-user from understanding the mark-ups on the foreign exchange spread.

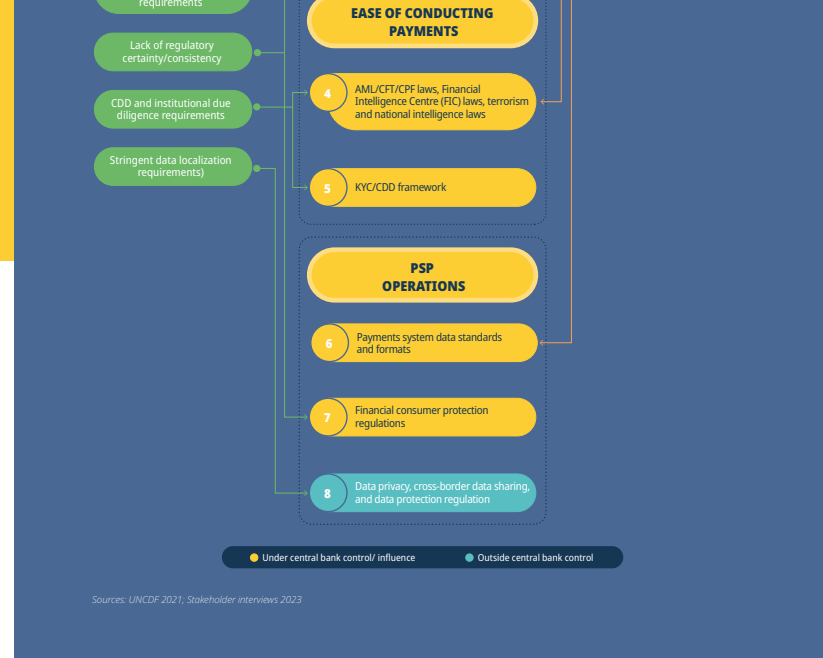


Regulatory harmonization is defined as the alignment of disparate regulatory processes and services or mutual recognition of policies and regulatory frameworks and standards (adapted from UNCDF 2022).

Harmonization is based on the three principles: cooperation, trust, and co-recognition. Cooperation ensures regulators work together to promote the development of good practices or use of a common denominator in payment regulations and policies (OECD 2020). With trust, regulators believe their peers will act consistently with their expectations. Regulators reach a state of co-recognition when the respective jurisdictions acknowledge one another's regulatory regimes, and align domestic regulation, guidance, processes, licensing, and reporting requirements based on their mutual undertakings and consistent with their common goals.

⁷³ For example, 67% of cross-border bank PSPs in Eastern Africa cited that their presence in different jurisdictions has allowed them to gain the scale necessary to introduce financial products that would not have been feasible in a single country (World Bank 2011).

⁷⁴ For example, in the Intergovernmental Authority on Development region, Djibouti has no stand-alone consumer protection legislation. Instead, Law No. 28/AN/08/Gème relates to protection, repression, and fraud. In addition, different aspects of consumer protection are found in other laws depending on the subject matter. In contrast, there are specific financial consumer protection guidelines in Uganda and South Sudan (UNCDF 2022a).



Sources: UNCDF 2021; Stakeholder interviews 2023

three categories: access to market, ease of conducting payments, and PSP operations—as follows:

Access to markets



PSP licensing and supervision regimes—

These could encourage risk-proportionate licenses that can be met with substantially similar requirements to serve lower-income end-users. This includes alignment in license requirements for PSPs to engage in the transfer of cross-border payments, and a prudential and supervisory risk-based approach adopted for varying entity types. Across Union du Maghreb Arabe (UMA) countries, for example, divergent approaches to the regulation of e-money across the region (CCAF 2021b), have resulted in stringent requirements for smaller, non-bank payment providers. Harmonization would lead to increased innovation in cross-border payments products as more PSPs would be able to offer cross-border payments services and not be restricted by onerous licensing requirements.



Foreign trade laws—These define how negotiated trade positions apply to existing laws and regulations. Trade agreements and associated foreign trade law could create a single standard for the mutual acceptance of e-money for cross-border application. Harmonization of high-level payments frameworks, acceptance principles and

in addition, over the longer term, authorities must scale down the use of hard currencies like the US dollar for cross-border settlements, lower the cost and administrative burden that comes with correspondent banking due diligence, and reduce additional and burdensome exchange margins on PSPs.⁷⁵ The administrative, investigative, and reporting burden that US and EU revenue authorities place upon PSPs, even those with no US or EU citizens among their customer base, is onerous and costly. Finally, monetary policy needs to address the current system of floating exchange rates at every correspondent bank per currency pairing for PSPs, which leads to mounting expenses.



Aligned KYC/CDD frameworks, especially around

eKYC—For end-users and institutions, this could limit the burden of gathering and confirming identity documents during onboarding and re-certification processes. For instance, COMESA Business Council (CBC), in their model policy framework on digital retail payments, recognized that across member states, proportionate risk-based approaches for KYC/CDD were incoherent or completely absent (Comesa Business Council 2021). Different frameworks must align requirements related to documentation and proof of identity used for KYC, permissions for the use of customer data via eKYC, ongoing monitoring required for CDD, and institutional due diligence standards between PSPs. PSPs are more willing to interoperate when CDD processes can be trusted across institutions and the KYC document burden decreases. Harmonization would lead to appropriate KYC requirements for end-users and risk-based CDD processes for PSPs, in line with national or regional risk assessments.



⁷⁵ When it comes to the tradability of currencies, central banks often have rules that prevent people from taking notes from other countries and holding foreign currency accounts. This directly affects how monetary policy is carried out. Central banks need to manage their foreign reserves well, especially when it comes to buying important things like fuel. This is why they often require dollars from neighboring countries.



Consumer protection laws—By harmonizing them, regulators can simplify reporting and operational requirements for PSPs at the regional level and ensure customer disputes are handled appropriately. Complaint and dispute resolution processes, as well as disclosure and transparency, should be prioritized for alignment. Harmonization would give end users and PSPs assurance that transactions have equal protection regardless of origin.



Data-related regulations—Aligning provisions related to data localization requirements and the use of data required for cross-border payments can reduce operational costs and complexities for PSPs. Existing regulations create complicated and burdensome compliance requirements, especially where domestic server deployments are required. For example, all licensed banks in Rwanda are required to maintain their primary data within the country's borders. Similarly,

zones such as CEMAC and WAEMU, together with their respective partners, have been working to harmonize payment-related laws and regulations, with the aim of creating a more seamless and efficient payment environment within their respective regions. Box 5.3 provides an overview of the ten combined RECs, monetary unions, and zones. These regional institutions are central to the harmonization of cross-border retail payment regulation and policy. Table G.1 in [Annex G](#) lists the various associated executive bodies. The regional bodies have mandates to foster cooperation and collaboration, including in cross-border payments.⁷⁶ They regularly convene with central bank representatives, setting regional policy goals that the central banks then execute domestically. Table G.2 in [Annex G](#) shows where research has noted divergence in the regulatory areas mentioned throughout this chapter, while Table G.3 lists the various initiatives that are ongoing.

⁷⁶ The SADC Payment System Oversight Committee (PSOC), for instance, was established to provide cooperation and coordination among central banks with regards to a cross-border payment strategy (Committee of Central Bank Governors, 2021). Likewise, ECOWAS was created with the main objective of promoting cooperation and integration to create an economic monetary union, which implies alignment in financial sector-related regulations and policies (Zoma & Wondjanga, 2022).



- **COMESA** is a regional economic community formed in 1994 with Comoros, Djibouti, Egypt, Kenya, Libya, Madagascar, Malawi, Mauritius, Rwanda, Seychelles, Somalia, Sudan, Tunisia, Zambia, Zimbabwe.
- **IGAD**, established in 1996, comprises eight Eastern Africa countries—Djibouti, Ethiopia, Kenya, Somalia, South Sudan, Sudan, and Uganda (Eritrea is currently inactive).
- The member states of **Economic Community of Central African States (ECCAS)** are Angola, Burundi, Cameroon, Central African Republic, Chad, Congo, Democratic Republic of the Congo, Equatorial Guinea, Gabon, Rwanda, and São Tomé and Príncipe. ECCAS was formed in 1983.

Monetary zones and unions:

- REC and monetary union: **CEMAC** is an organization of states established by Cameroon, Central African Republic, Chad, Republic of Congo, Equatorial Guinea, and Gabon established in 1994. CEMAC formed a monetary union with the Central Africa CFA franc as the common currency in 1999.
- Monetary union: Established in 1994, **WAEMU** is an organization of eight mainly francophone West African states, including Benin, Burkina Faso, Côte d'Ivoire, Guinea-Bissau (non-francophone), Mali, Senegal, and Togo. WAEMU member states share the West African CFA franc as a common currency.
- Monetary zone (no shared currency): Formed in 2000, the **WAMZ** is a group of six countries within ECOWAS—The Gambia, Ghana, Guinea, Nigeria, and Sierra Leone—who founded the organization together in 2000, and Liberia who joined in 2010.

Effective cross-border payments require sequencing and balance of domestic regulatory changes and regional initiatives need to be carefully considered. Regional regulatory payment landscape can guide domestic regulators on where to direct their efforts.



ENTRENCH IN TRADE AGREEMENTS

Trade agreements can realize longer-term harmonization outcomes.

Time to complete: Between five and ten years

Three overlapping and iterative phases of harmonization are typically required for payment-related laws and regulations, as seen in Figure 5.3.



Building block 1, policy formulation—This revolves around the development of policy at both the regional and domestic levels. Policy at the regional level provides the roadmap via the development of joint objectives and principles. The formalization of these policies and frameworks can take between one and five years depending on the consultation processes.



Building block 2, alignment of the regulatory framework—This requires the adaptation of regional objectives into the existing domestic policy environment and legislative frameworks. It also includes the actual change of regulatory frameworks. This process can be iterative, as domestic realities influence the regional policy set out under building block 1. The regulatory and legal reforms that are foundational to regional agreements can take between two and ten years, depending on the complexity of the subject and clarity of regulator mandate.



Building block 3, entrenchment in trade agreements—This involves the reform of regional agreements with a strong payment link, e.g., trade agreements, and the implementation of corresponding domestic regulation and law. This embeds the objectives supported by regulatory and policy reforms into other sectors that impact cross-border payments. Trade agreements can take between years and decades to fully implement, depending on the number of jurisdictions and granularity of agreement.

5.3.1 Building block 1: Policy formulation at the regional and domestic levels

A well-defined and clear regional policy is necessary to establish goals for regulators across jurisdictions to cooperate and coordinate in the cross-border payment space.



Regional research and landscaping activities can identify harmonization gaps and priorities, helping policymakers understand the local context and direct their efforts.

This research is typically led by external research organizations, such as AfricaNenda, Alliance for Financial Inclusion (AFI), the World Bank, or the United Nations Capital Development Fund (UNCDF), to name a few, to provide a neutral perspective.

AFI research on cross-border retail payments and remittances in the Eastern Europe and Central Asia (EECA) region provides an overview of the cross-border retail payment context,



Establishing a regional blueprint/roadmap with a financial integration endpoint provides clarity for regulators from different jurisdictions.

Several regions have developed comprehensive regional blueprints for financial integration. The examples below show how countries can develop regulatory approaches to financial integration. Buy-in and involvement from all member countries at the outset is an important precursor to regional blueprints.

77 The EECA region is made up of 22 countries, namely Albania, Armenia, Azerbaijan, Belarus, Bosnia and Herzegovina, Bulgaria, Georgia, Kazakhstan, Kosovo, Kyrgyzstan, Latvia, Moldova, Montenegro, Poland, Romania, Russia, Serbia, Tajikistan, Turkmenistan, Ukraine, and Uzbekistan.

payments.⁷⁸ Each sector involved in the ASEAN integration efforts prepared a Strategic Action Plan (SAP) to guide its efforts. The comprehensive SAP was approved by the ASEAN finance ministers and central bank governors in 2016.⁷⁹ ASEAN countries have since undertaken bilateral and multilateral initiatives to link their domestic real-time gross settlement systems and adopt a standardized messaging format (i.e., ISO 20022). Singapore and Thailand connected their fast payment systems in April 2021. According to a March 2022 article published by the information resource centralbanking.com, the connected system had processed 200,000 cross-border transactions, valued at \$44 million (Central Banking 2022).



Regulatory regional working groups or forums can establish a common purpose and address specific constraints, coordinating the development of payment system standards and elevating topics of mutual interest.

Forums allow regulators to discuss oversight approaches for domestic and cross-border PSPs, promote peer learning among regulators, and accelerate the development of common approaches to supervision.

2019, the main supervisory cooperation framework for the region, the SADC Regulatory Cooperation, Coordination, and Trust with Regulators from different jurisdictions.

- Based on the ASEAN Economic Community Blueprint 2025's goal for financial integration, ASEAN established the Working Committee on Payment and Settlement Systems and the e-Payments Coalition, a public-private partnership initiative (World Economic Forum 2020). Several cooperation agreements have been signed as a result, including an agreement between the monetary authorities of Singapore and Hong Kong that facilitates referrals of innovative businesses between the two jurisdictions, information sharing, and exchange of expertise for the purposes of financial innovation (Dunn and Scanlon 2017).
- The Committee of Central Bank Governors (CCBG) is a group of central bank governors in SADC.⁸⁰

cooperation, coordination, and trust with regulators from different jurisdictions.

- The SADC region has a set of model laws that describe the convergence state for regulators. The

- The EAC has a model policy on electronic payments and intends to create a uniform enabling framework for the region (East African Communications Organization 2017).

5.3.2 Building block 2: Alignment of regulatory frameworks

The frameworks and policy goals developed under building block 1 serve as a guide to implement regulatory changes at the domestic level. Across regions, several learnings have emerged, such as:



Proportional regulatory requirements around domestic licensing set consistent standards for cross-border PSPs.

While several regulatory areas are important to tackle domestically, the main operational barrier for cross-border payments, according to literature and stakeholder interviews, is the lack of proportionality in PSP licenses. Simply put, when new, innovative entrants such as non-deposit-taking fintechs are regulated according to the same standards as deposit-taking banks or other structurally important organizations, it discourages them from participating in the market. Proportionate licensing frameworks, in contrast, encourage entry by alternative players, fostering innovation and making it more

cost-effective for providers to serve formerly excluded and lower-income consumers. Risk-proportional licensing reduces cost by lowering the compliance burden for PSPs, and thus reducing cross-border transaction charges for end-users. There is some precedence for this.

- Licensing frameworks adopted by South Africa often become a de facto standard for central banks from other jurisdictions in SADC. The South Africa Reserve Bank grants licenses to authorized dealers in foreign exchange with limited authority (ADLAs) based on tiers associated with types of payment activities. The ADLA license, implemented in 2014, allows non-banks to offer cross-border payment services, encouraging remittances, based on tiered capital requirements. After the introduction of this regime in 2014, competition intensified, and prices fell. The services of ADLAs are aimed at low-value remittances and are more competitively priced for low-value transactions (IFAD 2022). Other countries in the Common Monetary Area (CMA), Eswatini,

⁷⁸ ASEAN consists of ten countries, including Brunei, Cambodia, Indonesia, Laos, Malaysia, Myanmar, the Philippines, Singapore, Thailand, and Vietnam.

⁷⁹ The financial sector integration vision for 2025 encompasses three strategic objectives—namely financial integration, financial inclusion, and financial stability—and three crosscutting areas—capital account liberalization, payment and settlement systems, and capacity building. The blueprint includes the development of guidelines to establish harmonized regulatory regimes and has led to the establishment of the ASEAN Payment Connectivity Initiative (2019) and the ASEAN Financial Inclusion Framework (2018).

⁸⁰ The CCBG serves as a platform for central bank governors to collaborate and coordinate their policies and activities related to monetary and financial issues. Functions include promoting regional cooperation, providing a platform to exchange information, developing policies and strategies, facilitating capacity-building and technical assistance, and representing the interests of the SADC region in international forums (Committee of Central Bank Governors n.d.).

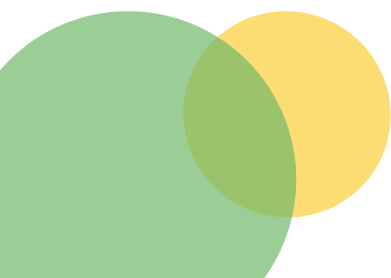
payment app, and printing of non-cash payment instrument services (Bank of Ghana n.d.).

- Regulators in the Philippines and Malaysia streamlined the licensing processes for non-bank payment service providers through the adoption of risk-proportionate regulatory requirements. They did this by licensing only those remittance service providers with a clear value proposition for end-users. Smaller players unable to comply with the streamlined rules were encouraged to become agent networks instead. The Philippines introduced a remittance platform provider licensing category that only requires registration and basic reporting. The cost of sending remittances via the Malaysia-Philippines corridor has dropped to the United Nations target of 3%, aided by simplifying compliance for non-bank providers (CGAP, 2019c).

Recognition and harmonized prudential measures (European Banking Authority n.d.). Within the supranational structure of the European Central Bank's Single Supervisory Mechanism and according to PSD2, supervision and regulatory requirements imposed on less risky institutions are less onerous than larger ones (European Central Bank 2019). The implementation of a single passport approach under the PSD2 laws has resulted in lower charges for end-users across the EU: cross-border person-to-person fees have decreased by approximately 5% for bank customers and 2.5% for non-bank customers (BIS 2022c).

in remittance in these corridors. The initiative includes a set of bilateral agreements among central banks to use their own currencies for cross-border settlements and mutual trade through commercial banks appointed as cross-currency dealers, also known as payment-versus-payment. The framework allows bilateral transactions to be done in local currencies to reduce overreliance on the US dollar (Muhammad 2023; Ito, Hiro, Kawai, and Masahiro 2021). The LCSF cooperation has expanded to include China and Japan (through memoranda of understanding). Although trade in the region is still dominated by the US dollar, total trade transactions through the LCSF using the Thai baht and Malaysian ringgit increased from 1.4% in 2018 to 4.1% in 2020 (Phoebe 2022). In addition, the Thailand-Indonesia remittance corridor has seen remittance cost reduction of 7% since 2016 (World Bank 2022a).

- There is a concerted effort on the continent for capacity building around the implementation of FATF standards and harmonization of AML/CFT/CPF frameworks, which help bring domestic regulators up to an adequate regional standard. ESAAMLG and GIABA work with their member countries to consolidate and combine efforts around AML/CFT/CPF regulations and laws to promote the adoption of the 40 recommendations made by FATF.⁸¹ Mutual evaluations are central to monitoring the implementation of FATF standards across member countries. After identifying the deficiencies raised by the mutual evaluations, both institutions work toward strengthening capacity to address detected gaps. For example, since becoming an ESAAMLG member in 2012, Angola went through a FATF action plan to ensure technical compliance standards were met.



81 ESAAMLG's 16 member states are Angola, Botswana, Ethiopia, Kenya, Lesotho, Malawi, Mauritius, Mozambique, Namibia, Seychelles, South Africa, Swaziland, Tanzania, Uganda, Zambia and Zimbabwe. Rwanda has been formally admitted as a member of the ESAAMLG but has not yet fully taken up its role as an active member of the organisation. GIABA's 17 members are Benin, Cabo Verde, Comoros, Côte d'Ivoire, The Gambia, Ghana, Guinea, Guinea-Bissau, Liberia, Mali, Niger, Nigeria, São Tomé and Príncipe, Senegal, Sierra Leone, and Togo.

5.3.3 Building block 3: Entrenchment in trade agreements

In addition to policy objectives and regulatory alignment in payments, agreements outside of the payments sector, such as in trade, can also reinforce payments goals. Trade agreements are existentially dependent on the effectiveness of payments rails. Therefore, payments regulation and trade agreements need to be aligned to be mutually reinforcing. Formal trade agreements, especially at a continental level, can take decades to be fully implemented and therefore must include longer-term visions around reinforcing the principles rather than the specific reforms in payments.



Trade agreements can be used to promote core harmonization principles

Although the timeline for formalizing trade agreements is long, they are key tools to attain long-term harmonization in payments. Trade agreements can embed harmonization principles around co-recognition, trust, and cooperation, to provide an overarching goal for domestic regulators.

While trade agreements cannot include process-level requirements and specifics for payment reforms due to the rapid pace of technical advancement, these agreements cement high-level standards, objectives, and policies.

AfCFTA came into force in 2019; as of 2023, there are 54 signatories (African Business 2022). The goal of the AfCFTA is to foster a common market in Africa and make it easier for people to conduct business across and within the continent. The AfCFTA uses the RECs to facilitate payments and trade integration between members of that region (African Union 2018). The AfCFTA's digital trade protocol covers data governance, data flows, and electronic transactions. Although it is too soon to gauge the ultimate effects of the AfCFTA on cross-border retail payments, several benefits are evident already today. Payment service providers are rethinking their strategies and expanding to more countries to prepare to reap gains from the AfCFTA's boost in intra-regional e-commerce.

The Digital Economy Partnership Agreement (DEPA) was signed between Chile, New Zealand, and Singapore

in 2020. It is the first agreement of its kind focused exclusively on trade in the digital era. It includes an entire chapter dedicated to digital payments, with an emphasis on international standards. DEPA Article 2.7, for instance, encourages parties to work together to create a consistent regulatory framework for payments. Since its implementation in 2021, DEPA has ensured that payment service providers in respective countries have adopted internationally accepted standards, like ISO 20022 (Ministry of Trade and Industry Singapore n.d.; New Zealand Foreign Affairs and Trade 2020; World Economic Forum 2022). The DEPA addresses various regulatory barriers that hinder digital trade, such as data localization requirements and restrictions on cross-border data flows, which are applicable in Africa, too.

The roll-out of the Single Euro Payments Area (SEPA) in the EU and the payment service directives PSD1 and PSD2—which have shaped the access, security, and consumer trust that underpin the overall effectiveness of the SEPA system—can serve as an example for Africa. The EU has solid footing and a supportive forum for the development of common law among EU participants; but nonetheless, PSD1 and PSD2 took, on average, five years each to formulate and pass, and then another two years

to become embedded within national law. PSD3, currently being formulated, aims to address issues that have arisen since PSD2, including right of access, recognition of licensed providers, and the development of technology.

The SEPA example highlights the long timescale necessary, even where there is an underlying union of national states and a currency union. It also highlights the iterative nature of the directives that drive access and usage and keep the platform relevant. For instance, PSD2 required banks wishing to participate in SEPA to recognize and open accounts for licensed third-party payments providers and share data, per customer consent. PSD3 now seeks to regulate the arbitrary closing of those accounts thereafter as well as develop provisions for innovative payments providers.

The key learning is that inclusive and effective cross-border payments systems are eminently possible, but the time and effort required to put them in place are significant. Examples like SEPA provide insights on how to shorten development cycles for cross-country frameworks, which can be tailored for the African context, potentially through a payment service directive for Africa to support ongoing initiatives such as the AfCFTA.

level to foster a secure and predictable environment for cross-border transactions.

- To overcome obstacles, regulators in one jurisdiction must be able to trust the capabilities and authority of their peers in foreign jurisdictions, and they must follow general guiding principles of co-recognition, trust, and cooperation.
- RECs, monetary zones, and monetary unions have important responsibilities in promoting cross-border payment harmonization efforts within their regions. They bring together representatives from central banks, set goals and clarify incentives for local-level implementation, and encourage cooperation and collaboration among members. Domestic central banks retain the power to change and adapt regulation in line with regional policies.

Regional goals: Areas of reform include proportionate payment licensing regimes, CDD/KYC regulatory reforms (including eKYC), and adjustment to foreign exchange laws and other supporting regulation, among others. Capacity building can help regulators adapt domestic-level regulations and approaches. Mutual recognition can be achieved through cooperative oversight arrangements.

→ **Multi-jurisdictional tools like trade agreements underpinned by effective payments solutions.** Having these in place can provide stability over the long term, reducing complexity in cross-border payments. These tools are better suited as principles of engagement as opposed to detailed operational guidance. Harmonization through a continental payment service directive for Africa (like PSD 1 and 2 in SEPA) can complement trade agreements such as the AfCFTA.

From this spotlight on cross-border payments and the benefits of regulatory harmonization, AfricaNenda turns to examine several trends that may affect how the availability of inclusive instant payments may evolve in Africa.



TABLE 6.1 | Market, system, and consumer trends

 Market trends	• Agents will cement their position as enablers within the digital payments value chain. • Fintechs will continue to launch innovative products and increase their networks/market share in the mobile payments market. • Regulators are revising payments and e-money laws to foster innovation and strengthen consumer protection mechanisms. • Digital ID rollouts will increasingly allow for additional proxy ID options. • Virtual assets for cross-border retail payments have the potential to divert scale from IPS.
 System trends	• Banks continue to be crucial participants of IPS. • Fintechs continue to provide front- and back-end services in partnership with established PSPs rather than becoming direct IPS participants. • Open Finance is emerging. • CBDCs are emerging as decentralized instant settlement and interoperability mechanisms, but there is more demand for building them than there is supply of the technical assistance needed to do so.
 Consumer trends	• Consumers are increasingly aware of the risks of fraud and cybercrime. • End-users are persistently price sensitive. • Consumers are steadily adopting smartphones, yet data access is increasing more slowly.

6

...ing in their agent networks to better compete in the retail payments space.⁸²

- Agents are increasingly performing activities beyond airtime and cash-in and cash-out services. However, the business case for their expansion in more rural and more sparsely populated locations, where demand is not as high as in urban areas, remains difficult. A potential solution to this challenge is to allow tiered agents, so that differentiated services can be provided by rural- and urban-based agents, to boost the business case for rural operations and for agent network expansion in these areas.
- A larger network of agents associated with an IPS via a PSP will bring scale and utility to the system. Some countries have facilitated expansion of agent networks by mandating interoperability of mobile money agents. In many countries, Senegal and Tanzania, for example, agent exclusivity is not permitted and, as such, mobile money providers share agents. In Kenya, a directive by the Communications Authority of Kenya in 2014 mandated that the dominant provider, Safaricom, open its network of agents to competitors. In Nigeria, agents can serve multiple PSPs (BIS 2023).

- Attracting fintechs to join IPS is vital to ensure that these popular services and solutions can be integrated into these central systems, thereby promoting inclusivity. Firms with significant capital stock are moving rapidly to launch in countries with smaller populations, such as Wave is doing by expanding into The Gambia. Incumbent fintechs are also forging strategic partnerships to further expand. MFS Africa, for example, has partnered with Nigeria's largest bank, Access Bank. The fintech recently entered a partnership with Western Union as well. Accelerated adoption of technologies such as QR codes is also enabling easier consumer and merchant transactions, as well as new business models and offerings.⁸³



Regulators are revising payments and e-money laws to foster innovation

- Regulatory and legal reforms in the payments sector, including updating payments and e-money laws, are underway in several African countries. These include the WAEMU and WAMZ countries, and Uganda (Stakeholder interviews 2023). At least 38 countries in Africa have financial inclusion strategies, which prioritize reducing cash usage overall and pathways to increased digital financial services use (AFI 2022b). Envisioned regulatory reforms aimed at licensing non-bank players, expanding agent networks, and enabling the use of technology in onboarding, among others, can lower access barriers. They can

also increase the number of providers in retail payments by updating regulations on wallets, digital onboarding, electronic signatures, agent licensing, consumer and data protection, account and payment license tiering including for fintechs, and the implementation of a risk-based approach.

- While these reforms are evolving slowly, and thus will only result in material changes in the medium term, the gradual updating of the regulatory and policy environment is sending positive signals to prospective providers and investors in African markets. They also increase the opportunity for regulatory and policy harmonization for cross-border payments, which will create more opportunities for IPS scalability.

⁸² Fawry in Egypt and SANEF in Nigeria have more than 100,000 access points and are not led by telecoms.

⁸³ Smartphone apps for merchant payments are increasingly available, especially in Kenya, Nigeria, and South Africa. For offline and online payments, iPay incorporates MasterCard and Visa, as well as M-Pesa and other mobile money services. (Katana 2021). SnapScan and Zapper on the back of bank cards are two payment apps in South Africa that have been successful in gaining a large user base. Nigeria's Paga mobile wallet users are expected to reach 37.6 million in 2025, more than its competitors KongaPay, MokoPay, Opay, and PalmPay combined (de Best 2021).

It supports by simplifying customer due diligence and favorable digital onboarding regulations, digital ID schemes have potential to propel users toward digital payment usage.



the spectrum, there are countries that have bans on virtual assets. On the other, there are countries that are considering formal licensing. In the meantime, private cryptocurrencies are playing a role by facilitating informal cross-border payments.

- There is no sign of increased licensing for formal cryptocurrency providers. Nonetheless, the use of informal cryptocurrency is on the rise. IPS will have the potential advantage over cross-border virtual asset schemes given established networks of agents to facilitate the vital step of cash in and cash out. The trend therefore points toward IPS increasingly needing to consider whether competing with or combining crypto technology will best enable them to scale.

⁸⁴ The adoption of crypto assets is increasing in Africa, driven by high costs and inefficiencies associated with traditional financial services, high inflation, political instability in some countries, and the potential for cryptocurrencies to offer a lower-cost and more efficient way to transact.

book and provide value-added services to lower income customers who typically would not be able to access them through the mobile money provider or MFIs that serve them. That ability could give bank participants an edge in the IPS market.⁸⁵ Offering these services would be consistent given the new focus from some banks on historically excluded populations. This is partially a response to fintechs and mobile money operators offering mobile wallets and other products that challenge bank market share across segments in both account provision and retail payments.

- While telecom companies continue to dominate retail payments via closed-loop systems in some countries, such as Kenya and Rwanda, a supportive regulatory environment and/or a competitive mobile money ecosystem can lead to cross-domain IPS with strong bank participation.

⁸⁵ While wallets will continue to see the highest take up, key markets such as Egypt and South Africa will remain card driven. Card technology is sophisticated and secure, enabling offline solutions and value-added services to the extent that mobile solutions cannot yet, partly due to regulatory restrictions.

and cross-border transactions remains a challenge across the continent. Natswitch in Malawi allows for direct fintech participation, though none have integrated yet. The capital that a fintech must tie up in an IPS for security purposes is likely too high to make it more viable than a simpler bank partnership.

- The increasing number of cross-domain and bank IPS will eventually allow fintechs to innovate on the front end of payment products for direct participants. In addition, fintechs could enable IPS to diversify the participant pool by providing integration layers and other shared services, such as government-to-person (G2P) matching and distribution, eKYC interfaces, and proxy identifiers. The stronghold that international card networks have on providing innovative virtual solutions, coupled with SWIFT's renewed focus on instant retail payments, will open more opportunities for fintechs to provide front-end innovation in the banking sector.



Open Finance is emerging

- Data is a powerful enabler of innovation: it helps IPS and PSPs to better understand and serve participants and end-users, as well as improve how they manage their own risks and business models. In response to unequal access to data among PSPs, more countries have started to implement, or consider implementing, an open finance regime. Nigeria, for example, has issued an Open Banking regulatory framework to enhance financial inclusion and improve competition and efficiency in the financial services sector (Central Bank of Nigeria, 2021).
- Open Finance consists of sharing consumer data between PSPs and/or third-party providers with consumer consent. Whereas Open Finance encompasses all financial data, including mortgages, pensions, and insurance, in addition to banking data, Open Banking is a subset of Open Finance that relates specifically to the exchange of transactional and bank payment financial data (Cenfri 2022). Payment initiation with Open Finance has the potential to reduce PSP dominance in consumer payments, because it allows third parties to initiate transactions on an end-user's behalf from an account the consumer holds with another institution.



CBDCs are emerging and interoperability mechanisms, but central banks need help with them, resulting in more demand than supply for technical assistance

- Central bank-backed digital currencies (CBDCs) are consolidated instruments that can be accepted everywhere since it is sovereign currency. They are also single channel and therefore have a much-reduced risk profile from an IPS perspective. As such, CBDCs offer simplified interoperability.⁸⁶
- CBDC schemes across the continent are currently in proof-of-concept, particularly in countries with deep experience in promoting digital adoption.⁸⁷ From a central bank perspective, these proofs of concept and pilots require significant skill, investment, and effort, and are often in direct competition with ongoing non-CBDC IPS innovation projects. As a result, no country apart from Nigeria has launched a CBDC scheme; and even in Nigeria, the design is not yet finalized.
- The value proposition for direct-to-consumer CBDCs and the resulting new ecosystem is not compelling in the short term. Yet the technical assistance programs

by the International Monetary Fund and World Bank remain in high demand from African countries. As a result, they are oversubscribed. While progress in CBDC development in Africa has been slow, projects that eventually emerge will interoperate with and thereby expand the degree of interconnectedness and hence the utility of IPS systems, leveraging existing ecosystems rather than duplicating or competing with them.

- Emerging cross-domain IPS projects will benefit from CBDC mechanisms through simplified processes, lower settlement risk (for IPS that do not settle in real time), higher degrees of interoperability, and ultimately lower consumer cost and greater choice. Current testing of domestic functionality indicates there will be more offline capabilities emphasized, with a focus on synthetic CBDCs.⁸⁸ African countries are looking abroad to observe the learnings from cross-border retail CBDC pilots (Stakeholder interviews 2023).

⁸⁶ IPS participants can more efficiently clear and settle in real time with multiple PSPs without having either direct or indirect access to a traditional central bank account. PSPs would not need to change any of their distribution channels or end-user-facing services, which is a considerable shift from engineering a completely new CBDC consumer ecosystem. This configuration represents an advantage for PSPs and end-users alike.

⁸⁷ Some notable examples are the Digital Real of Brazil and the development of cross-border multi-currency decentralized settlement in the Bank for International Settlements (BIS) Innovation Hubs Project eBridge, linking the Hong Kong Monetary Authority, the Bank of Thailand, the Digital Currency Institute of the People's Bank of China, and the Central Bank of the United Arab Emirates (BIS Innovation Hub-Hong Kong Centre).

⁸⁸ In synthetic CBDC models, private sector payment service providers issue liabilities matched by funds held at the central bank. See references item BIS 2018a for more details.

second is the degree of systemic cybersecurity risks. The prevalence and sophistication of fraud increases as consumers increase their use of digital payments. Less digitally literate end-users are more susceptible to fraud and scams.

USSD services, as implemented today, are riskier than other channels and merit extra monitoring to detect cybercrime. One reason is that USSD is not encrypted, which allows criminals to divert payments or conduct SIM swaps by tampering with command requests and responses.

Consumers have a limited risk tolerance when it comes to payments. They will replace their digital activity with cash-based transactions if fraud and cyber risks go unmitigated by IPS and PSPs. For IPS stakeholders, this highlights the need for increased monitoring of transactions, end-user awareness-raising and education, and timely and efficient recourse mechanisms.



primary market is the slow rate of smartphone penetration and continued unaffordability of smartphones. Yet smartphone ownership and use across the continent is on the rise and SSA smartphone connections are predicted to rise to 87% of mobile connections by 2030, up from 51% in 2022 (GSMA 2023b).

Some governments, such as Rwanda, are helping to facilitate that transition by subsidizing smartphone rollouts to rural areas and for vulnerable households. Several mobile money providers are lowering the costs of smartphones across West and East Africa. Most MNOs in SSA, however, continue to charge among the highest prices for mobile data globally (Cable 2022). Apps tend to be more user friendly due to the ability to embed QR codes or NFC. They are also more secure than USSD, providing momentum for IPS and customized payment products. The enhanced coverage by 3G/4G/5G services underpins the successful rollout of smartphones used for products beyond USSD.

The following chapter offers some summary observations to conclude the SIIPS 2023.

7

As this report has laid out, instant payment systems (IPS) in Africa have made tangible progress in the last 12 months. Three new systems have been launched since last year and payment volumes and values continue to climb steadily. More participants have signed on to integrate with IPS as the number of systems rises, and more of those systems are enabling all-to-all interoperability. Data on volumes and values of payments is becoming more widely available as more systems go live and more central banks and switch operators share their IPS data. More data means we have a greater ability to evaluate IPS progress toward inclusivity.

Yet the findings also reveal that more needs to be done to ensure that IPS are inclusive, and thus fulfill their potential as digital public infrastructure (DPI). We see opportunities for IPS to expand inclusivity by offering more use cases and channels that more fully address the needs of underserved groups; to enable cross-border trade; to enhance interoperability within markets; and to directly integrate with non-traditional providers like fintechs. For example:



Use cases and channels. Most IPS still offer a narrow set of use cases or do not yet provide access to the channels people prefer. Only by facilitating all retail use cases via an expansive set of channels and instruments, offered at minimal cost, can IPS support retail flows including personal remittances, government



A design focus on underserved groups, including women and small businesses. One way to address some

of the inclusion criteria is for IPS to become more intentional in their design. This includes putting more focus on design factors related to gender and small businesses, given persistent gaps for these groups. Moreover, the trajectory of a given domestic or regional IPS is context-specific; it depends as much on digital development in the market it serves as it does on social and cultural norms. IPS stakeholders with an inclusivity objective and mandate can therefore draw valuable insights from local data and user voices to design systems that best serve their populations.

market development. Whether an IPS leverages third-party or multilateral integration varies by the size of the relevant economy and its state of development in digital financial services. Trade-offs include bilaterally integrating PSPs versus investing in a third-party switch to deliver all-to-all interoperability. A switch may not capture scale if dominant providers with significant market share process a large share of the market's payments as on-us transactions. Bilateral integration can become complex and risky as the market grows. Implementors have already built enough IPS switching capacity to accommodate the sharing of infrastructure between countries and regions, if desired.

Cross-participation would be particularly advantageous for the IPS in situations where a cross-border payment fintech has capitalized on a lack of regional IPS to establish a multi-nodal network in closed-loop solutions. These systems are in direct competition for scale with the existing and planned IPS but could instead be brought in with a complementary value proposition. To incentivize usage, IPS need to create that compelling value proposition—not only for the growing fintech sector but also for the benefit of end-users. Fintechs who operate as technical service providers also provide value to open-loop IPS by providing such services as messaging integration layers.

AfricaNenda is committed to helping IPS stakeholders build DPI to serve all Africans. The annual State of Inclusive Instant Payment Systems report is one of our efforts to spread knowledge on inclusivity in finance through digital public infrastructure. We strongly believe in the promise of interoperability to drive inclusivity in digital payment systems and, together with the World Bank and the United Nations ECA, we are excited to help advance IIPS on the African continent.

- **Involving end-users** from the design of the architecture of the IPS is amenable to an interoperability and increase the value proposition for PSPs.
- **Moving beyond the USSD channel over time**, given its limitations, can be achieved through responsible, inclusive innovation. Unintended consequences need to be managed with care, such as exclusion if USSD access is not offered in the medium term or loss of trust if offered without safeguards.
- **Developing pricing models that can compete with cash and closed-loop solutions** in order to incentivize adoption by PSPs and end-users, including small businesses.
- **Rolling-out a portfolio of scale- and value-driving use cases**, especially person-to-business (P2P), business-to-business (B2B), and government-to-person (G2P) in order to increase network touchpoints and perpetuate a cycle of digital funds.
- **Promoting effective agent channels and recourse mechanisms** through engagement with participating PSPs can help to build awareness, capability, and trust, especially for women.
- **Delivering value-added services and shared infrastructure** can improve IPS efficiency and consumer access. The use of proxy ID, digital ID, electronic know your customer (eKYC), centralized fraud, and centralized cybersecurity facilities holds particular promise.

Regulators need to consider how to regulate and supervise to support the advancement of inclusive IPS:

- **Innovation and coordination** between regulators can build inclusive national payment ecosystems. Priorities include risk-based and harmonized licensing of PSPs to drive innovation and reduce reliance on cash; network upgrades to increase digital transaction quality and trust; sustained roll-out of agent networks for access and education, particularly in rural areas; and roll-out of smartphones to increase access channels, innovation on the USSD spectrum to increase convenience and improve security.
- **Principles-based regulatory frameworks** for consumer protection and data privacy, and a move towards risk-based supervision, are required to effectively mitigate risks and protect end-users.
- **Harmonization of regulation for cross-border payments** within regions is needed to enable funds to freely move, with specific emphasis on mutual recognition of identity, licensing requirements, and regional data sharing.



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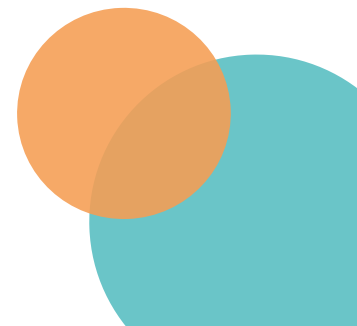
Information was provided on the following systems: eKash (Rwanda), National Financial Switch (Zambia), Kenya mobile money and PesaLink, Madagascar mobile money, GhIPSS Instant Pay and Ghana mobile money, NatSwitch (Malawi), Sociedade Interbancaria De Mocambique (Mozambique), and Real-Time Clearing (South Africa).

For this report, we conducted interviews with key stakeholders, including payment system experts, regulators, IPS providers, IPS operators, and PSPs between January and April 2023. These interviews provided valuable insights on trends, barriers, opportunities, and other pertinent information required for the analysis. [Annex B](#) provides the list of stakeholders interviewed.

b. Cross-border harmonization deep dive:

Comprehensive reviews of available resources, including literature from development partners, government publications, and publicly available sources from regional bodies, informed the cross-border retail policy and regulatory harmonization insights.

Interviews with key stakeholders, including cross-border PSPs, regulatory bodies, development organizations, and regional bodies, contributed to our understanding of barriers, work-to-date, and emerging lessons.



Senegal. It covered both low-income adults and micro, small, and medium enterprises (MSMEs).

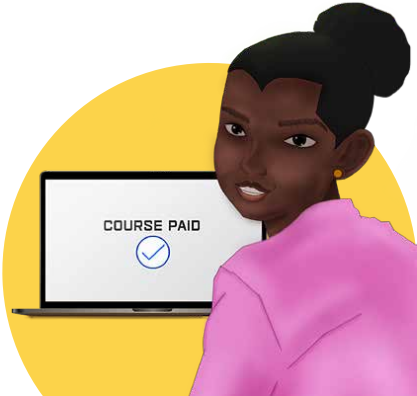
The qualitative research included 75 respondents for individual interviews, and 25 mystery shoppers. The quantitative surveys included 653 respondents across the study countries. All figures are cumulative. The sample is not nationally representative, but rather focuses on the experience of emerging consumers to identify constraints and drivers of access, adoption, and usage of digital payments in Africa and related implications to the design of IIPS. [Annex E](#) provides further insights into the methodology.

The case studies were conducted to align with the typologies adopted in the landscaping method, with emphasis on governance models and system development.

Interviews with case study participants provided a detailed understanding of the system history, data flows, operations, constraints, and expansion plans. All case studies are available in [Annex H](#).

d. Case studies:

The report showcases four IPS case studies, three domestic and one regional, to provide a closer examination of different origin stories, trajectories, and design components. These case studies include eKash (Rwanda), National Financial Switch (Zambia), Natswitch (Malawi), and GIMACPAY (CEMAC region).



GSMA	› Nadia Jeffrie
Independent consultant	› Andrea Bises › Arthur Cousins › John Kiff
MFS Africa	› Funmi Dele-Giwa › Patrick Gutmann
Mukuru	› Catherine Denoon-Stevens
Natswitch Ltd	› Gertrude Kadumbo
RSwitch Ltd	› Afazad Kalisa › Rachel Uwamahoro
Southern Africa Development Community (SADC) Bankers	› Maxine Hlaba
Trade Economics	› Paul Baker
United Nations Capital Development Fund (UNCDF)	› Eliamringi Mandari › Albert Mkenda › Bisamaza Mukankunga
West African Monetary Institute (WAMI)	› Adama Diakite › Clifton Garpeh › Isaac Osei Mensah › Dauda Mohammed › Souleymane Tall › Augustine Ujunwa › Abdulrasheed Zubair
Zambia Electronic Clearing House Limited (ZECHL)	› Morgan Chishala › Kabwita Kabwita › Francis Lwanga

GhIPSS Instant Pay (GIP)	Ghana	2015	Bank	52	76,483,008	4,921,361,258	Bank of Ghana	Bank of Ghana	GhIPSS	GhIPSS	Bank of Ghana
GIMACPAY	CEMAC	2020	Cross-domain	91	10,046,359	657,020,970	BEAC and commercial banks	BEAC	GIMAC	GIMAC	BEAC
InstaPay	Egypt	2022	Bank	27	16,000,000	2,588,728,675	Egyptian Banks Company	Central Bank of Egypt	Egyptian Banks Company	Egyptian Banks Company	Central Bank of Egypt
Kenya mobile money	Kenya	2018	Mobile money	3	9,932,247,314	96,551,482,669	None (bilateral agreements)	Central Bank of Kenya and Communications Authority Kenya	None (bilateral agreements between EMIs)	None (bilateral agreements between EMIs)	Central Bank of Kenya
Madagascar mobile money	Madagascar	2016	Mobile money	4	704,081,198	10,224,077,794	None (bilateral agreements between EMIs)	Central Bank of Madagascar	None (bilateral agreements between EMIs)	None (bilateral agreements between EMIs)	Central Bank of Madagascar
MarocPay	Morocco	2018	Cross-domain	23	-	-	Bank Al-Maghrib	Bank Al-Maghrib	The Moroccan Mobile Payment Group (GP2M)	HPS Switch	Bank Al-Maghrib
Mauritius Central Automated Switch (MauCAS)	Mauritius	2019	Cross-domain	14	-	108,336,838	Bank of Mauritius	Bank of Mauritius	Bank of Mauritius	Bank of Mauritius	Bank of Mauritius
NamPay	Namibia	2021	Bank	9	25,362,000	19,446,749	Payments Association of Namibia	Bank of Namibia	Payments Association of Namibia	Payments Association of Namibia	Central Bank of Namibia
National Financial Switch (NFS)	Zambia	2019	Cross-domain	30	61,250,000	2,088,883,903	ZECHL	Bank of Zambia	ZECHL Board of Directors	ZECHL	Bank of Zambia
Natswitch	Malawi	2015	Cross-domain	11	11,458,762	311,826,606	Bankers Association of Malawi	Reserve Bank of Malawi	Bankers Association of Malawi	Banking Payments Context	Reserve Bank of Malawi
NIBSS Instant Payment (NIP)	Nigeria	2011	Cross-domain	450	5,140,093,000	746,565,392,271	NIBSS	Central Bank of Nigeria	NIBSS Board of Directors	NIBSS	Central Bank of Nigeria
Nigeria mobile money	Nigeria	2013	Mobile money	21	714,597,976	42,014,940,067	NIBSS	Central Bank of Nigeria and Nigerian Communications Commission	NIBSS	NIBSS	Central Bank of Nigeria
Pan African Payment & Settlement System	Continent-wide	2022	Bank	36	-	-	PAPSS Governing Council	PAPSS Governing Council	PAPSS Governing Council	PAPSS Management Board	African Export-Import Bank
PayShap	South Africa	2023	Bank	4	-	-	BankservAfrica and commercial banks	South African Reserve Bank	BankservAfrica and commercial banks	BankservAfrica	South African Reserve Bank
PesaLink	Kenya	2017	Bank	35	3,850,359	3,879,324,621	Kenya Bankers Association	Central Bank of Kenya	IPSL	IPSL	Central Bank of Kenya
Real Time Clearing (RTC)	South Africa	2006	Bank	33	215,832,611	78,849,497,513	BankservAfrica	South African Reserve Bank	Payment Association of South Africa	BankservAfrica	South African Reserve Bank

Uganda mobile money	Uganda	2017	Mobile money	7	5,230,548,350	43,507,845,576	None (bilateral agreements)	Bank of Uganda	None (bilateral agreements between EMTs)	Pegasus	Central Bank of Uganda
Zimswitch Instant Payment Interchange Technology (ZIPIT)	Zimbabwe	2011	Cross-domain	27	25,938,579	828,510,618	Zimswitch	Reserve Bank of Zimbabwe	Zimswitch	Zimswitch	Reserve Bank of Zimbabwe
Virement Instantané	Morocco	2023	Bank	19	-	-	Bank Al-Maghrib	Bank Al-Maghrib	Not available	GSIMT	Bank Al-Maghrib
EthSwitch	Ethiopia	2022	Cross-domain	Not available	2,063,534	371,052,431	Not available	Not available	Not available	Not available	Not available

Madagascar mobile money	Basic	✓	✓	✓		
MauCAS (Mauritius)	Basic	✓	✓		✓	
MarocPay (Morocco)	Basic	✓	✓	✓		
SIMO (Mozambique)	Basic	✓	✓	✓		✓
NamPay (Namibia)	Basic	✓	✓		✓	
eNaira (Nigeria)	Basic	✓	✓			
NIP (Nigeria)	Basic	✓	✓	✓		✓
RTC (South Africa)	Basic	✓	✓		✓	
Taifa Moja (Tanzania)	Basic	✓	✓		✓	✓
TIPS (Tanzania)	Basic	✓	✓	✓		✓
Uganda mobile money	Basic	✓	✓		✓	
ZIPIT (Zimbabwe)	Basic	✓	✓	✓		✓
SYRAD (Djibouti)	Not ranked	✓		✓		✓
Kenya mobile money	Not ranked	✓			✓	
PesaLink (Kenya)	Not ranked				✓	
Virement Instantané (Morocco)	Not ranked	✓				✓
Nigeria mobile money	Not ranked		✓		✓	✓
eKash (Rwanda)	Not ranked	✓		✓	✓	✓
EthSwitch	Not ranked	✓		✓		✓
PayShap (South Africa)	Not ranked	✓			✓	
Somalia National Payment System	Not ranked		✓			✓
Tunisia mobile money	Not ranked		✓			✓
TCIB (SADC)	Not ranked	✓		✓	✓	✓
PAPSS (Continent-wide)	Not ranked		✓			✓

*The two systems in Ghana are considered as one aggregated cross-domain system

BOX D.1 | Examples of consumer protection guidelines for IPS participants

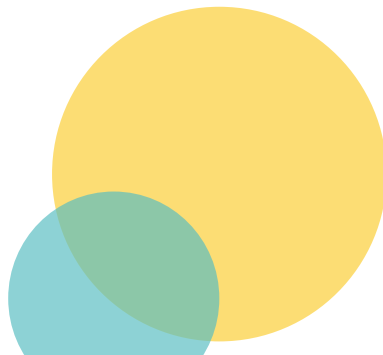
Country	IPS operator	Examples of consumer protection guidelines for IPS participants
 Ghana	The Ghana Interbank Payment and Settlement Systems Limited (GhIPSS)	According to the scheme rules, participants must adhere to the recourse requirements defined in the "Consumer Recourse Mechanism Guidelines for Financial Service Providers Framework" defined by the Bank of Ghana.
 Malawi	Natswitch Ltd	Natswitch Ltd has included provisions for dispute management within their operating rules and they have also developed detailed dispute resolution rules in line with Malawi's Consumer Protection Act (Stakeholder interviews 2023).
 Nigeria	NIBSS	Providers must, according to the scheme rules, follow CBN's website recourse mechanisms. They must have a helpdesk to handle consumer complaints. End-users report complaints to providers who then have two weeks to resolve the issue. If unresolved, complaints can be escalated to CBN's consumer protection office for recourse.

Competition and Consumer Protection Commission of Zambia, and the Bank of Zambia (Proto 2023). Through these platforms, end-users can engage with their banks, initiate the dispute management process, and if necessary, proceed with a charge-back procedure when the dispute is determined to be valid. Given that providers are mostly the first point of call for complaints from fast payments users, increasing automation is critical for ensuring transparent and speedy resolution. This also highlights the importance of a centralized complaint tracking solution that consumers can leverage to track the status of their complaints across institutions. In Nigeria, for instance, consumers can track complaints on the official Central Bank of Nigeria Consumer Complaint Management System using the reference/tracking number that was sent to them by the financial service provider where they handed in their complaint (CCMS 2020).

authorities responsible for supervising all aspects of financial service providers within specific sectors, like banking or insurance (Stakeholder interviews 2023).

- The twin peak model, found in South Africa, assigns one authority with a financial stability mandate (in this case, the Prudential Authority established within the South African Reserve Bank) and another authority (for South Africa, the Financial Sector Conduct Authority) with a focus on financial consumer protection and market conduct.

No model yet exists for a comprehensive consumer protection authority covering financial and non-financial activities (AFI 2022c).



regulator. Depending on the applicable laws, regulators may also lack the power to make legally binding rulings or award compensation. This may, ultimately, leave consumers without effective recourse. Hence, ombud schemes enhance accessibility and affordability of dispute resolution for consumers. An ombudsperson can be established as a voluntary body set up by industry associations, as a statutory body established by law, or as a hybrid where a voluntary body is entrenched in law, with oversight by the regulatory authority.

Financial ombudsmen are on the rise in Africa, but there are still gaps in reach and effectiveness. Only three countries in Africa have ombudsmen that are affiliated

establishing a consumer ombudsman as a party recourse mechanism requires careful consideration of the governance structure, funding model, and operational structure; the passing of legislation to give effect to the chosen structure (often a multi-year endeavor); and a plan for ensuring that it is accessible. Therefore, the journey to establishing broadly accessible and effective third-party recourse in Africa is ongoing.

The establishment of ombudsmen or similar structures for IPS is crucial to creating additional trust among end-users in instant payments. Ideally, IPS would provide additional recourse mechanisms in the absence of a financial sector ombudsman.

BOX D.2 | Example of the Ombudsman for Banking Services in South Africa (OBSSA)



In South Africa, banks that are members of the Banking Association of South Africa are signatories to the OBSSA, which provides a detailed process geared to effective and speedy resolution. The process deals with all payment types including both interbank FPS offerings. All member banks are required to maintain specialized dispute resolution departments through which clients are encouraged to lodge any complaints; in response, clients receive a written response and a complaint tracking number. The bank has a maximum of 20 days to respond and, failing any response or a rejected response within the 20 days, an OBSSA complaint may be lodged and tracked on the OBSSA website or

through the call center. Once a formal OBSSA complaint is lodged, the bank becomes responsible for the OBSSA costs of adjudication. Particularly with lower value retail FPS matters, there is a powerful cost incentive for banks to settle complaints promptly and before they reach the OBSSA. Banks typically choose their battles focusing on matters of systemic concern or based upon organizational norms and values (OBSSA 2019).



in which the perpetrator masquerades as a legitimate business or reputable person (NIST 2023).



Vishing: A type of cyberattack that uses voice and telephony technologies to trick targeted individuals into revealing sensitive data to unauthorized entities (TechTarget 2023).



Smishing: A social engineering attack that uses fake mobile text messages to trick people into downloading malware, sharing sensitive information, or sending money to cybercriminals (IBM 2023).



Spoofing: Faking the sending address of a transmission to gain illegal entry into a secure system (NIST 2023); or other fraudulent activities aimed at obtaining personal account details or transferring money (Stakeholder interviews 2023). Perpetrators exploit victims' fears, insecurities, and vulnerabilities, utilizing various online tactics. While the nature of these fraudulent activities has remained consistent over time, they are frequently repackaged to deceive users repeatedly. Mobile money and mobile banking platforms, particularly in Eastern Africa, are increasingly targeted by fraudsters, with 56% of surveyed users in Kenya reporting phishing attempts through phone calls or SMS in 2021 (Innovations for Poverty Action 2021).

Mobile money agents fall victim to fraud as well. Fraud types perpetrated on agents include the agent losing float in their account due to social engineering attacks or compromised PINs (CGAP 2017).

Mobile channels in Africa are particularly susceptible to subscriber identity module (SIM) swapping and hijacking. Mobile channels often rely on SMS-based two-factor authentication, which exposes them to SIM swapping and hijacking attacks. For instance, scammers can deceive customer service agents by falsely reporting a device as lost and requesting the activation of a new SIM card using the victim's phone number. Once the criminal has activated their device with the victim's phone number, they can bypass two-factor authentication through SMS or voice calls directed to that phone. In South Africa, the South African Banking Risk Information Centre reported a 63% surge in incidents between 2020-2021, from 2,686 cases to 4,386 cases (South African Banking Risk Information Centre 2021). In 2019, the Nigeria Security and Civil Defence Corps took action against SIM swap fraud in Nigeria by arresting and prosecuting 113 internet fraudsters (Nigerian Tribune 2019).

Signalling System 7 (SS7) vulnerabilities remain and extend to 4G wireless broadband systems. The lack of robust security layers makes SS7 susceptible to hacking and collaboration with mobile network operators.⁸⁹ 4G systems seem to face similar risks despite more structured defenses. Attacks on SS7 and 4G compromise layered security in IPS and occur most frequently in Asia and Africa (GSMA 2018b). SS7 attacks occur when the log-in for a virtual workstation at an MNO is hacked externally or hired out by actors within the MNO. The fraudsters then have access to many unsecured MNOs' consumer proxy identities and connections across the world, allowing them to eavesdrop or reroute data to alternative SIMs in a number of jurisdictions. IPS are particularly vulnerable as it is exceedingly difficult to stop the flow of fraudulent proceeds across multiple accounts and ATMs. These kinds of attacks put consumers at risk of unauthorized account withdrawals. The financial service providers accuse them often of having compromised their password or personal identification number. SS7 attacks are not always distinguishable from other types of attacks and are included across other types of fraud (Stakeholder interviews 2023).

⁸⁹ SS7 is a telecommunications signalling architecture traditionally used for the set up and tear down of telephone calls. It has a robust protocol stack that uses out-of-band signalling to communicate between elements of the public switched telephone network. In recent years it has been superseded by the Diameter signalling protocol (Techopedia 2017).

opportunities, resulting in increased risk of fraud against IPS participants (Stakeholder interviews 2023). However, more sophisticated financial systems like those in Kenya, Nigeria, and South Africa attract a significant number of such attempts, including cyber-enabled heist attacks. These attacks involve threat actors compromising bank networks and gaining privileged access to interbank payment systems, allowing them to issue fraudulent transaction requests and acquire substantial amounts of money. An organized crime group targeting South Africa's Standard Bank in 2016 managed to withdraw over \$19 million from ATMs in Japan using forged cards (Singh 2016). Similarly, the National Bank of Kenya suffered a loss of at least KSH 29 million (\$261,000) due to an attack on their internal network (PC Tech Magazine 2018). West African financial institutions have also faced cyberattacks aimed at compromising internal networks and conducting fraudulent transactions (Threat Hunter Team Symantec 2019).

Furthermore, internal fraud attacks that are enabled by insiders within IPS participant institutions pose significant challenges and result in substantial losses. These attacks, often unreported, involve current or former employees exploiting their privileged access to steal funds. Malware and phishing techniques are commonly utilized by scammers to facilitate fraudulent activities, such as setting up malicious domains. There also have been attempts to compromise public interfaces of financial institutions, as evidenced by the visibility and monitoring maintained in IPS environments like Zambia (Stakeholder interviews 2023).



With IPS, the Central Bank of Kenya and Central Bank of Nigeria published guidelines on cybersecurity for payment service providers (Central Bank of Kenya 2018; Central Bank of Nigeria 2018). However, while most African countries have developed national cybersecurity strategies, few have guidelines specifically tailored to IPS participants (ITU 2023). Regulations can also aim to combat one particular type of fraud, such as by introducing stricter SIM card activation/swap regulations (further outlined in Box E.1). In the absence of coherent regulatory provisions and guidance, the scheme rules remain the safeguard, but they are often more aligned with commercial liability than consumer protection. At an organizational level, IPS scheme rules set standards for procedures, rules, and technical requirements governing payment execution. These multilateral rules may address security, processing, or technical aspects of instant payments.

The implementation of centralized fraud monitoring and detection systems in IPS is limited. While the majority of IPS operators adopt a decentralized approach to fraud monitoring, a few have implemented different centralized systems or approaches, as further described in Box E.2. Many IPS operators recognize the need for centralized systems in addition to the decentralized measures implemented by individual participants to better support monitoring and detecting fraud risks, enable regular monitoring and analysis of transactional data, and to facilitate prompt reporting of potential fraud incidents. Operators are engaging partners to better understand their needs and challenges in implementing centralized



Nigeria case study. In 2022, The Nigerian Communications Commission prescribed a standard procedure for SIM replacement to be used by communications service providers. The guidelines oversee SIM replacement under certain conditions, stipulating that SIM replacement requests can be rejected by the communication service provider on reasonable grounds so long as the commission is notified within 48 hours of the rejection. Data on rejected cases must be reported by the providers. Rejection decisions are determined by subjecting the requesting subscribers to a screening and verification process, which requires them to answer security questions pertaining to their SIM activity. SIM swaps that are conducted by a subject proxy require an affidavit and passport photo, along with a national identification number. SIM replacement for deceased subscribers requires a certified true copy of the death certificate. The SIM card being replaced will only be activated upon verifying the validity of the information provided by the subscriber, and until then, only limited service is allowed on the SIM card (Nigerian Communications Commission 2022).



Nigeria

NIBSS

The integration of the bank verification number's biometric attributes with transaction monitoring helps to prevent fraudulent proceeds from being forwarded to accomplice accounts by syndicate groups (Central Bank of Nigeria 2013).



South
Africa

BankservAfrica

BankservAfrica's Integrated Cash Management System team has implemented an automated transactions monitoring system. This system allows BankservAfrica to have a comprehensive view of financial transactions across different data sources and formats. The system enables the early detection of any irregularities or issues that may arise during the processing of high-value cash transactions (Stakeholder interviews 2023).



Zambia

ZECHL

Every transaction processed through the switch undergoes fraud vetting. Advisory notes are sent to participants regarding potentially fraudulent transactions. Although the switch performs fraud analysis for each transaction, it does not block the transactions based on the advisory notes. Instead, it conducts screening. Currently, there are no universally adopted industry standards for actions to be taken in response to flagged transactions. It is the responsibility of the participants to determine their risk standards and take appropriate action based on the advisory notes (Stakeholder interviews 2023).



Zimbabwe

Zimswitch
Technologies
Private Limited

At the switch level, trends are monitored, and any concerning patterns are flagged to the impacted institutions. In cases related to money laundering or terrorist financing, the flagged information is shared with the Financial Intelligence Unit. The authorization process belongs to issuers, who track their own indicators. As a designated non-financial institution, Zimswitch is responsible for submitting suspicious transaction reports, allowing for a system-wide view of fraudulent activities (Stakeholder interviews 2023).



education programs, training sessions, and social engagements are crucial for raising awareness and promoting fraud prevention. This collaboration is important given the prevalence of end-user-affecting fraud. While IPS participants are primarily responsible for leading these programs, collaboration can increase their effectiveness. The Bankers Association of Zambia runs an annual fraud campaign that includes activities and visits to remote areas to educate people about different fraud schemes. In Zimbabwe, engagements are conducted by the IPS operator to educate both providers and end-customers, including merchant meetings, awareness sessions at outlets, engagements with colleges and schools, and media broadcasts through radio, television, and billboards, all aimed at spreading awareness and strengthening partnerships to combat fraud.

IPS operators across Africa see a strong role for artificial intelligence (AI) and machine learning in fraud prevention, detection, and mitigation. Centralized AI systems can flag suspicious transactions in real time. Depending on

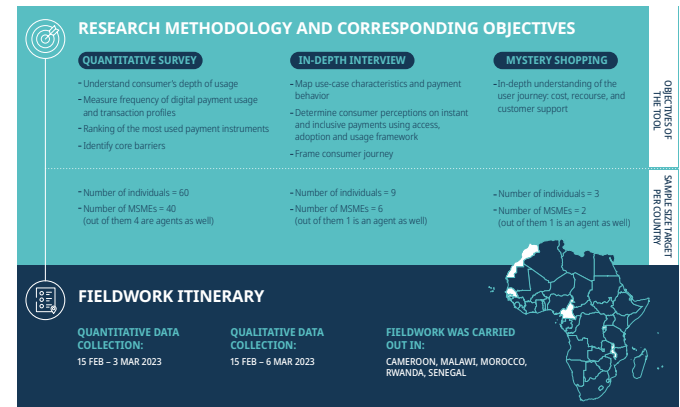
suspicious transaction detection to a behavioral-based approach (Stakeholder interviews 2023). Advancements in customer identification and authentication are also being explored by IPS participants across Africa, with voice recognition systems emerging as a notable consideration to detect distress and prevent fraud. By analyzing unique vocal patterns and characteristics, these systems aim to identify signs of distress during customer interactions, enabling timely intervention.

Some banks in the region have already implemented facial biometrics to identify and authorize customers. For instance, Access Bank in Nigeria has launched FacePay, a facial biometric payment system that enables customers to verify their identity and authorize retail transactions conveniently, without passwords or physical cards. Similarly, the Standard Bank of South Africa has partnered with iDentifi to integrate facial recognition into its mobile app, allowing customers to securely access their online banking accounts using facial biometrics. These advancements seek to provide fast, seamless, and secure payment and banking experiences and to prevent fraud.



qualitative research, as shown in Figure F.1.

FIGURE F.1 | Breakdown of quantitative and qualitative methods



community and/or on social grants.				
28%	28%	18%	26%	SAMPLE PROPORTION (quant. survey)
79% of the total sample for THE QUANTITATIVE SURVEY are DIGITAL PAYMENT USERS (individuals and businesses) and 90% of the total sample for the QUALITATIVE RESEARCH COMPONENTS are DIGITAL PAYMENT USERS (individuals and businesses).				
Within each of the four groups, an ADEQUATE COVERAGE OF WOMEN AND YOUTH was ensured.				
The study sample focuses on the "EMERGING MARKET" that is expected to use digital payments and thus only sampled those in URBAN AND PERI-URBAN settings. The focus was on low-income earners and MSMEs and the sample is therefore NOT NATIONALLY REPRESENTATIVE. Any inferences made on a country-by-country basis are with respect to the sampled respondents.				
* Country specific monthly turnover cut-off has been applied				

* Country specific monthly turnover cut-off has been applied.



Detailed sample breakdown. The breakdown of the quantitative component and exact sampling of each method for the qualitative component across the five markets are provided in Table F.1. In total, the sample included 653 respondents across the five markets. The collection of the quantitative data took place between February 15, 2023 and March 3, 2023. For the qualitative component, the sample included 100 respondents across the IDIs and mystery shopping methods. The collection of the qualitative data took place within these five countries between February 15, 2023 and March 6, 2023.

	Small business	57	4	1
	Total	156	15	5
	Percentage of sample that are digital payment users	73%		
Morocco	No/infrequent income earners	41	4	1
	Low frequent income earner	36	4	2
	Micro business	19	3	1
	Small business	28	4	1
	Total	124	15	5
	Percentage of sample that are digital payment users	77%		
Rwanda	No/infrequent income earners	29	4	1
	Low frequent income earner	46	4	2
	Micro business	33	3	1
	Small business	17	4	1
	Total	125	15	5
	Percentage of sample that are digital payment users	83%		
Senegal	No/infrequent income earners	39	4	1
	Low frequent income earner	28	4	2
	Micro business	29	3	1
	Small business	44	4	1
	Total	140	15	5
	Percentage of sample that are digital payment users	81%		

Data security and cyber security	MENA Financial Crime Compliance Group (MENA FCCG)	ECOWAS Commission	CEMAC Commission	SADC Secretariat	IGAD Secretariat	EAC Secretariat	COMESA Secretariat	ECCAS Commission	Information unavailable	
Licensing and authorization regime	UMA Secretariat	WAMI	BEAC	SADC Payment System Oversight Committee (PSOC)					WAMI	BCEAO
Consumer protection	MENA Financial Crime Compliance Group (MENA FCCG)	ECOWAS Regional Competition Authority (ERCA)	Commission Bancaire de l'Afrique Centrale (COBAC)	SADC Secretariat		EAC Competition Authority	COMESA Competition Commission		ERCA	WAEMU Commission
Foreign exchange controls	UMA Secretariat	West African Monetary Agency (WAMA)	BEAC	PSOC, Committee of Central Bank Governors (CCBG)		East Africa Business Council (EABC)	COMESA Secretariat		West African Monetary Agency (WAMA)	BCEAO
Prudential supervision			COBAC						Governor of Central Banks of ECOWAS	
Payment system standards	Union of Maghreb Banks	Governor of Central Banks of ECOWAS	BEAC							Governor of Central Banks of ECOWAS

	and non-deposit-taking FIs	✓	✓	✓						
Financial integrity (AML/CFT/CPF)	Inconsistency in agent guidelines	✓	✓	✓	✓	✓	✓			
	Differing CDD/KYC documentation requirements and transaction limits	✓	✓	✓	✓	✓	✓			
	Divergent implementation and interpretation of FATF standards (for example, risk-based approach)		✓	✓	✓	✓	✓			✓
	Reliance on extra territorial correspondents in hard currency jurisdictions and not within the continent; inappropriate corresponding banking compliance standards				✓	✓	✓			✓
Consumer protection	Unclear mechanisms for external complaints handling and dispute resolution		✓	✓			✓			
Foreign exchange controls	Reliance on hard currency settlements introduces foreign and inappropriate compliance issues				✓					
	Variations in compliance and reporting		✓	✓						
Prudential supervision	Differences in accounting standards, regulatory reporting requirements, and minimum capital adequacy requirements		✓					✓		✓
	Incompatibility of tax regimes			✓		✓				
Data security and cybersecurity	Inconsistent data security frameworks and risk management procedures		✓				✓			
	Different approaches to flows and local data storage requirements		✓				✓			
Payment system standards	Lack of uniformity in data and messaging standards implementation		✓	✓			✓			✓

Source: Finmark Trust (2014); UNCDF (2022a); WAMI (2022); IMF (2017); COMESA Business Council (2021); UNCDF (2022b); CCAF (2021a); CCAF (CCAF, 2021b); Finmark Trust (2021a); SADC Secretariat (2021); Stakeholder interviews (2023)

✓ Regional divergence identified

CEMAC	UNCDF and BEAC		Regulatory environment for digital financial services	2023
SADC	SADC Payment System Subcommittee, Committee of Central Bank Governors (CCBG)		Regional payments a focus in updated financial inclusion policy	2021
	ESAAMLG		FATF recommendation implementations	1999
IGAD	UNCDF and IGAD Secretariat		Harmonization of remittance policies	2022
EAC	EAC Secretariat		Harmonization of payment systems	2022
	World Bank and EAC Secretariat		Financial sector development and regionalization project	2021
COMESA	COMESA Business Council (CBC)		Digital payments policy for micro, small and medium enterprises	2021
ECCAS	UNCDF and the ECCAS Commission		Harmonization of remittance policies	2022
	UNCTAD and ECCAS Secretariat		Regional integration through the World Trade Organization (WTO) facilitation agreement	2020
WAMZ	WAMI		Harmonization of regulatory and prudential framework	2022
	West African Institute for Financial and Economic Management (WAIFEM)		Use of international payment standards and licensing regimes	2021
WAEMU	BCEAO		Alignment of payment regulatory frameworks	2022
	UNCDF and BCEAO		Remittance payments technical assistance	2023





RWANDA: eKASH

Origin story



Challenge

The payment ecosystem in Rwanda was previously characterized by low levels of interoperability for retail payment transactions. Mobile money operators (MMOs) and banks had developed bilateral relationships, enabling some limited interoperability, but retail transfers between different banks were not possible (National Bank of Rwanda 2018). The bilateral arrangements were not up to international payment system standards and ill-equipped to handle increasing digital traffic. As a result, the industry faced increased risks of failure and inefficiencies (National Bank of Rwanda 2020).



Adding value

Given that half of the adult population did not use digital payments, the National Bank of Rwanda (BNR) wanted to catalyze ongoing maturation toward digital payments and away from cash. It created the National Payment System Framework and Strategy 2018 to 2024—referred to as the ‘payment strategy’—with the goal of achieving a cashless economy by promoting digital payments.

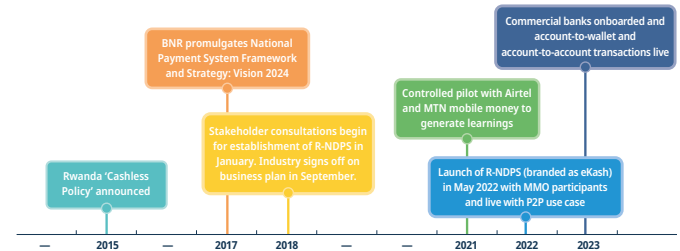
The BNR and the payments industry are intent on promoting all-to-all interoperability of retail payments to



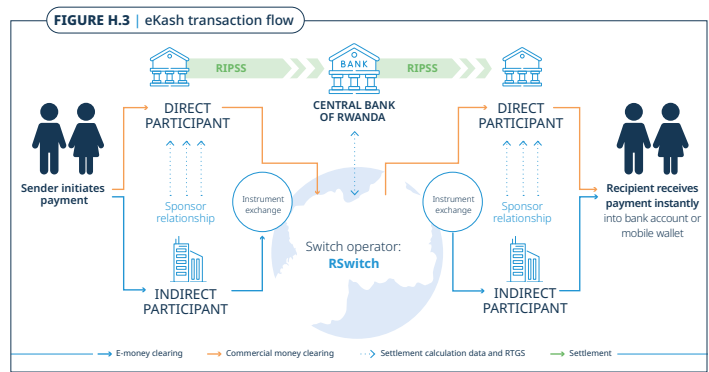
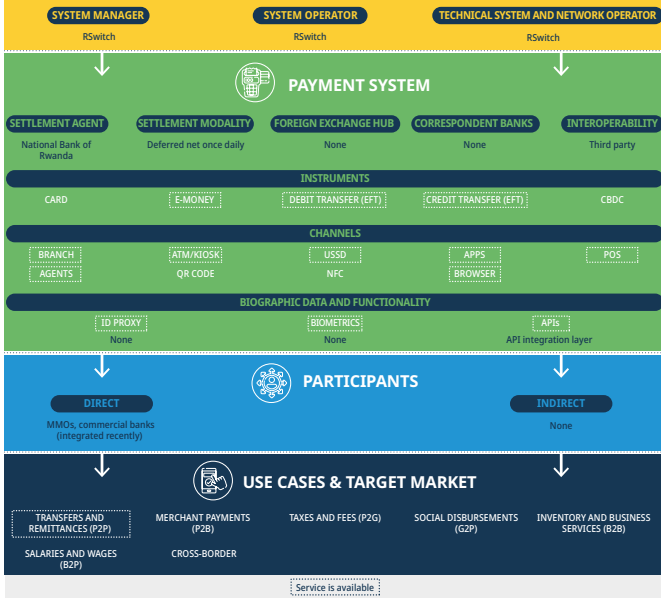
Creating a national payment policy was essential to the creation of an interoperable payment landscape. The BNR and Rwandan government were motivated to establish a payment strategy that could serve an efficient cashless economy centered around robust, modern, and inclusive payment systems. A pivotal aspect of BNR’s strategy was to achieve interoperability among payment service providers (PSPs) as opposed to the existing bilateral set-ups. The policy also aimed to ensure a financially included population through enhanced monitoring of access to and use of digital funds.

BNR’s initial strategy was to establish a controlled pilot with two mobile money operators—Airtel Mobile Commerce and Mobile Money Rwanda Limited. The pilot was used to test the security and reliability of the system’s infrastructure, which were enhanced before eKash’s launch. eKash went live in May 2022 with mobile money participants and for P2P transactions. In 2023, commercial banks completed system integration tests and were onboarded to the system, making it an all-to-all, cross-domain IPS. The system will explore integrations with regional switches, such as COMESA’s payment system in development, once eKash reaches its intended goals of serving the Rwanda public efficiently and inclusively.

FIGURE H.1 | eKash timeline



Source: National Bank of Rwanda 2023





Governance

eKash has embraced a democratic and inclusive governance structure to give voice to the broader payment industry.

eKash is a participant-owned system that follows a private association governance model. All financial institutions (banks, MFIs, fintechs) are association members. RSwitch is the system operator and supports the participant-owned system in a governance role. RSwitch is privately-owned according to a shareholding structure and is the system manager. Investments in RSwitch in the wake of COVID-19 have resulted in changes to its board of directors, which now includes members of civil society, of the banking association, MMOs, and fintechs (New Times Rwanda June 2023). The board of directors form the governance body and there is limited distinction between the system and operator given the representative board. The BNR has non-voting observer status at the board. All other board members have equal voting rights and collectively elect new members. Actors who have an ability to vote include the commercial bank and MFI representatives, fintech representatives, and a government representative. Participant representation on the board ensures that the industry has a voice in governance.

Though eKash is privately owned, the Rwandan government is substantially involved in its governance structure through agencies such as the Ministry of

Finance and the Ministry of ICT, and through oversight activities by the BNR. Working groups manage operations, legal and compliance, technology, and products and services. Group members are participants of the system. Members are nominated by their respective companies. Participants receive the meeting minutes so they can contribute comments and a validation meeting is called to confirm decisions with the broader membership. Any decision made by a working group is preferably by consensus. Only in limited cases when consensus is not possible, and the matter is pressing, does the committee defer to a majority decision. For instance, the participants were involved in establishing the pricing principles and pricing metrics within eKash's scheme rules.



Functionality

The system supports multiple channels and both commercial bank and e-money instruments (RSwitch n.d.). eKash is channel-agnostic, in that the system supports any channel endorsed by a participant PSP. The participant PSPs include eKash as a consumer-facing option within their menu of payments. Payment instruments supported by the system include card, debit EFT, credit EFT, and e-money. Payment channels include ATM, POS, USSD, and mobile app (National Bank of Rwanda 2020). Discussions regarding the role of CBDC in the system are underway, but are not finalized.



Technical standards and use cases

RSwitch has endorsed a staged approach to rolling out use cases. eKash currently only supports P2P payments, although

it is possible that some transactions are P2B payments to MSMEs through the personal accounts of the business owners. RSwitch plans to implement the remaining use cases, in the following order: merchant payments (P2B), bulk disbursements (B2P), social transfers (G2P), and business inventory payments (B2B; National Bank of Rwanda 2020).

The system uses ISO 20022 as its native message format for integration, which RSwitch expects will better prepare the system for future integration with regional and international payment systems. RSwitch also offers an API integration to allow banks and PSPs with systems running on other messaging standards to integrate. The layer enables non-traditional players, such as fintechs and aggregators, to participate securely in the system and bring new services to consumers (National Bank of Rwanda 2020). While P2B payments are incorporated through existing channels, the industry will re-convene to decide whether to introduce a scheme-level QR code.



Business model

AFR funded the initial costs, particularly around the creation of the RNDPS blueprint and scheme rules. RSwitch

and the industry absorbed the remaining start-up costs. The association opted to recoup these initial costs to avoid putting an undue burden on end-users. As such, the eKash scheme is a for-profit system that aims to operate in a way that enables cost-recovery by incentivizing efficiency and innovation. The system charges participants switching fees, which amount to 0.3% of the transaction value up to a maximum of \$0.38 (RWF 420). Provisions exist for a nominal return to its investors. Participants also pay a one-time integration fee of \$893 (RWF 1,000,000). The system was designed to be affordable for all participants, which should ensure affordability for end-users.



Scheme rules

eKash is a cross-domain system and therefore incorporates the same rules for e-money as for commercial money switching. The scheme rules stipulate prudential obligations, security, liability demarcation, and settlement modalities (i.e., direct or indirect settlement processes). The scheme rules state that every organization licensed by the National Bank of Rwanda is eligible to become a participant and can be directly connected to the switch; only members of the clearing house are directly settled through RPPS. Organizations which do not participate in the clearing house must have a settlement agent. The dispute resolution module can reconcile the status of a transaction between



Since its launch, specific payment system laws have been introduced to create an enabling environment, which creates the framework within which eKash functions. The laws and regulations define the structure and functioning of the payment system, payment service providers, operators, instruments, consumer protection, and overall governance. BNR's mandate to supervise and regulate payment and banking systems derives from *Law No 061/2021 of 14/10/2021 Governing the Payment system*, and the *Regulation N°06/2010 of 27/12/2010 of the National Bank of Rwanda relating to the Oversight of Payment Systems and the Activities of Payment Service Providers* (National Bank of Rwanda 2023, 2010a and 2010b). These laws cover regulation and supervision of payments-related activities, including the licensing of supervised pilots for innovation. Other pertinent laws and regulations include the *Law on AML/CFT-2020*, establishing CDD procedures, and the *Financial Service Consumer Protection Act* (2021).

efficient and effective regulatory framework, including a reliable and efficient payment system. The Rwanda National Payment System Strategy 2018–2024 (the “payment systems strategy”) guides the Government of Rwanda and payment system stakeholders toward developing a cashless economy and ensuring financial services reach all.

On the side of inclusivity, however, eKash is the only IPS in Africa to publish its scheme rules. This serves as an exemplar for other systems to enhance transparency, allowing participants to assess the system provisions.

Rwanda's National Payment System Strategy also centers on a transition toward a cashless society. The policy's principles have permeated throughout the payments industry, resulting in prospective participants who are invested in supporting a modern,

play a key role in developing the blueprint, and drove an inclusive consultation process between industry and the BNR. AFR and the BNR were responsible for drafting the Rwanda interoperability blueprint that set the foundation for the implementation of eKash. During the drafting of the blueprint, industry was repeatedly invited to share feedback and to approve draft language. AFR's role in developing the blueprint illustrates how independent third parties can be effective in design consultations of an IPS.

A layered governance model with industry inputs from different angles guarantees the voice of participants. Further, eKash has a transparent governance structure. This includes executive management, working groups and forums for industry input. These various layers of governance provide outlets and representation for both the regulator and the industry.



ZAMBIA: NATIONAL FINANCIAL SWITCH

Origin story

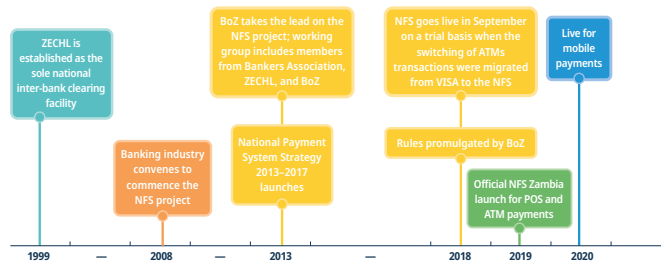


Challenge

The payment ecosystem in Zambia was characterized by limited interoperability, with both bank and non-bank payment service providers (PSPs) relying on bilateral agreements, resulting in a fragmented ecosystem. End-users were unable to consistently transfer payments digitally between PSPs of different types, leading to underutilization of digital retail payment methods by

end-users and low velocity of funds flowing through the national payment systems. In addition, all domestic card transactions switched outside of Zambia at international rates. Cash-based payment dominated and financial inclusion levels remained low (ZECHL n.d.). In 2009, before the commencement of the National Switch Project, 36% of Zambian adults had access to a financial account (World Bank 2014).

FIGURE H.4 | Zambia NFS timeline



Sources: Bank of Zambia 2020; Bank of Zambia 2019b; Bank of Zambia 2014; Bank of Zambia 2012

development. The banking industry attempted to develop its own interbank interoperability initiative in 2008 to create scale. However, the high costs involved with such a project and resistance towards the participation of non-bank PSPs led the BoZ to take the project over in 2013. The BoZ partnered with the Zambia Electronic Clearing House (ZECHL), a jointly owned entity by the banking industry and the BoZ, to spearhead the NFS project with the goal

ZECHL plans to introduce agent banking and a QR code standard and channel by 2024. ZECHL also has plans to launch a centralized eKYC platform within NFS's services using existing proxy ID functionality (i.e., mobile phone numbers and Bank Identification Number (BIN)) along with limited biographic data. Integration with SADC's TCIB has begun and integration with COMESA's planned IPS is in the pipeline.

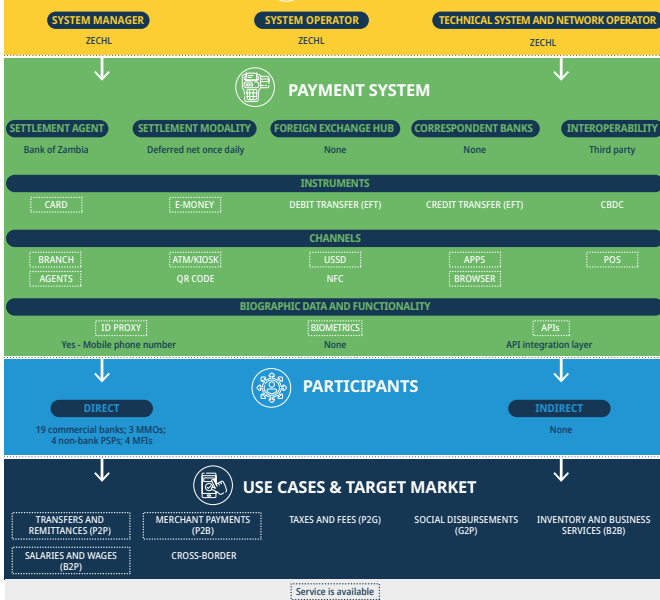
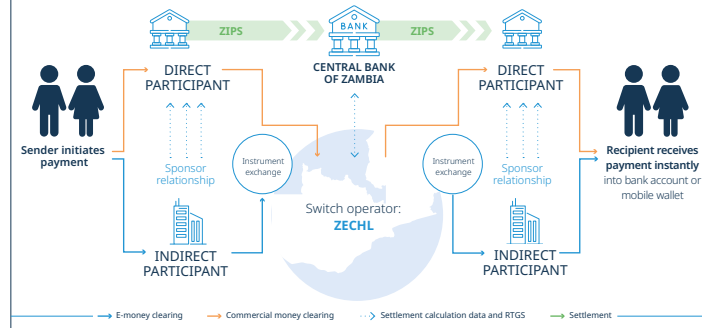


FIGURE H.6 | Zambia NFS e-money transaction flow





Governance structure

The NFS is operated by ZECHL, a non-profit entity jointly owned by the BoZ and the banking industry. ZECHL was established in 1999 as the country's first and only national inter-banking clearing facility. The organization is overseen by a board of directors comprised of member banks, which collectively own a 50% share. BoZ chairs the NFS and owns the remaining 50% share. Shareholding proportions between member banks are not equal and larger, more established, banks hold more shares (ZECHL n.d.). The board is chaired by the BoZ, however, and each shareholding board member has an equal voice.

Revisions or amendments to the ZECHL scheme rules go through a review process that involves the ZECHL management and ZECHL board. If ZECHL stakeholders agree, the amendments go to the Bankers Association of Zambia (BAZ) for vetting. Thereafter, BoZ reviews and agrees to the changes; they must be signed by each of the participants (ZECHL General Rules 2022). It is unclear if each participant has effective veto power over the rules should they refuse to sign or whether the purpose of the signature is to acknowledge changes, not secure agreement. However, input from non-banks comes through their representation in the Payments Association of Zambia, of which the BoZ is also a member. Through this channel, non-banks are able to voice concerns and the Governor of the BoZ can ascertain whether those views have traction in the market. ZECHL consults

stakeholders in introducing or changing fees incurred using the NFS. The BoZ conducts oversight of the NFS and new developments in the payment landscape through a dedicated payments systems department (Stakeholder interviews 2023).



Functionality

The system supports e-money and card payment instruments. The system is card-based and does not include EFT, which is available on a different clearing system (EFT system). ATM functionality was rolled out first, followed by POS and interoperable e-money (Bank of Zambia 2019). For e-wallet transactions, users can use their mobile numbers as an ID proxy. Participant PSPs map proxy IDs to accounts. ZECHL plans to incorporate national ID numbers as a secondary form of proxy ID. The NFS is channel-agnostic and participant PSPs determine the channels they offer to end-users.



Technical standards and use cases

The NFS currently supports several use cases, including P2P and P2B. The technical functionality also supports social disbursements (G2P), however, no social disbursement programs are routed through ZECHL yet, as of June 2023 research deadline. Bulk payments through donor-based agencies flow through the platform (B2P).

The system currently supports the ISO 8583 messaging standard and participants are required to ensure their technical interfaces with the NFS comply with the standards as outlined in the system's interface specifications document. The NFS is evaluating a transition to the ISO 20022 messaging standard, which will depend on whether the system determines the switch will be useful and value-added for participants.



Business model

ZECHL is a non-profit corporate entity established to provide low-cost services to the payments industry. The revenue of the NFS comes from a one-time application fee, annual participation fees, and a per transaction fee, all charged to participants. Revenue from transactions covers approximately 60% of the operating costs and the remaining 40% is covered by ZECHL. The NFS is not currently amassing enough revenue to cover its expenses. ZECHL has embarked on a review of the current business model with the aim of making the switch self-sustaining. ATM, POS, and e-money transactions face a minimal flat switching fee, with e-money transactions only charged for transaction amounts above a certain threshold (ZECHL 2020). In terms of their membership requirements and in accordance with the scheme rules, each system participant is liable to pay or is entitled to receive the fees specified by ZECHL (ZECHL 2018). These fees refer to the agreed tiered interchange fee,

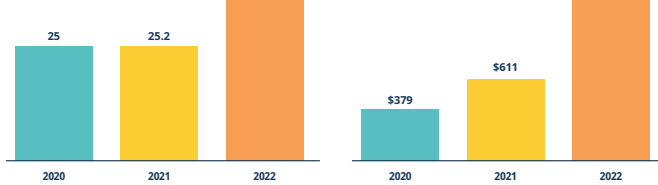
which is based on the various use cases of the system and between all PSPs. The exact fee structure is not publicly available.

Scheme rules



The scheme includes one set of rules for ATMs and POS, and another set for electronic money payments. For the latter, all relevant payment providers are required to follow them if they want to remain a member of the system.

To join, each participant must be designated by the BoZ to provide financial services and meet all general requirements for participation. Participants must either have a settlement account with the BoZ or provide proof of a settlement sponsorship arrangement with a settlement bank (ZECHL 2018; ZECHL 2020). The scheme rules also contain AML/CFT/CPF requirements for participants, including reporting suspicious activities to the Financial Intelligence Centre and maintaining updated customers/cardholder verification data, among others. All participants are required to adhere with the BoZ's AML directives and other applicable AML/CFT/CPF regulations (ZECHL 2018; ZECHL 2020). Moreover, the scheme rules require participants to adhere to Payment Card Industry Data Security Standards (PCI DSS; Bank of Zambia 2020).



Source: Stakeholder interviews 2023



Regulation

Prior to the release of the NFS, the BoZ passed the National Payment System Act of 2007 which gave the BoZ authority to regulate and oversee the nation's payment systems (Bank of Zambia 2007). Moreover, the strategy sought a payment system that would reduce transaction costs and cash use (Bank of Zambia 2013). The system was instrumental in realizing the vision of the National Payment Strategy of 2013–2017, evidenced by the BoZ issuing regulations in 2017 which regulate banks and non-banks, including e-money issuers. For example, the Banking and Financial Services Act of 2017 stipulated

prudential and supervisory requirements, such as minimum capital requirements and the prohibition of unsafe and unsound practices for banks and financial service providers. The National Payment Systems Directives on Electronic Money Issuance governs the e-money issuers' operations (Bank of Zambia 2017; Republic of Zambia 2018).

Zambia has several regulations on AML/CFT/CPF, which include the Prohibition and Prevention of Money Laundering Act of 2001, Anti-Terrorism Act of 2007, the Financial Intelligence Centre Act of 2010, Anti-Corruption Act of 2010, and others.

→ **Explicit demonstration of the value of interoperability to both the banking sector and non-bank PSPs.** Commercial banks were opposed to the idea of interoperating with prospective non-bank members due to fears about added competition and diminished market power. However, ZECHL and the BoZ showed that cross-domain interoperability would lead to better commercial prospects for the banking sector, particularly by deepening the reach of entry-level deposit accounts and stabilizing the existing retail deposit base. Cross-domain interoperability has led to a greater array of payment modes that are fit-for-purpose for end-users.

→ **A phased approach to non-bank integration reduced the complexity of multiple simultaneous configurations, which encouraged bank participants to join.** The NFS launched with bank-only channels and extended to mobile money after two years. This phased approach allowed the system to successfully pilot bank-specific channels and demonstrate the market-wide benefits of interoperability. This motivated participation by a diverse group of PSPs and significantly expanded the reach to end-users, resulting in NFS uptake.

→ **Deliberate steps ensure industry voices are heard and considered, specifically those of non-bank participants.** The BOZ and most banking participants have a vote through their shareholding membership. Non-bank PSPs have a voice through the Payments Association of Zambia. These mechanisms of communication and decision-making allow for effective oversight from the BOZ and ensure that the entire industry gets a say, deeming it as inclusive as possible.

→ **Cross-border PSP integration a challenge.** Despite a regulatory environment conducive to inclusive payments, regulations around data localization are a challenge to the integration of entrant PSPs based outside of Zambia.



MALAWI: NATSWITCH

Origin story

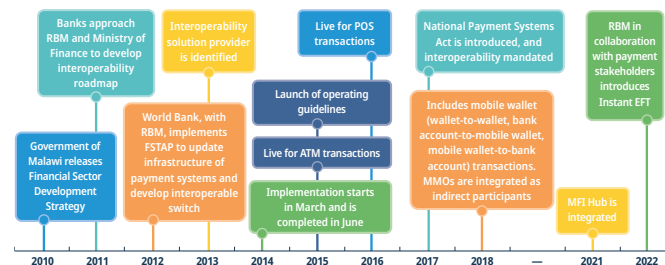


Challenge

In 2014, 75% of all transactions in the Malawian economy took place using cash and roughly half of the adult population were financially excluded (UNCTAD 2014). Malawi residents had limited access to financial services, exacerbated by a lack of coordination between public and private initiatives seeking to promote financial

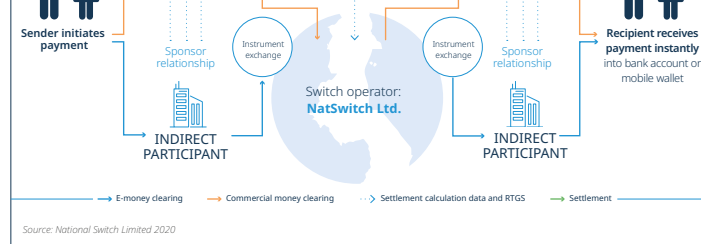
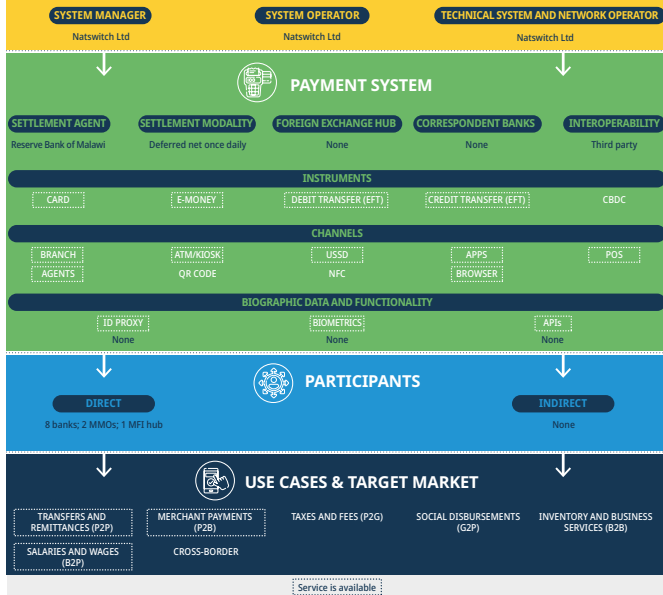
inclusion. The use of electronic payments infrastructure in Malawi was low, and the retail and business sectors lacked a convenient means of making and receiving payments (Reserve Bank of Malawi 2008). The absence of interoperability between payment service providers (PSPs), high infrastructure costs, and low fee revenue on transactions were the primary barriers to reaching additional scale with electronic payments.

FIGURE H.8 | Natswitch timeline



The RBM formed a steering committee at the outset of the project that included the Ministry of Finance, RBM, and the banks. The steering committee was responsible for providing recommendations on project scope, procurement, and the business case for the digital

payments infrastructure. These events resulted in 2018 in the introduction of non-bank PSPs, including e-money providers, as indirect participants of the system. The microfinance hub (MFI Hub) integrated with Natswitch in 2021. In 2022, Natswitch started processing EFT transfers in real-time. Natswitch Ltd is working with the Malawian finance ministry to implement an e-gateway under the 'Financial Inclusion and Entrepreneurship Scaling' project. By the end of 2023, Natswitch Ltd hopes to start testing the gateway and integrating government ministries, departments, and agencies. Natswitch Ltd plans to integrate the system with SADC's TCIB by December 2024.



Governance structure

Natswitch, the switch, is owned by Natswitch Ltd, the company, with established consultations with the participants and RBM. Natswitch Ltd is a privately-owned, not-for-profit limited liability company. It sets the scheme rules and technical standards, a process in which all shareholders participate. Natswitch is run by Natswitch Ltd under a defined service-level agreement (SLA) with its members. Owners include all eight commercial banks and two MMOs. Each shareholder has equal voting rights. The Natswitch board is the governing body, with ultimate authority over the system's operations, subject to the oversight of the RBM. The CEOs of all Natswitch

member banks hold board seats. Participants are either members or non-members, according to their ownership role and contribution to the facility's set-up costs. Natswitch's board operates with multiple committees, including a technical committee that creates task forces to deal with technical issues, and a policy committee that considers governance issues related to the national switch ecosystem. RBM has no direct decision-making power in Natswitch beyond oversight and regulation. Natswitch permits fintechs to join, however they first require a letter of no objection from other existing participants. The MFI Hub is integrated with Natswitch as one participant representing the affiliated SACCOs and MFIs.



Functionality

Natswitch offers instant clearing of low-value transactions through ATMs, POS systems, branches, internet-based payments (browser and apps), and USSD (Reserve Bank of Malawi 2022a). There are plans to include a QR code standard and proxy ID guidelines under an e-payments gateway project in development. In the future, mobile numbers and other PSP-maintained proxy IDs will be supported and routed using debit card payment standards combined with institutional BIN routing. Most transactions are single message. The exception are POS transactions, which are dual message and PIN/password authorized. Natswitch similarly provides a range of instruments, including e-money, card, credit EFT, and debit EFT (Natswitch Limited 2020).



Technical standards and use cases

Natswitch supports a set of use cases, including merchant payments (P2B), P2P, and B2B. There are plans to develop a national e-payments gateway that will facilitate social disbursements and government payments (G2P), as well as P2G, G2B, B2G, and G2G payments. Natswitch also plans to develop an e-commerce service that will facilitate B2B, B2P, and P2B payments. The gateway will be

structured such that the agency initiating a payment has a contractual relationship with the RBM or commercial bank through which the funds will be disbursed.

The switch currently runs on the ISO 8583 messaging standard. Discussions are underway regarding the creation of a translation layer to enable the conversion from ISO 8583 to ISO 20022 (and vice versa). Natswitch is also working on an agent banking functionality project to allow participant end-users to process interoperable cash-in/cash-out transactions through the switch.



Business model

Natswitch is set up as a public utility and operates on a cost-recovery basis. The World Bank, through the FSTAP, funded the start-up costs for the National Switch with a five-year loan facility of \$28 million (MK28.7 billion). \$2.2 million (MK2.7 billion) was disbursed to acquire and install the national switch.

The Bankers Association of Malawi supplemented the costs of implementing Natswitch through equity investments (Malakata 2023). To cover operating costs and maintain a not-for-loss business model, all participants of Natswitch are required to pay a joining

fee; only non-member participants pay an annual membership fee (Chiphwanya and Orama 2016). In addition, Natswitch assesses an interchange fee for all transactions. All fees, other than for POS transactions, can be passed onto the originator. The exact amounts are not publicly available. Under the Retail Payment System Interoperability Directive, PSPs cannot charge end-users to initiate POS transactions (Reserve Bank of Malawi 2017). The interchange fee is therefore covered by acceptance fees charged to the merchant. PSPs are required to consult the RBM before raising ATM and instant EFT charges. There is an explicit mandate against fee collusion among PSPs.



Scheme rules

Natswitch Ltd launched the operating guidelines for the National Switch in 2015, with revisions in 2020, to govern the operations of the Natswitch ecosystem and relationships among its members. The guidelines include member obligations, fees, and roles and responsibilities within the switch. To be considered a participant of Natswitch, a PSP applicant must be licensed by RBM as a bank or financial institution, and possess technical infrastructure and the requisite up-to-date security and integration software. The Natswitch board is responsible for formal

approval of new participants. Non-compliance with the requirements outlined in the guidelines results in penalties or fines. The scheme rules also set out security requirements based on PCI-DSS and EMV standards and SWIFT international security guidelines.

The scheme rules include mechanisms for dispute management between PSPs. Disputes are filed via an online portal available to PSPs. As per the payment systems law, if a dispute is not resolved via the dispute management portal within 72 business hours, the offending PSP receives a penalty (Natswitch Limited 2020). Natswitch follows conciliation, mediation and arbitration processes when resolving disputes between PSPs, in line with payment systems law.



Volumes and values processed by the payments system

End-user transaction volumes between 2018 and 2022 had a 51% CAGR, while values saw a 53% compound annual growth rate (CAGR) (Figure 64). Average annual transaction values were \$17 in 2020 and \$27 in 2022, indicating increased adoption of higher-value transaction use cases. Overall, only around 20% of transaction values that go through Natswitch are off-us transactions.



Regulation

Prior to the launch of Natswitch, in 2001, the National Payments Council together with RBM and the Bankers Association of Malawi endorsed the Malawi National Payments Systems Vision and Strategy framework. The strategy framework was subsequently updated in 2008 and governed the future of payment systems development until 2013.

The RBM also implemented several laws and regulations targeted at Malawi's payment ecosystem and its

respective actors. This includes the Malawi National Payment System Bill, drafted in 2014 and circulated in the Malawi Gazette in 2015; and the 2017 Interoperability of Retail Payment Systems Directive, which mandated the connection of all licensed PSPs to Natswitch (Reserve Bank of Malawi 2017).³ Other notable regulations include the 2011 Mobile Payment Systems Act, 2016 Payment Systems Act, and 2019 directives around e-money schemes (IFC 2021). Supporting regulations include the AML/CFT/CPF law (2006), and the Electronic Transaction and Cyber Security Act (2016).

Several payments-related laws created the enabling environment for Natswitch and provided clarity to the industry about what is legally permissible within the system's operations. A strong legal framework supports the roll-out of an IPS that can drive inclusion.

→ **Central bank and development partner-led initiatives are key catalysts to driving implementation.** Although private sector-owned and governed, Natswitch was originally led by the RBM with input and funding from the World Bank. The RBM established committees with commercial players to ensure that industry views were incorporated during the system's design phase, ultimately leading to greater participant buy-in.

→ **Including multiple channels, instruments, and participant types enables scale.** An array of payment channels supported by the IPS and facilitated by different participant types, provides end-users with a variety of choices that suit their means and contexts. Natswitch supports important mobile money and banking channels. Further incorporating higher-volume use cases like G2P payments will provide the foundation for a digital value circulatory system. The cycle will only be complete with comprehensive P2B and B2B payments.

→ **Mandated interoperability ensures successful integration of non-bank and bank PSPs, but it is not a panacea.** Commercial bank participants may be reluctant to allow non-bank PSPs to integrate with the system, for reasons of market control and fear of losing competitive advantage. Directives around interoperability in Malawi ensured that private-sector resistance could not override the interoperability goals for the system. Interoperability mandates are effective tools by regulators to bring industry onboard. However, they do not guarantee scale for the switch, especially if there are local dominant PSPs processing a significant volume of on-us transactions. In the case of Natswitch, the switch continues to be under-utilized because most domestic retail traffic is routed as on-us transactions.

→ **An ISO 20022 API integration layer supports inclusion.** The enhanced interoperable technical features and improved governance aspects of ISO 20022 have the potential to enable Natswitch to reach greater scale, but the downside is the cost of system upgrades. An efficient ISO 20022 API layer creates an inclusive compromise, as it does not require an industry-wide upgrade and can support smaller institutions that focus on vulnerable market segments.



CEMAC: GIMACPAY

Origin story



Challenge

The Central African Economic and Monetary Community (CEMAC) includes Cameroon, the Central African Republic, Chad, the Republic of Congo, Equatorial Guinea, and Gabon. The region has a large proportion of financially excluded adults. Only 23% of the adult population in CEMAC had a financial account in 2017 (World Bank 2018). As a monetary union, CEMAC has a single currency,

Central African Franc (XAF), and a single regional central bank, Banque des États de l'Afrique Centrale (BEAC). Given that the majority of retail transactions, both domestically and cross-border, were conducted in cash, the regional central bank faced difficulties controlling capital flows and monitoring money laundering and terrorist financing within the region. Likewise, the reliance on cash constrained merchant retail trade activity between countries within the region.

a representation of the digital payment ecosystem. GIMACPAY reinforces the BEAC's mission to create a cashless society and facilitate capital flows in the region (Agence Ecofin 2020).

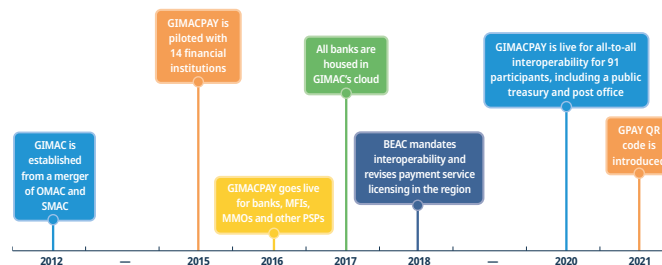


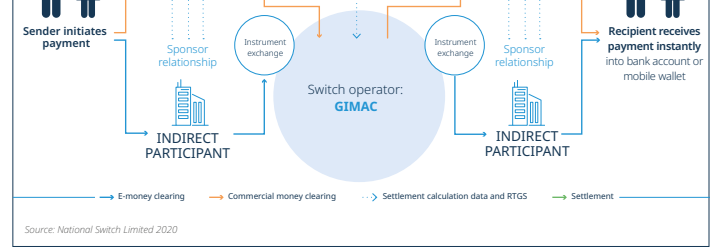
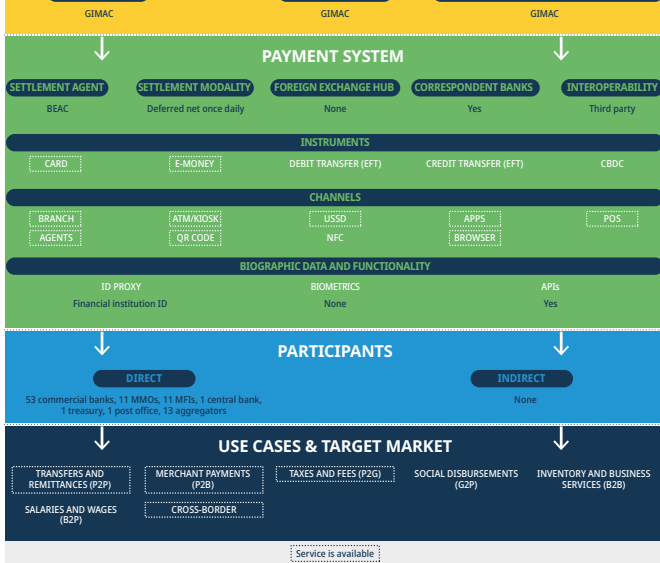
Timeline

The objective of forming GIMAC was to facilitate regional card payment integration, to interconnect all banks, MMOs, and

be the entity to operationalize the interoperability vision. GIMACPAY, the system, was officially launched in July 2020. Twenty-nine participants had undergone the onboarding process by the official launch date. The remaining PSPs in the region onboarded in subsequent years. GIMACPAY introduced the QR code channel, called GPAY QR Code, as a way to boost digital payments in 2021 in response to the region's COVID-19 measures (GIMAC 2021).

FIGURE H.12 | GIMACPAY timeline





Governance structure

GIMACPAY was established by commercial banks and the BEAC as a private company dedicated to digital payments. It follows a joint ownership model and public-private partnership governance model, where BEAC owns a 99.2% share and the remaining 0.8% is divided between the commercial bank members in proportion to their size. BEAC contributed the majority of the start-up funding, as commercial banks were reluctant to invest in another regional interbank project at the time, SYSTAC. BEAC plans to transfer the majority shareholding to GIMACPAY's members in the future, though no timeline has been set. In addition to owning the system, BEAC is also the overseer and settlement

agent. GIMAC performs the role of system governance and technical operator.

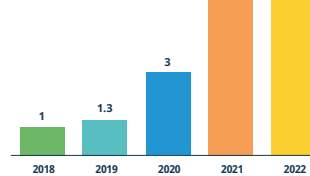
Direct participants include commercial banks, MMOs, MFIs and payment system aggregators, post offices, BEAC, and the treasuries.³ The treasury of Cameroon is the only one connected to date. In 2020, at the CEMAC ministerial committee, it was recommended that treasuries be linked to GIMACPAY in order to benefit from secure solutions for expenditure and collections (GIMAC 2021). Participants in GIMACPAY sign membership agreements with GIMAC and BEAC. GIMAC hosts meetings between its members and participants to collect comments and proposals on the adjustments to scheme rules and creation of new guidelines (GIMAC 2021).

population (CEMAC interview 2023). Regarding standards, GIMACPAY currently uses ISO 8583, but the ISO 20022 protocol will also be available soon.

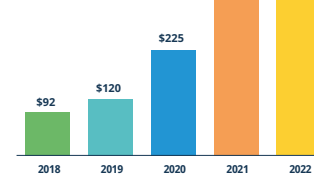
As of 2017, all commercial banks in the sub-region were housed in GIMAC's cloud. Cloud hosting was always part of the specifications of the system. For GIMAC, it was important to provide a hosting option for banks and MMOs to facilitate quick integration with the system. Those PSPs that did not yet have the infrastructure could be hosted on the GIMAC platform.



GIMAC's governance framework defines levels of availability, participant assistance, response time, incident resolution, and an escalation plan, as well as penalties in the event of a service failure (GIMAC 2021). The scheme rules are not publicly available.



Source: Stakeholder interviews 2023



Regulation

BEAC fulfills the statutory sovereign function of regulating market structures for payments, including processing technical approvals, requests or authorizations, and payment solutions. Banking regulations set by BEAC apply to the six member states (BEAC 2021). There are four regional institutions that assist BEAC—namely, the regional banking sector regulator (COBAC), the securities regulator, the Central African Financial Market Supervisory Commission (COSUMAF), the task force against money laundering (GABAC), and the Development Bank of Central African States. BEAC's Strategic Plan 2017-2023 included the modernization of the CEMAC's payment system within the framework of policies aimed at increasing the proportion of the adult population with accounts (BEAC 2022). The principal laws on anti-money laundering in the

CEMAC zone are Regulation No. 01/CEMAC/UMAC/CM of 2016 and the CEMAC Currency Exchange Regulation n°02/18/CEMAC/UMAC/CM (BEAC 2016; BEAC 2019). The latter imposed more stringent criteria on the transfer, payment, and settlement of routine business transactions (Norton Rose Fulbright 2019).

In the CEMAC there is a regional central bank, a regional banking sector regulator, as well as country-specific regulations. Although COBAC sets financial services regulations for the region, other regulations, such as data protection and consumer protection regulation, are drafted and implemented at the domestic level by different regulators. This may result in uncertainty, overlap, or conflict between national regulations within the region (Stakeholder interview 2023; Cooper 2018; World Bank 2018).

- **The proliferation of multiple overlay services and channel offerings enhance access and adoption of digital payments.** The availability of a broad spectrum of channels owes much to the widespread participant ecosystem. The emergence of GPay QR code overlay service has further streamlined merchant transaction, eliminating the upfront incentives in POS hardware for merchants to accept payments. Moreover, the increase in transactions along with the new additions to the system indicate that end-users are increasingly using the system for lower value transactions.
- **The regional central bank played a critical role in establishing the IPS.** BEAC holds a majority share in the IIPS and largely funded its development. This was necessary given competing payments projects at the time that made PSPs reluctant to contribute. The central bank ownership creates a steady foundation for the IIPS participants across the region and contributes to its status as a public utility for end-users.
- **Providing domestic and regional instant payment functionalities enhances system scalability.** Although termed a regional IIPS, GIMACPAY also provides domestic IIPS functionality to end-users in CEMAC member states. This fills an important functionality gap in the Central Africa region and allows for a consolidation of scale, ultimately leading to the lowest possible per unit transaction costs.

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